Alkenes and Alkynes

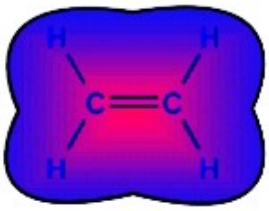
DISCLAIMER

"The content provided herein are created and owned by various authors and licensed to Sorting Hat Technologies Private Limited ("Company"). The Company disclaims all rights and liabilities in relation to the content. The author of the content shall be solely responsible towards, without limitation, any claims, liabilities, damages or suits which may arise with respect to the same."

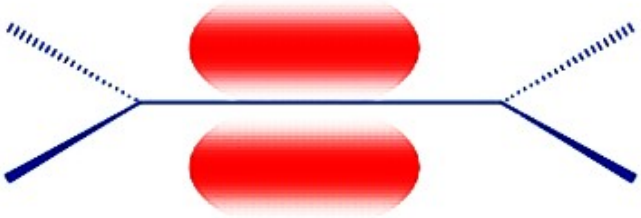
Alkenes and Alkynes

Why do Addition Reactions Occur?

1. Electrons present an exposed region of electron density in a molecule that is above and below the σ bonding framework. π electrons are therefore more available for reaction than σ electrons.
2. Electrophiles are attracted to the exposed electron density of π bonds.



An electrostatic potential map of ethene shows the higher density of negative charge in the region of the π bond.



The electron pair of the π bond is distributed throughout both lobes of the π molecular orbital.

Alkenes and Alkynes

Electrophilic Addition Reaction

The Regio or Stereochemistry of addition reactions of alkenes processes that can be add molecules of water, halogens, carbon and other method across alkenes events that cleavage double bonds and provide more highly oxidized compounds alkyne reactions that are analogous to alkene reactions.

Addition Reactions of Alkenes

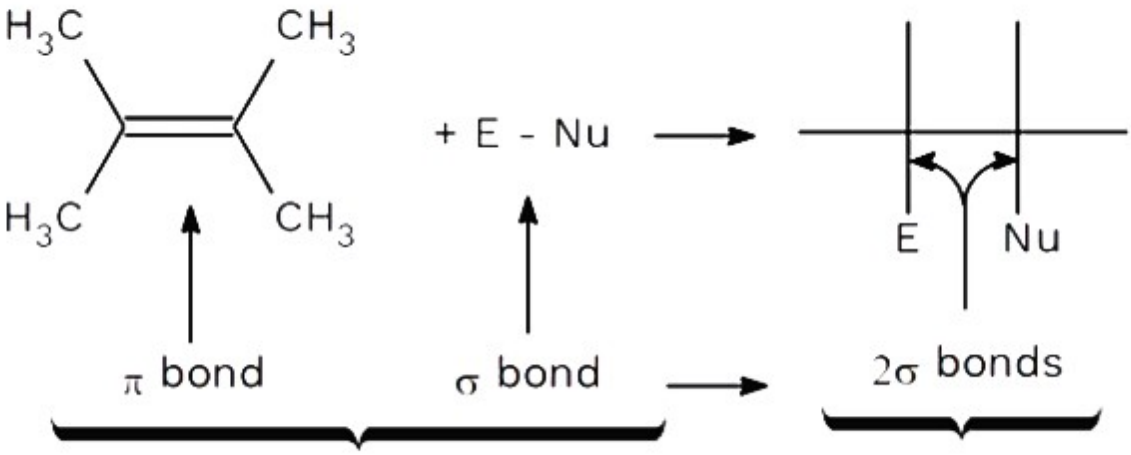
$E^{\oplus}$ use for an electrophilic portion of a reagent and $Nu^{\ominus}$ use for a nucleophilic portion, as follows.



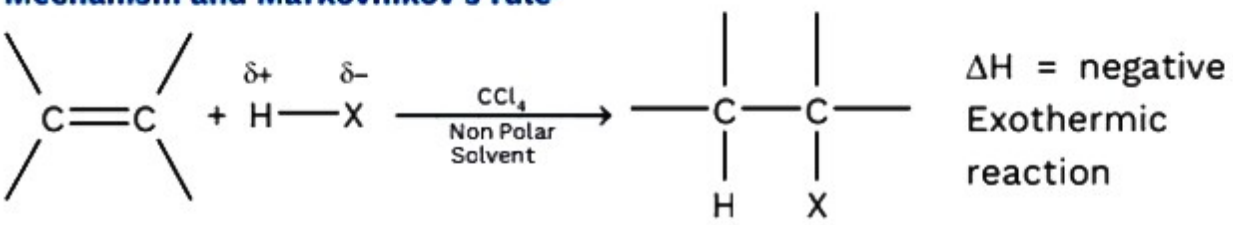
Points to remember

- Electrophiles seek electrons. They are attracted to sites of negative charge and have the property of being electrophilic.
- Nucleophiles are electron donors. They are attracted to sites of positive charge and have the property of being nucleophilic.

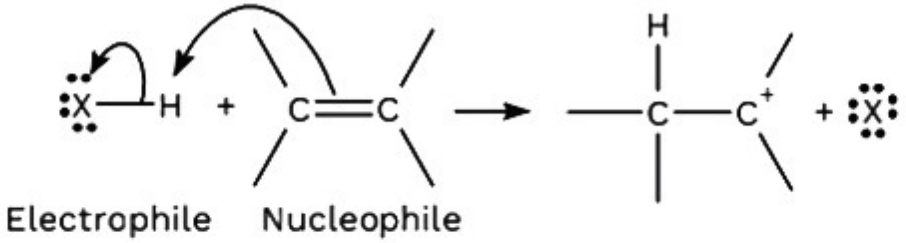
π electrons are in higher energy orbitals than σ electrons, and π bonds are weaker than σ bonds. Thus, reactions that convert π bonds to σ bonds are usually energetically favorable. In an alkene addition reaction, one π bond and one σ bond are converted to two σ bonds, lowering the energy of the system.



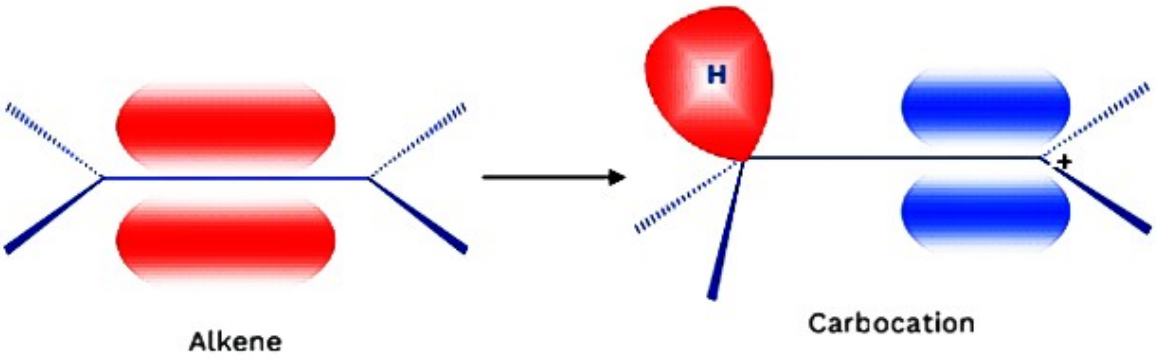
Electrophilic addition of hydrogen halides to alkenes
Mechanism and Markovnikov's rule



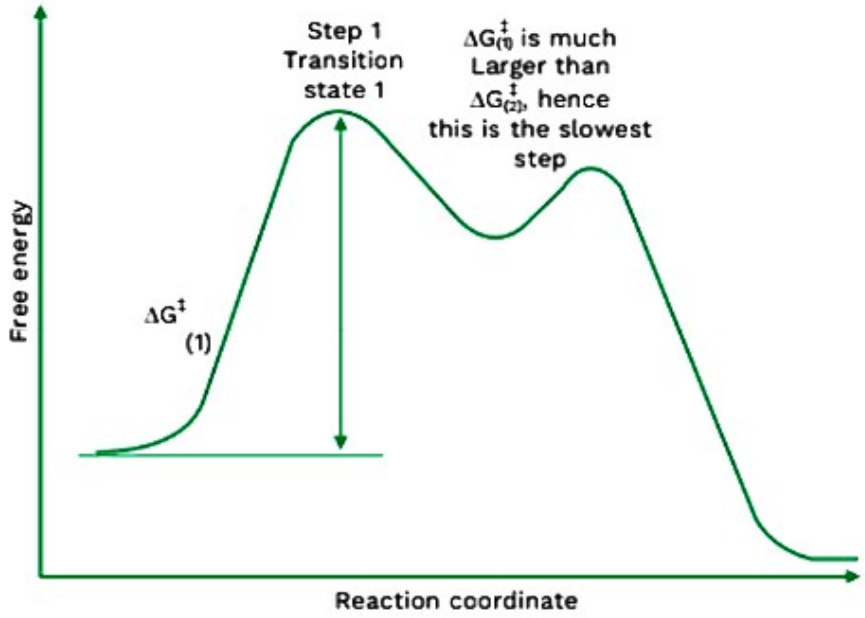
Step 1 :



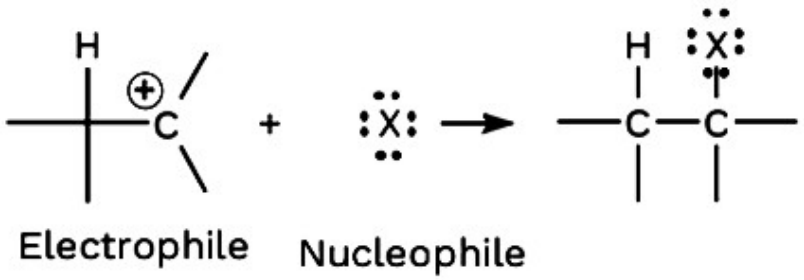
Orbital Diagram of addition reaction of halogen acid at alkene

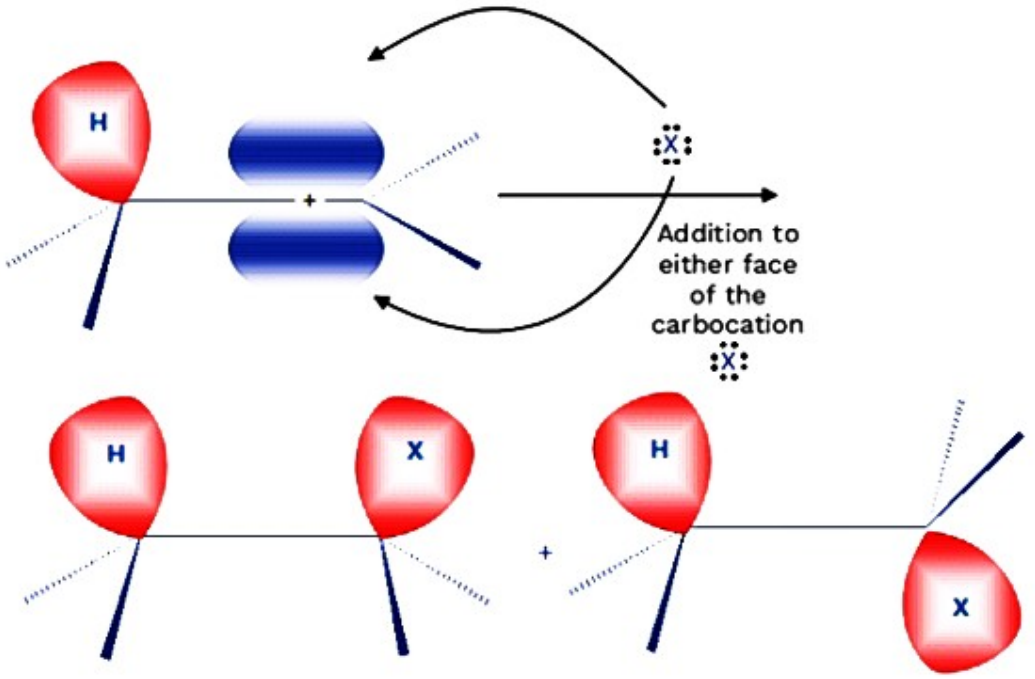


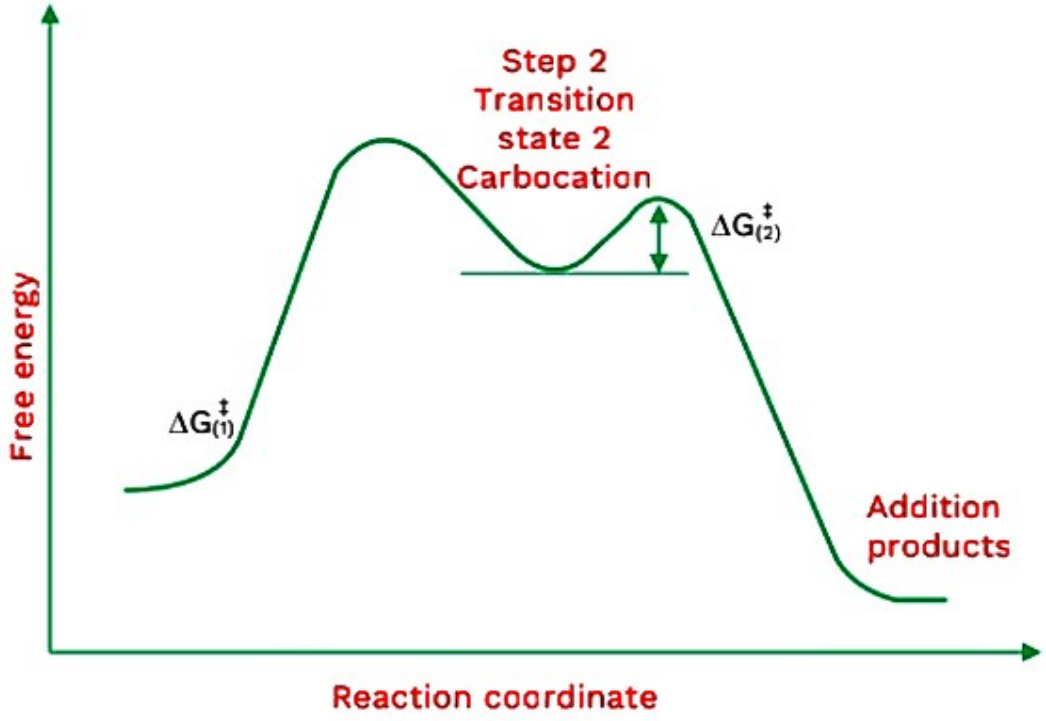
The halide ion then acts as a nucleophile, donating an electron pair to the highly reactive carbocation.



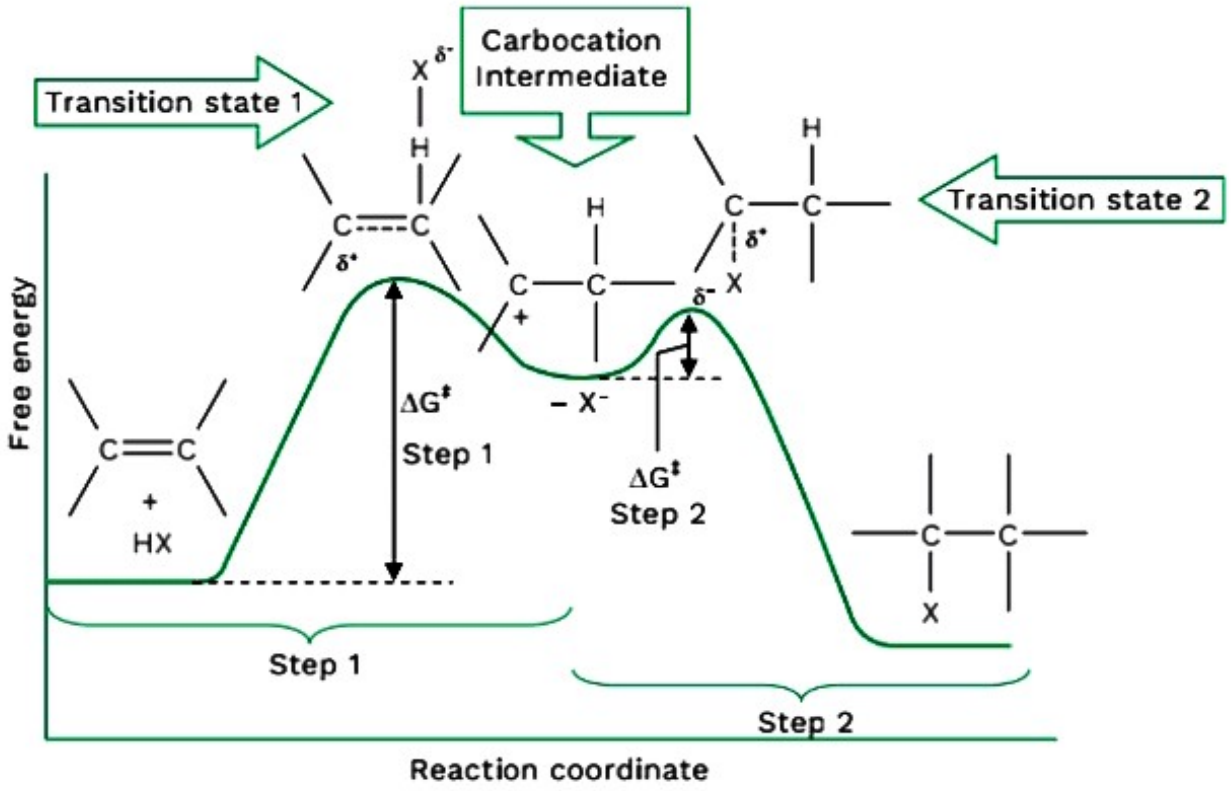
Step 2





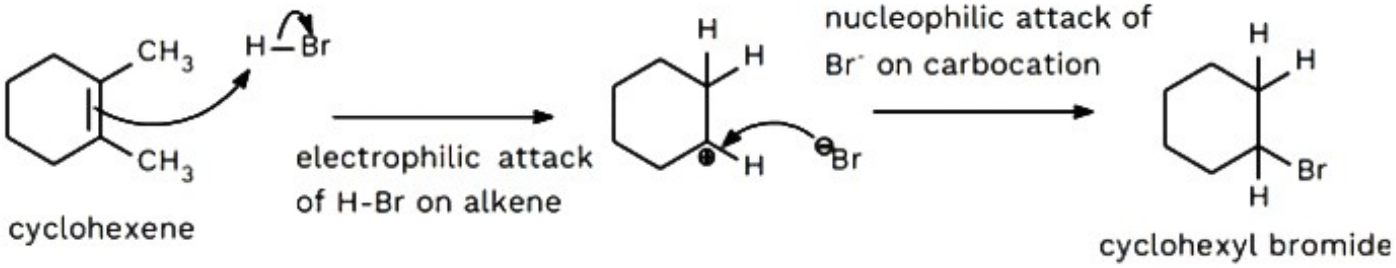


Complete Energy Profile Diagram



Addition Reaction of HBr to Symmetrical Alkenes

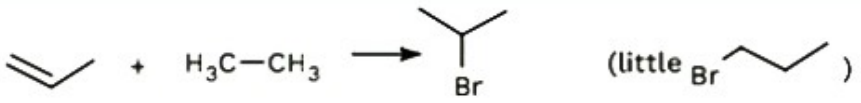
Addition reactions of HBr to symmetrical alkenes s follow electrophilic addition mechanism.



$$CH_2 = CH_2 + HBr \longrightarrow CH_3 - CH_2 - Br$$

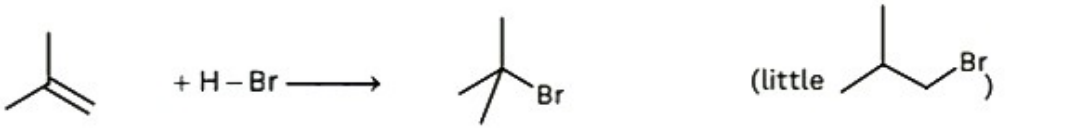
$$CH_3 - CH = CH - CH_3 + HBr \longrightarrow CH_3 - CH_2 - \underset{\displaystyle Br}{\underset{|}{CH}}CH_3$$

Addition reaction of HBr to Unsymmetrical Alkenes (Markovnikov Rule)



2-Bromopropane 1-Bromopropane

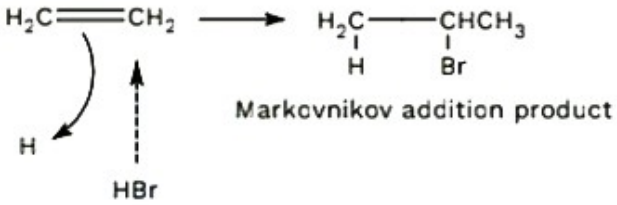
When 2-methylpropene do reaction with HBr, the main product is 2-bromo-2-methylpropane, not 1-bromo-2-methylpropane:



2-Methylpropene 2-Bromo-2-methylpropane 1-Bromo-2-methylpropane

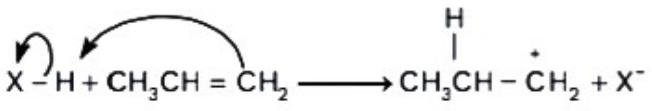
According to Markovnikov's rule the addition of HX to an alkene, the hydrogen atom connect to the carbon atom of the double bond that already has the greater number of hydrogen atoms.

The alkene carbon atom with the greater number of hydrogen atoms receives the hydrogen.

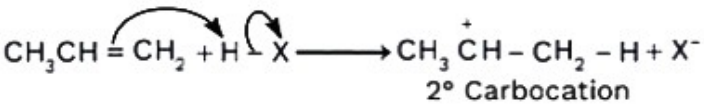


Markovnikov addition product

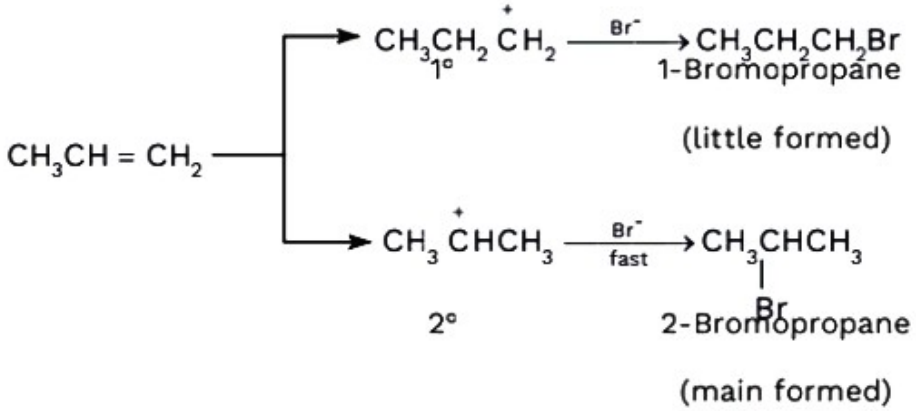
Advance Markovnikov's Addition of $E^{\oplus}$ in such manner that formation of most stable carbocation is possible.
Hydrogen bromide (HBr) provides an electrophile, H+ which attacks the double bond to form carbocation.



1° Carbocation
(less stable)



2° Carbocation
(more stable)



(i) Formation of Carbocation is Rate determining step.

(ii) Rearrangement of carbocation to more stable carbocation can occur to form a more stable product.

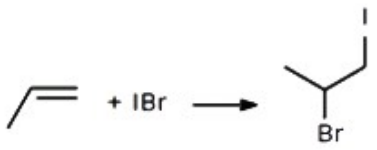
(iii) Regiochemistry-Morkovnikov's Addition of HX.
Addition of HX on unsymmetrical alkenes ($R-CH=CH_2$) takes place according to Markovnikov's rule which states that, "the negative part of abbendum is added on the carbon atom carrying lesser number of hydrogen atoms".

(iv) Reactivity order of different hydrogen halide towards addition
HI > HBr > HCl > HF

ILLUSTRATION

Q67 **Draw the structure and name of the product that would be obtained from the ionic addition of IBr to propene..**

Sol.



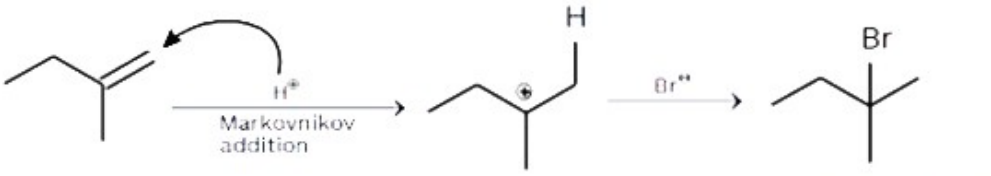
2 Bromo-1-iodopropane

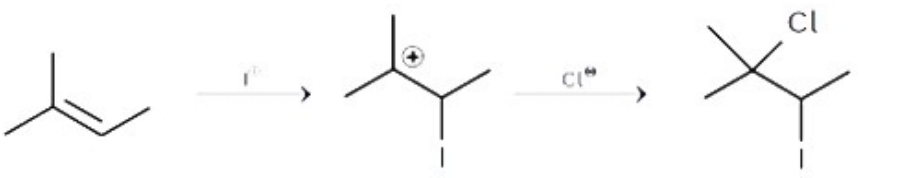
Q68 **Write mechanisms for the following addition reactions:**

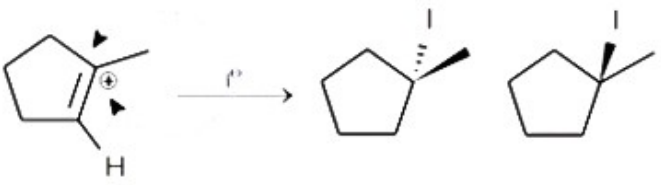
(a) 

(b) 

(c) 

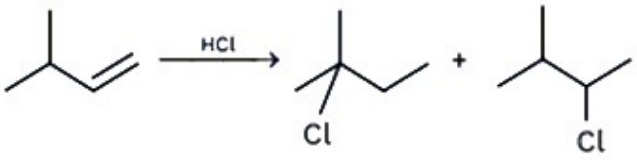
Sol. (a) 

(b) 

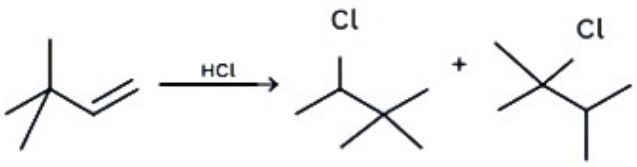
(c) 

Q69 **Provide mechanistic explanations for the following observations:**

(a)

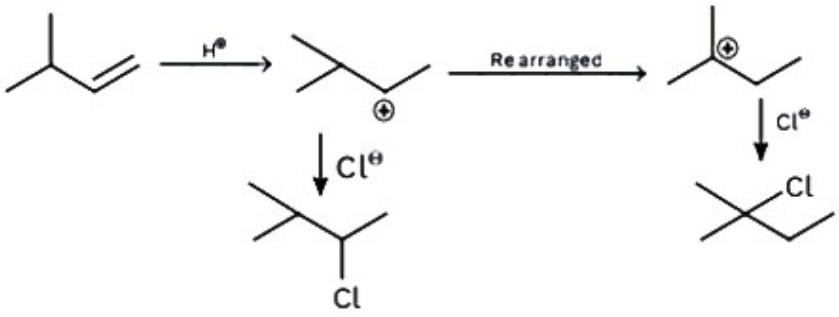


(b)

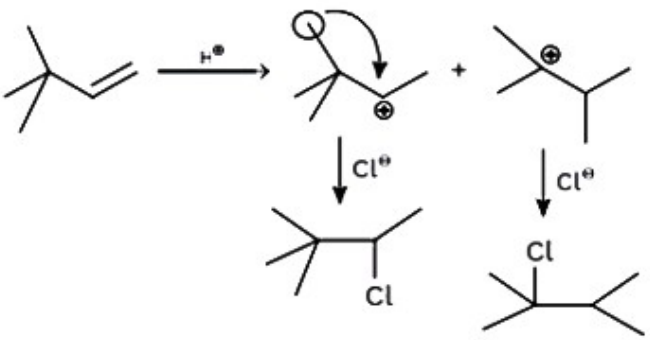


Sol.

(a)

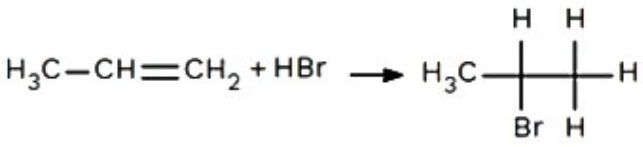


(b)

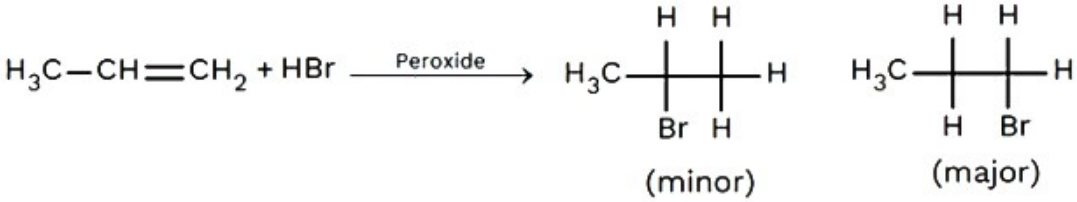


Anti Markovnikov's Principle / Kharasch Effect / Peroxide Effect–

To understand anti Markovnikov's principle let us consider the following reactions :



Markovnikov's rule



Anti Markovnikov's rule

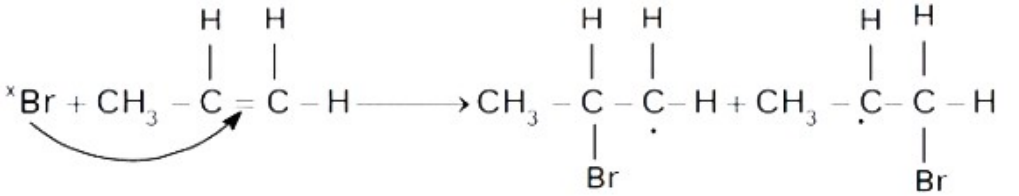
It is based on free radical mechanism

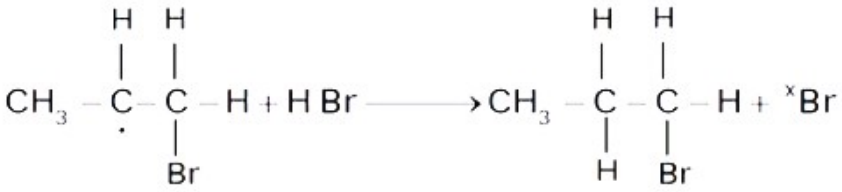
(i) $R-O-O-R \longrightarrow RO^{\cdot} + {}^{\cdot}OR$

Peroxide

(ii) $RO^{\cdot} + H-Br \longrightarrow ROH + {}^{\cdot}Br$

(iii)



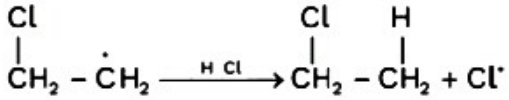


$^{x}Br + {}^{x}Br \longrightarrow Br_2$

Points to remember

It is interesting to note that anti markovnikov addition in the presence of peroxide is not applicable for HCl and HI.

(i) In the case of H – Cl, the step



is endothermic (as $\Delta H^{\circ} = +12.6 KJ\ mol^{-1}$)

Points to remember_continued

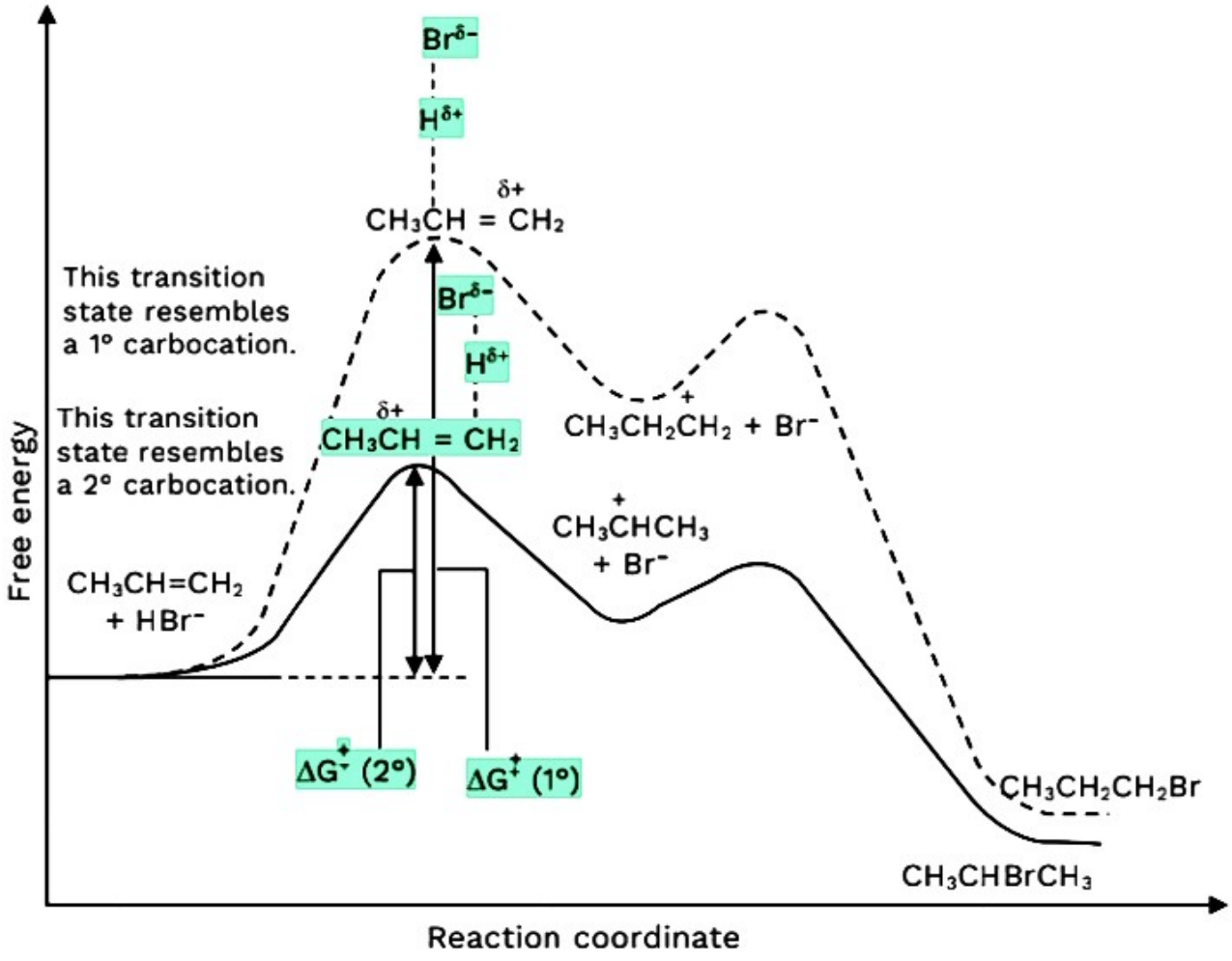
(ii) In the case of H – I, the step

$$CH_2 = CH_2 + I^{\cdot} \longrightarrow \dot{C}H_2 - \underset{|}{CH_2}\ I$$

is endothermic (as $\Delta H = 46\ KJ\ mol^{-1}$)

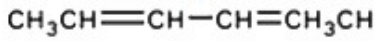
But in the case of HBr both steps are exothermic, which results spontaneous reaction.

$CCl_4, CBrCl_3$ etc. can also be added to alkene in anti markovnikov manner.



Electrophilic addition to Dienes

double bonds are separated by one single bond



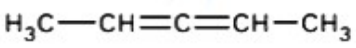
a conjugated diene

double bonds are separated by more than one single



an isolated diene

double bonds are adjacent



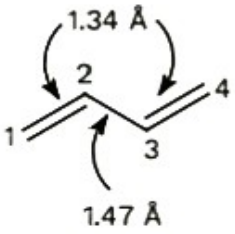
a cumulated diene an allene

Electron Delocalization in 1,3-Butadiene

Bond lengths of 1,3-Butadiene

The carbon-carbon bond lengths of 1,3-butadiene have been determined and are shown here:

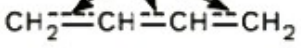
Double bonds are shorter then single bonds



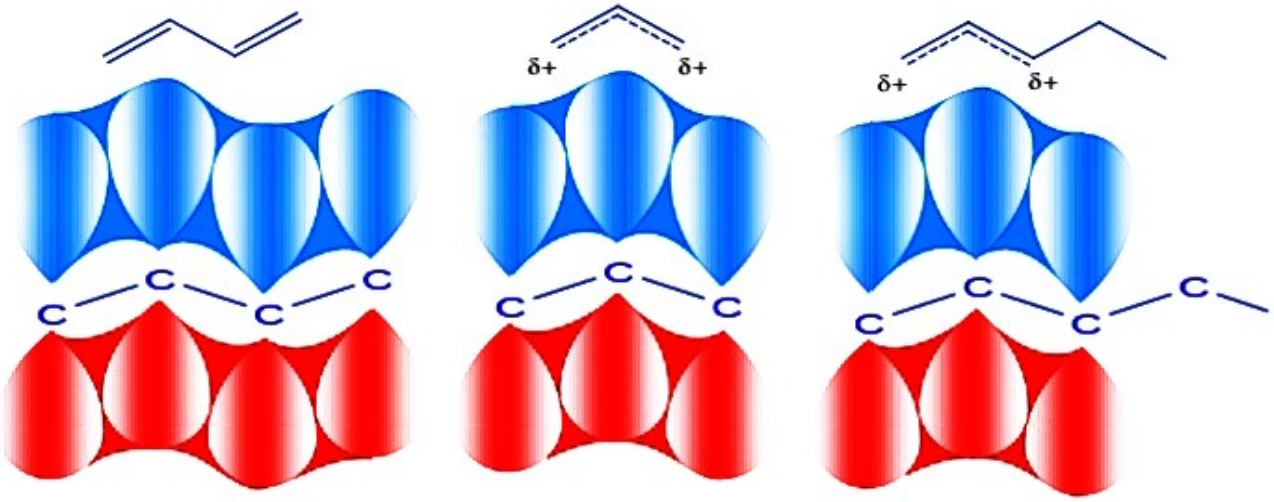


resonance contributors

delocalized electrons



resonance hybrid



1,3-butadiene
(a conjugated diene)

The allyl radical

An allylic carbocation

Relative stabilities of dienes

most stable > conjugated diene > isolated diene > cumulated diene > least stable

Q70 Rank the following dienes in order of stability

(A) $CH_3CH = CHCH = CHCH_3$

(B) $CH_3CH = CHCH = CH_2$

(C) $CH_3C(CH_3) = CHCH = CCH_3(CH_3)$

(D) $CH_2 = CHCH = CH_2$

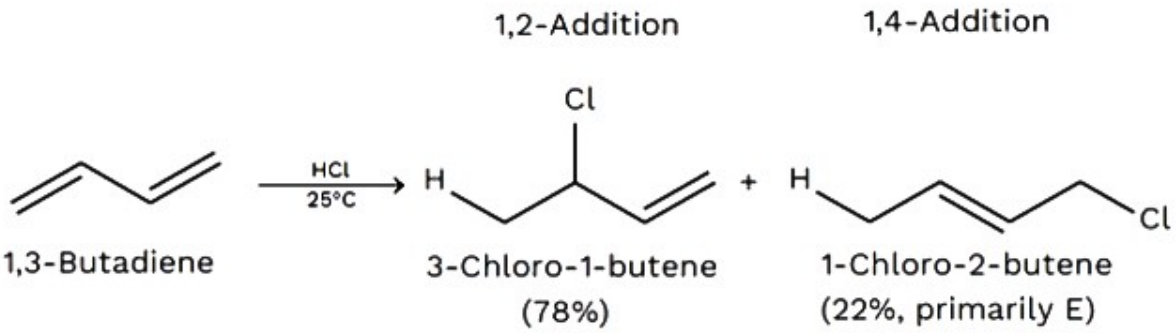
Sol. C > A > B > D

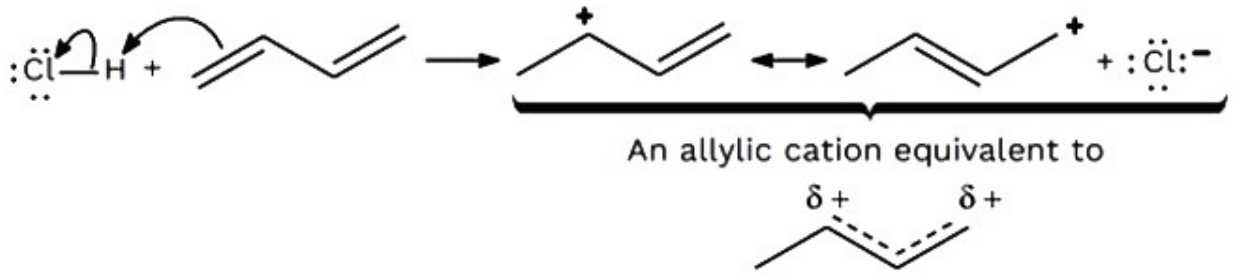
Resonance + α–H numbers

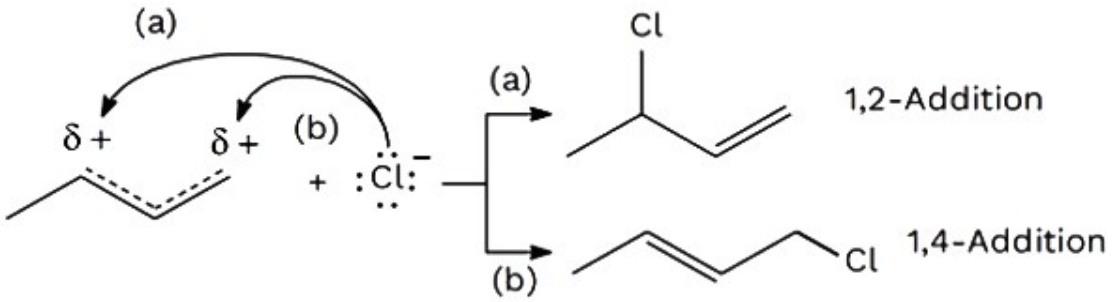
How Dienes React?

Dienes, like alkenes and alkynes, are nucleophiles because of the electron density of their π bond. Therefore, they react with electrophilic reagents. Like alkenes and alkynes, dienes undergo electrophilic addition reactions. Addition at the 1- and 2-positions is called 1,2-addition or direct addition.

- Addition at the 1- and 4-positions is called 1,4-addition or conjugate addition.

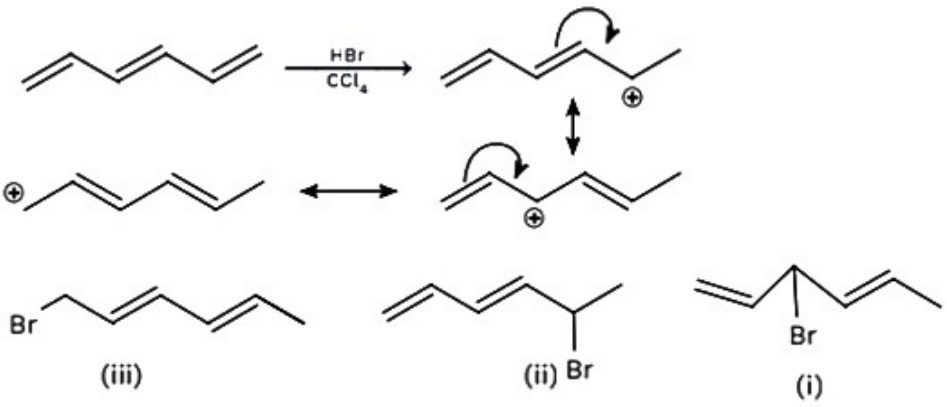






Q71 **Which products is obtained from the reaction of 1,3,5-hexatriene with one equivalent of HBr? Ignore stereoisomers.**

Sol.



Q72 **Write the products of the following reactions, ignoring stereoisomers (equivalent amounts of reagents are used in each case):**

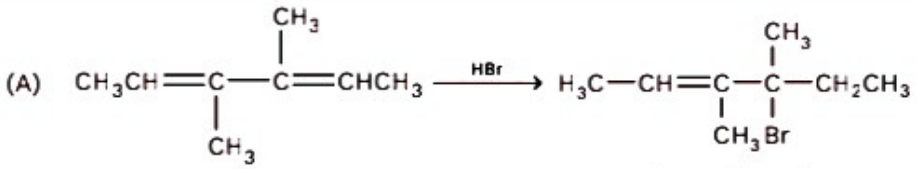
(A) $CH_3CH=C(CH_3)-C=CHCH_3 \xrightarrow{HBr}$

(B) $CH_3CH=CH-C(CH_3)=CHCH_3 \xrightarrow{HBr}$

(C)  $\xrightarrow{HCl}$

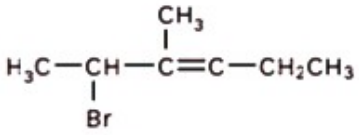
(D) $CH_3CH=CH-C(CH_3)=CHCH_3 \xrightarrow{HBr}$

Sol.

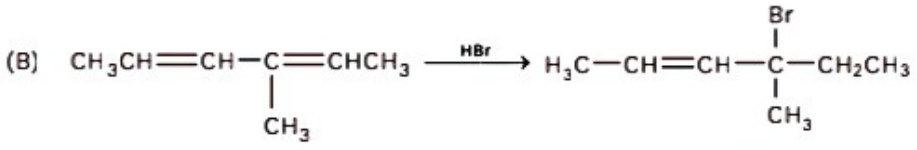


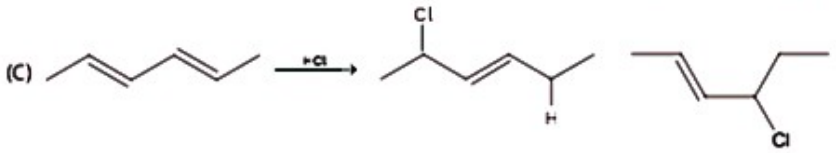
(1,2 addition)

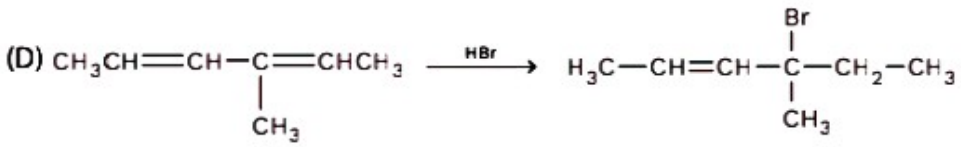
+

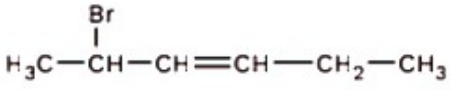


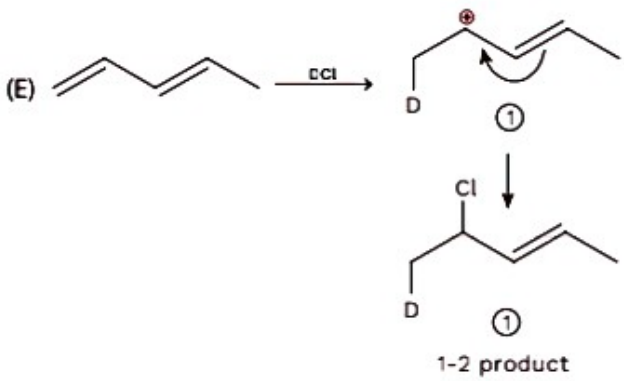
(1,4 addition)



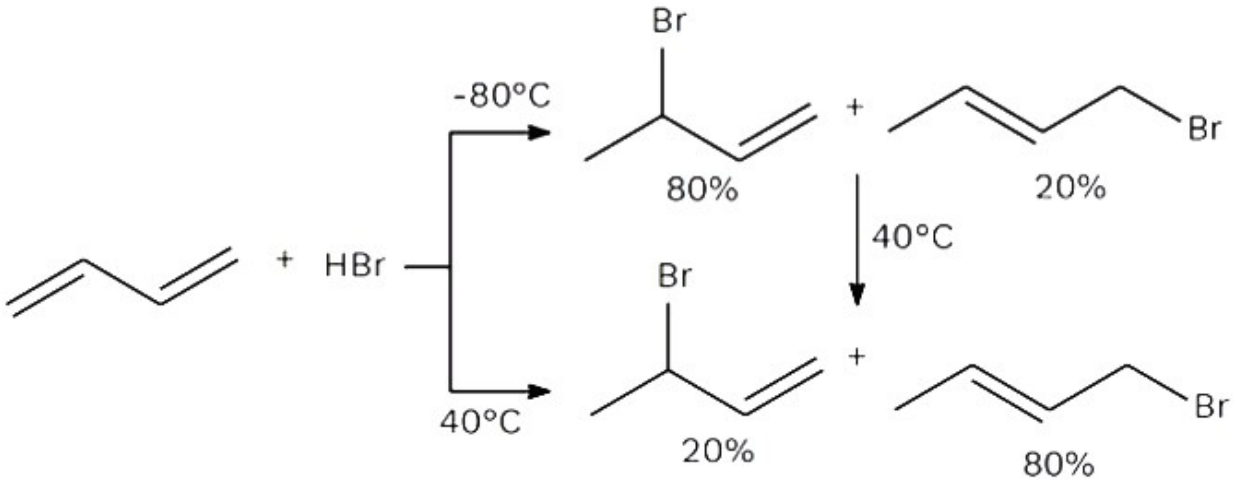






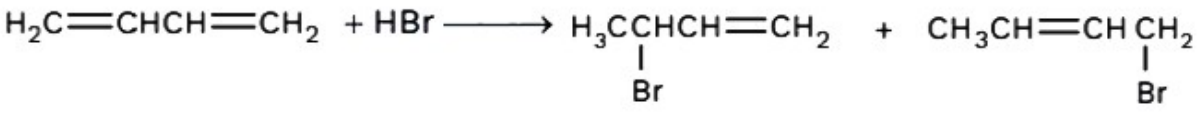


At different temperature, different major Products (KCP and TCP)



Comparison of Kinetic Control and Thermodynamic Control of a Chemical Reaction

If in a reaction produces more than one product, then that product that is formed most rapidly is called the kinetic product, and the most stable product is called as thermodynamic product. Reactions which produce the kinetic product as the major.

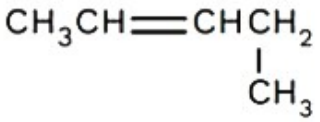


1,3-butadiene

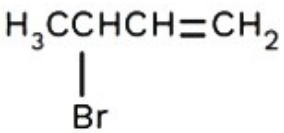
1,2-addition product
Kinetic product

1,4-addition product
Thermodynamic product

The favoured products at higher temperature in a reversible reaction are those that are most stable. In this case the reaction is said to be under thermodynamic (or equilibrium) control, (TCP), and the major products are called the thermodynamic (or equilibrium) products (TCP)



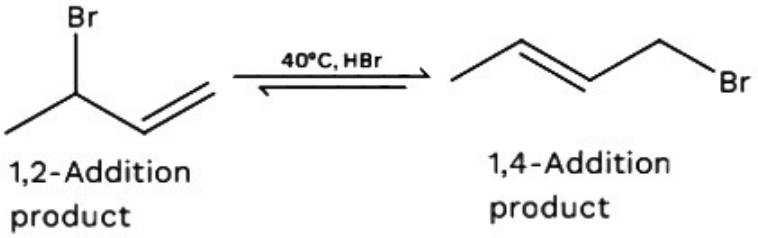
1,4-addition product
thermodynamic product



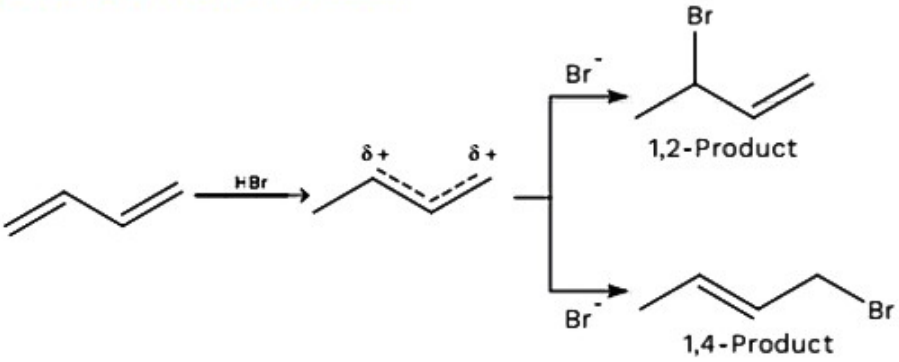
1,2-addition product
kinetic product

The favoured products in a reaction at low temperature are those formed by the pathway having the small energy of activation barrier. The reaction product considered under kinetic (or rate) control (KCP)product, and the

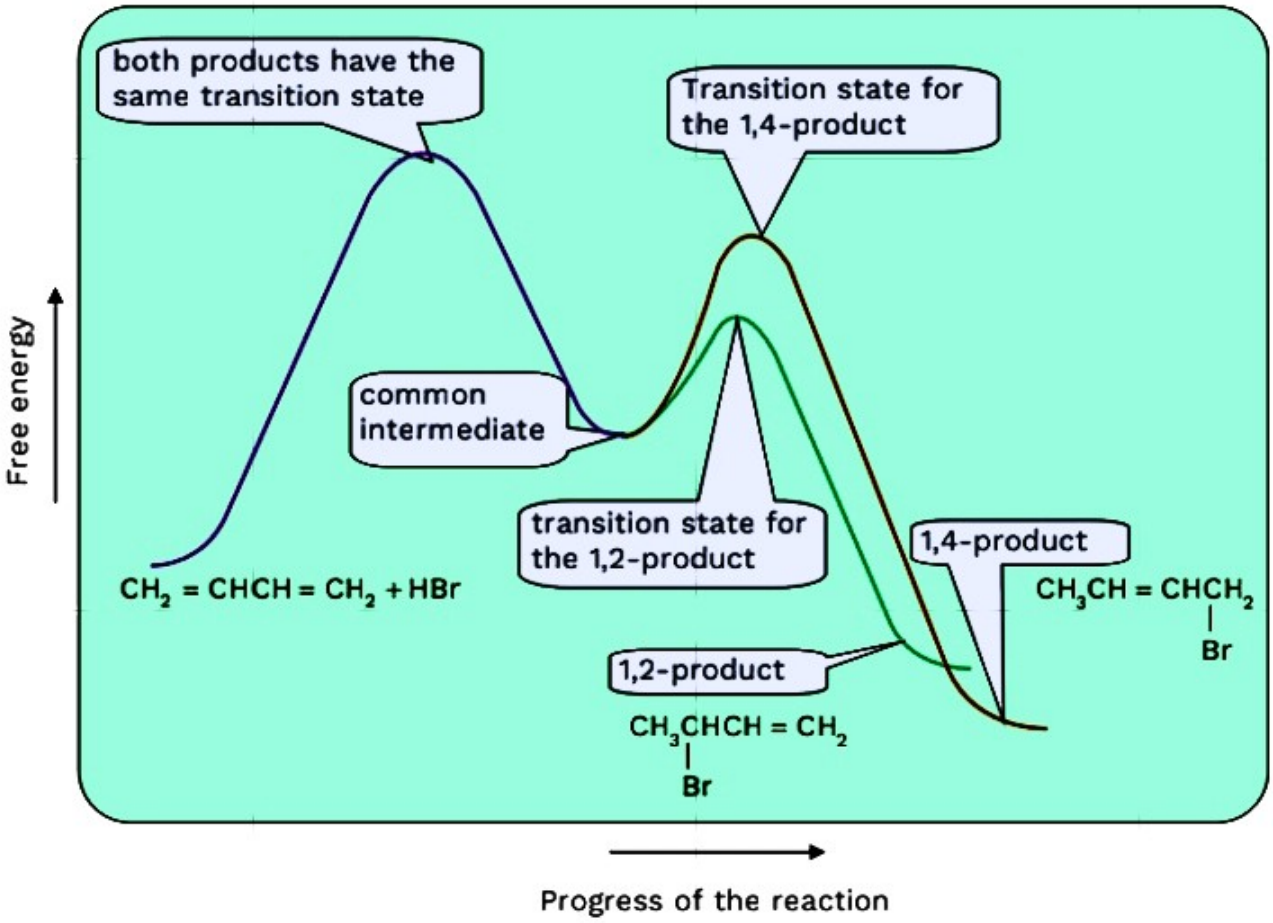
predominant products are called the kinetic products.



Regio selective Nature of reaction

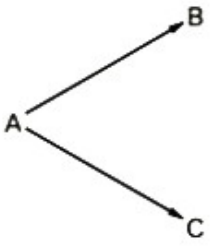


The reaction of the bromide anion with the allylic cation determines the regioselectivity of the reaction.



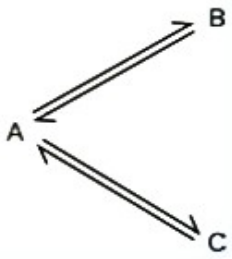
kinetic control:
both reactions are irreversible

the major product is the one formed most rapidly



thermodynamic control:
one or both reactions are reversible

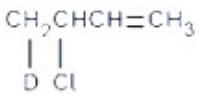
the major product is the one that is the most stable

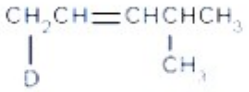


Q73 CH_2 CHCH $CHCH_3$ + DCl $\xrightarrow{78\,^\circ C}$

1,3 pentadiene

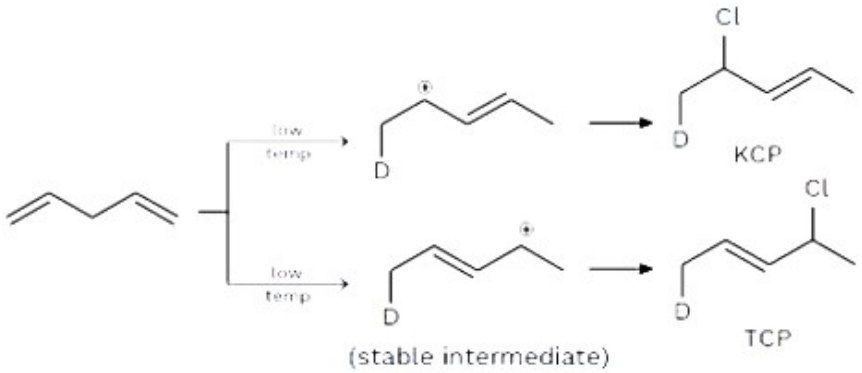
Products Formed, Explain the mechanism.



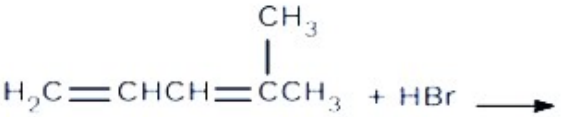


1,2-addition product 78% 1,4-addition product 22%

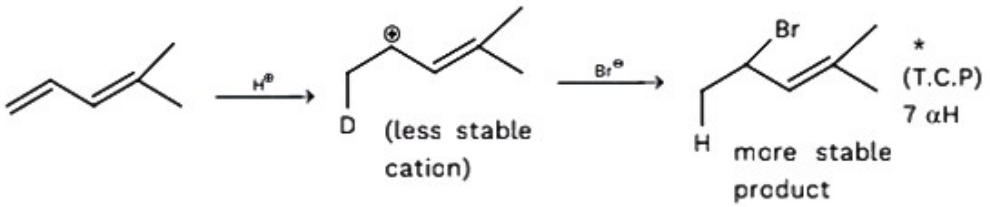
Sol.

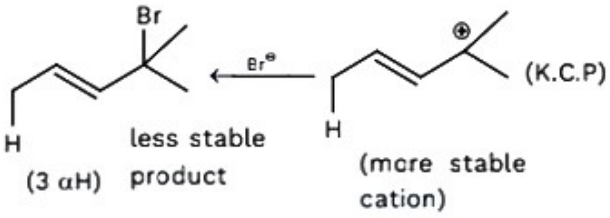


Q74



Sol.

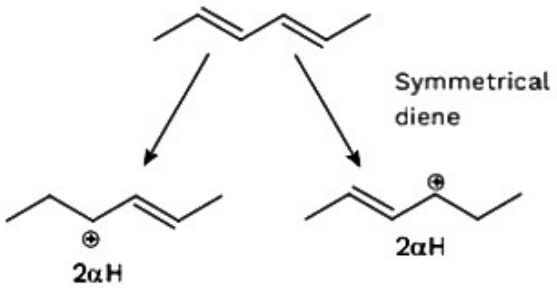




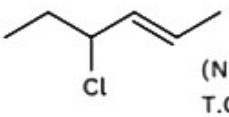
Q75 $CH_3CH=CHCH=CHCH_3 \xrightarrow{HCl}$

2,4-hexadiene

Sol.

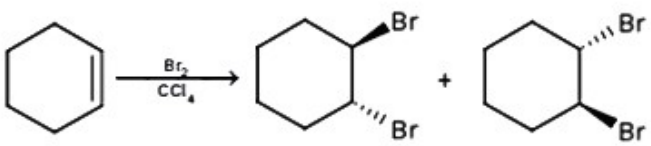


Same product will be formed



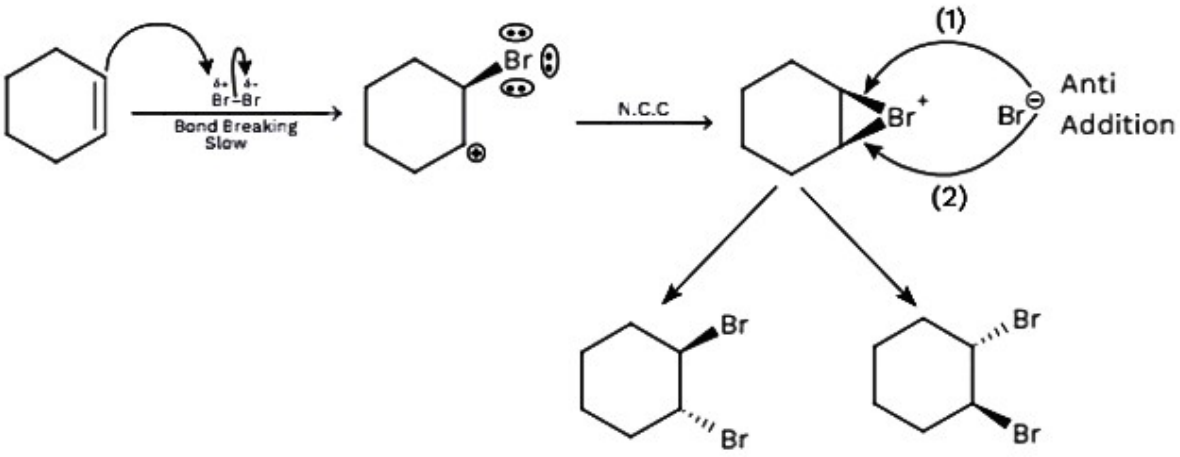
(No K.C.P. and T.C.P. applicable)

Halogenation at Alkene & Alkyne

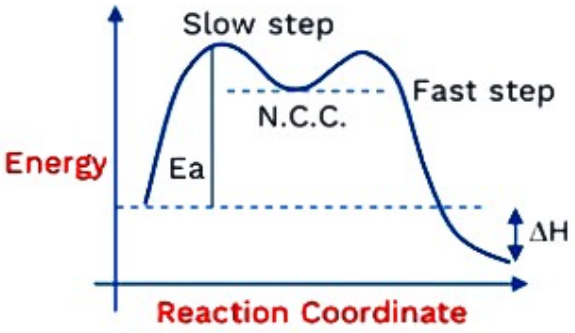


Enantiomeric pair

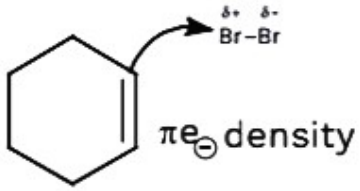
Based on Non-Classical Carbocation Formation (Anti Addition)



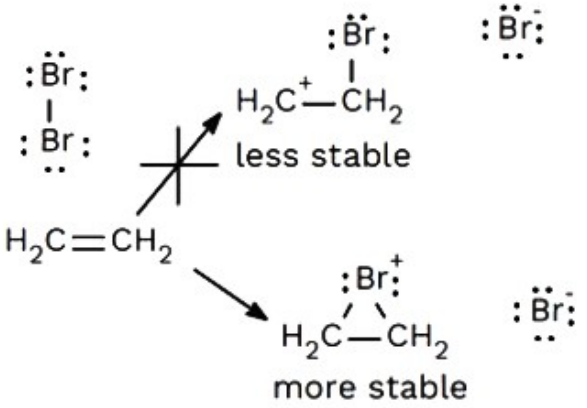
Energy Profile Diagram :



Halogenation is an electrophilic addition because an electrophile involved in slow step. Bromine is a non polar but alkene attacks then due to induced dipole or inductomeric effect.



In the first step, product is not a carbocation; rather, it is cyclic bromonium ion because bromine's electron cloud is close enough to the other sp^2 carbon to engage in bond formation. The carbocation is less stable than the carbocation would have been, since all the atoms (except hydrogen) in the bromonium ion, where as the positively charged carbon of the carbocation have incomplete octet.



ILLUSTRATION

Q76



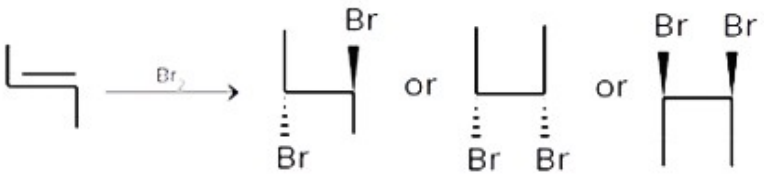
Sol.



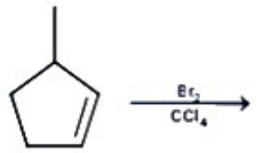
Q77



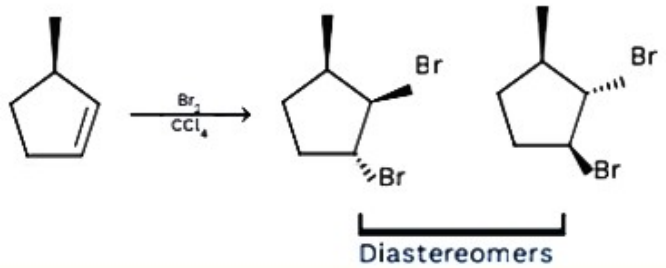
Sol.



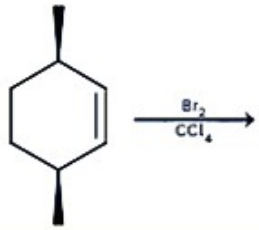
Q78



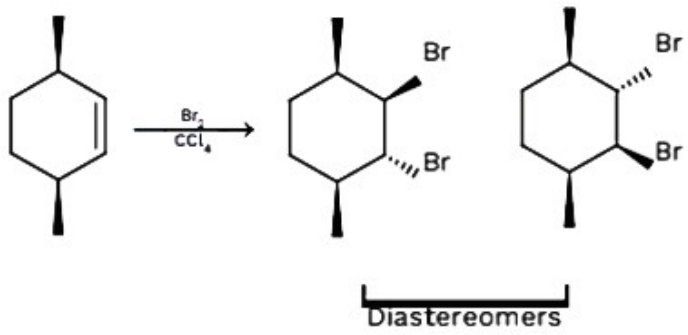
Sol.



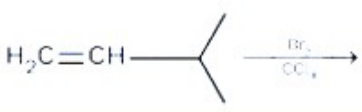
Q79



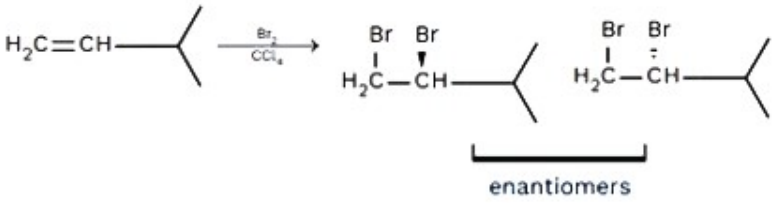
Sol.



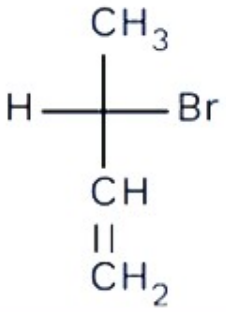
Q80



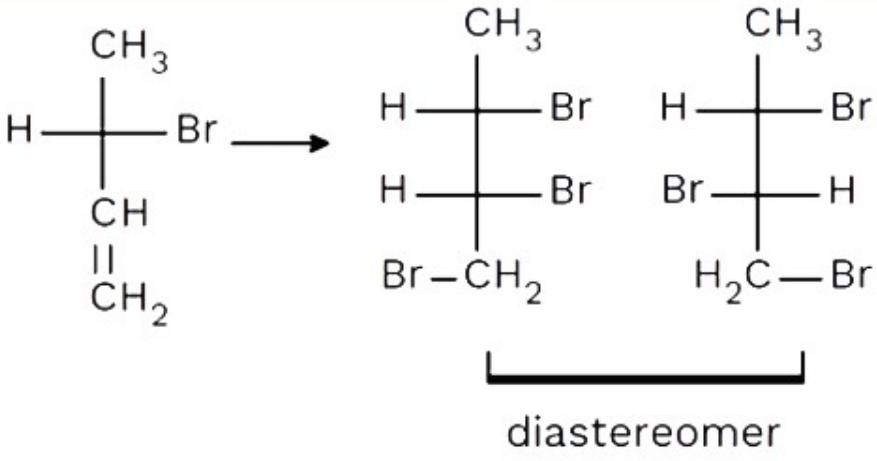
Sol.



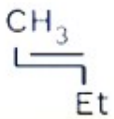
Q81



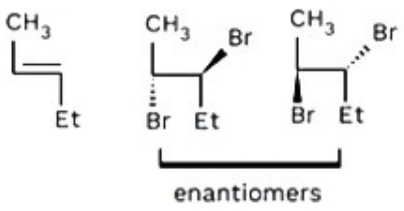
Sol.



Q82



Sol.



More electron donor alkene more the rate of Halogenation

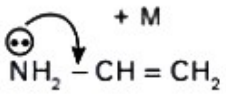
Q83 (a) $NH_2 - CH = CH_2$

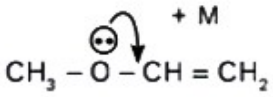
(b) $CH_3 - O - CH = CH_2$

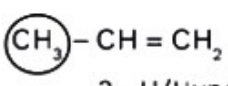
(c) $CH_3 - CH = CH_2$

(d) $NO_2 - CH = CH_2$

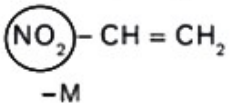
Sol. A > B > C > D

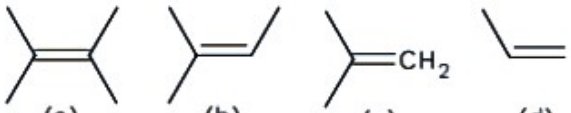
(a) 

(b) 

(c) 

3 αH/Hyper conjugation

(d) 

Q84 

Sol. More αH, more rate of reaction

F_2 and I_2 are halogens, but they do not act as reagents in an electrophilic addition reactions. Because Fluorine reacts explosively with alkenes, so the electrophilic addition of F_2 is not a synthetically useful reaction. The addition of I_2 to an alkene is a thermodynamically

unfavourable reaction: The vicinal diiodides are mostly unstable at room temperature, decomposing back to the alkene and I_2.

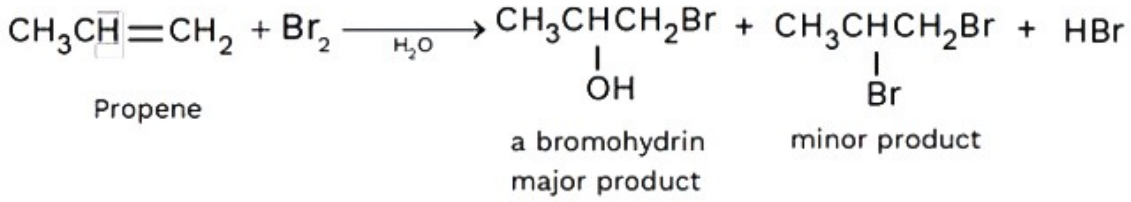
* If we are polar protic solvent then it behaves like nucleophile
* Use advance mark effect.

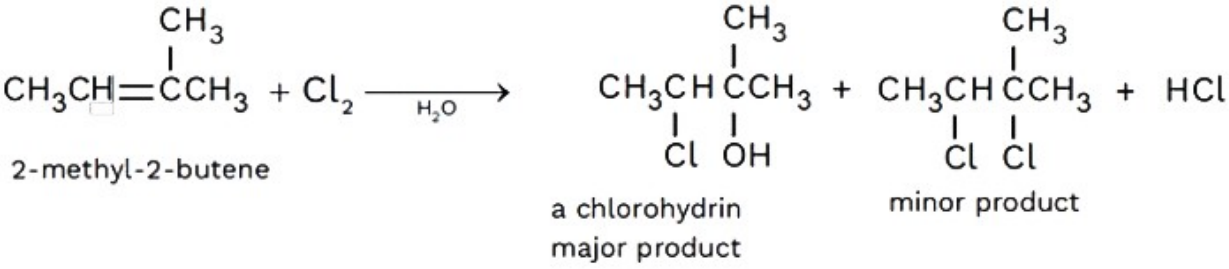
Example: $Br_2 + H_2O$

Concept of Solvent as Nucleophile

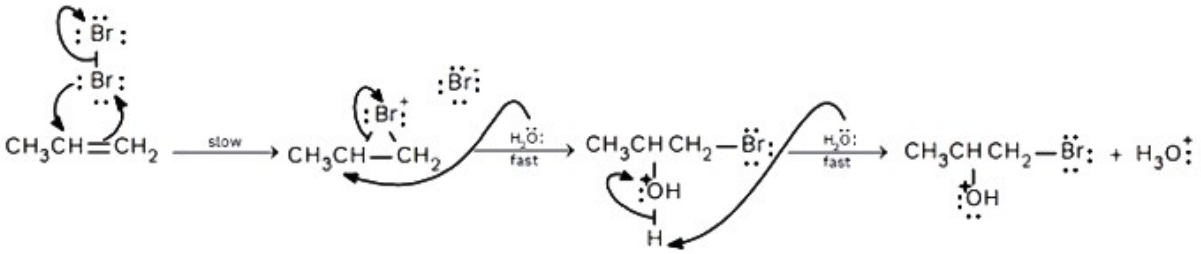
Concept of solvent :

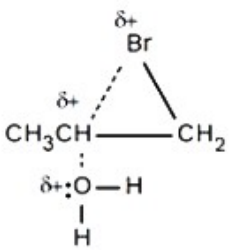
If CH_2Cl_2 is used instead of H_2O as the solvent, the major product of the reaction will be a vicinal halohydrin. A halohydrin (or more specifically, a bromohydrin or a chlorohydrin) is an organic molecule that contains both a halogen and an OH group and in a vicinal halohydrin, the halogen and the OH group are bonded to adjacent carbons.



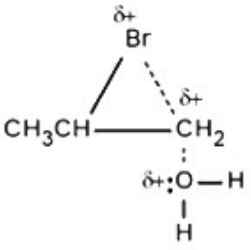


Halo Hydrine Formation :





more stable transition state



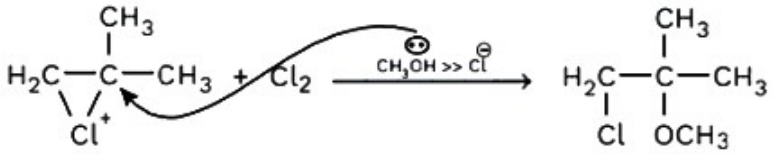
less stable transition state

Q85 **There are two nucleophiles in each of given reactions, Give explaination why there is a greater concentration of a particular nucleophile than the other for each reaction. What will be the major product of every reaction?**

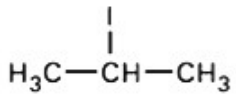
(A) $CH_2 = \underset{\displaystyle CH_3}{\overset{|}{C}} - CH_3 + Cl_2 \xrightarrow{CH_3OH}$ (B) $CH_2 = CHCH_3 + 2NaI + HBr \rightarrow$

(C) $CH_3CH = CHCH_3 + HCl \xrightarrow{H_2O}$ (D) $CH_3CH = CH_3 + HBr \xrightarrow{CH_3OH}$

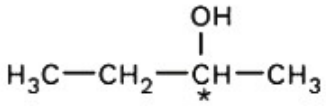
Sol.

(A) 

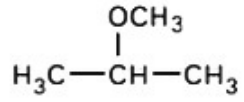
solvent concentration is high comapre to $Cl^{\ominus}$

(B) 

High concentration of NaI

(C) 

Solvent effect of H_2O

(D) 

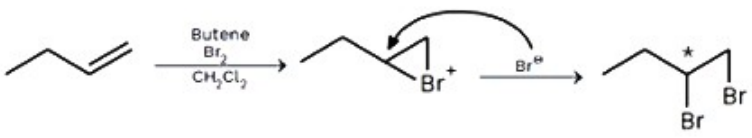
CH_3OH is solvent

Q86 **What would be the major product produced from the reaction of Br_2 with I-butene if the reaction were carried out in**

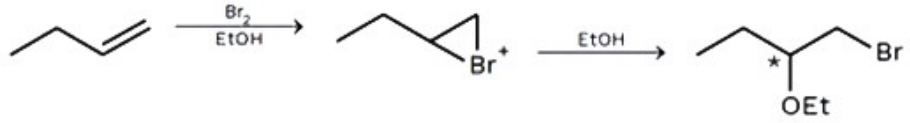
(A) dichloromethane? **(B) ethyl alcohol?**

(C) water? **(D) methyl alcohol?**

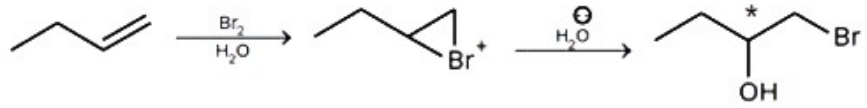
Sol.

(A) 

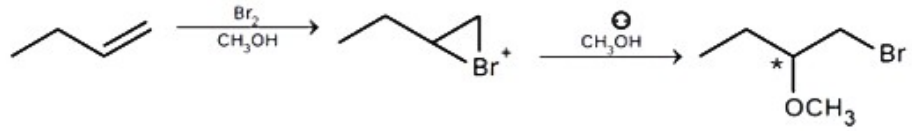
(B)



(C)

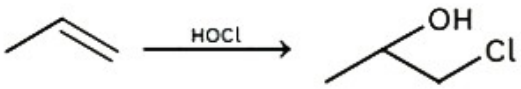


(D)

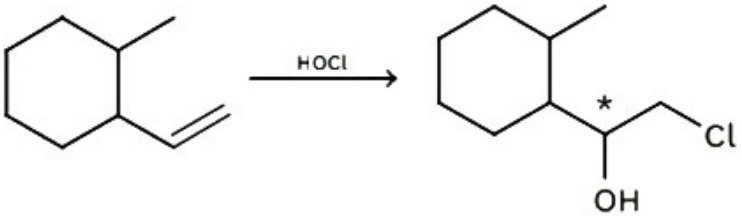


Other Reagents Used Frequently

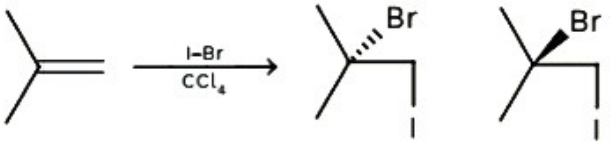
(i)



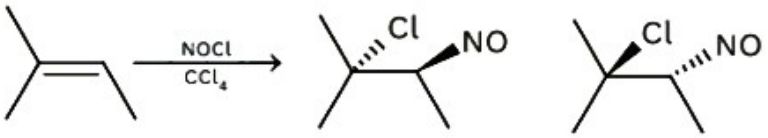
(ii)



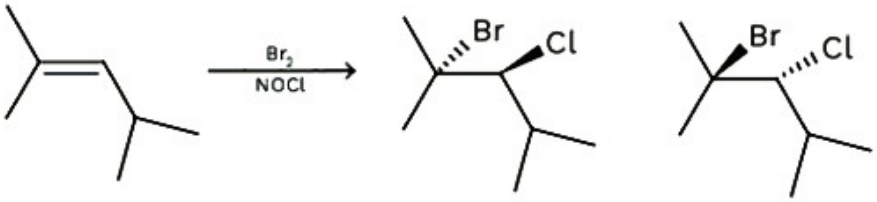
(iii)



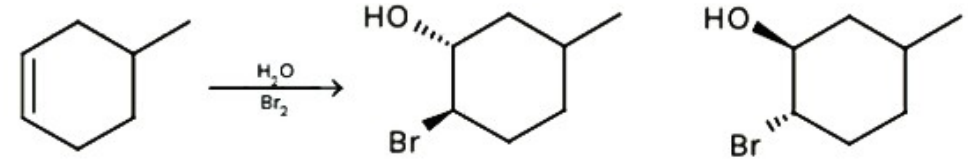
(iv)



(v)

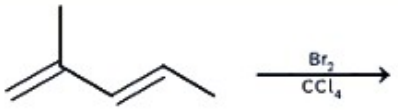


(vi)

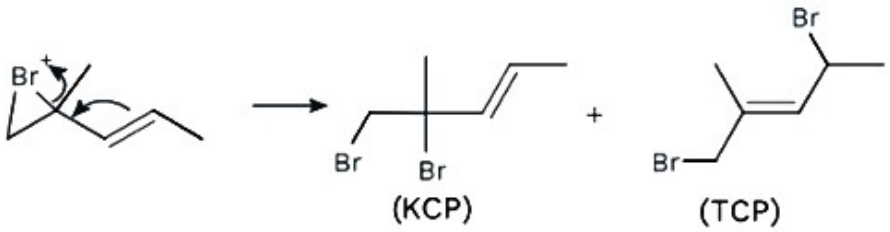


KCP and TCP Product

Q87

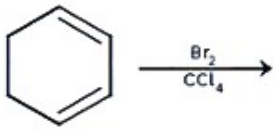


Sol.

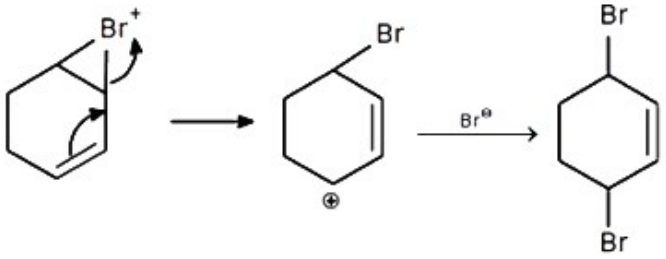


(KCP) (TCP)

Q88

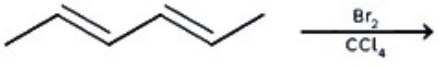


Sol.

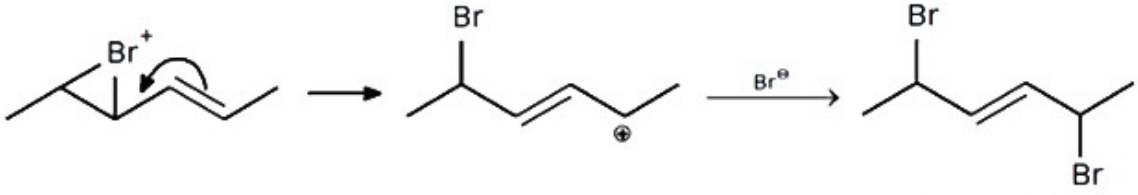


(1, 4 addition)

Q89



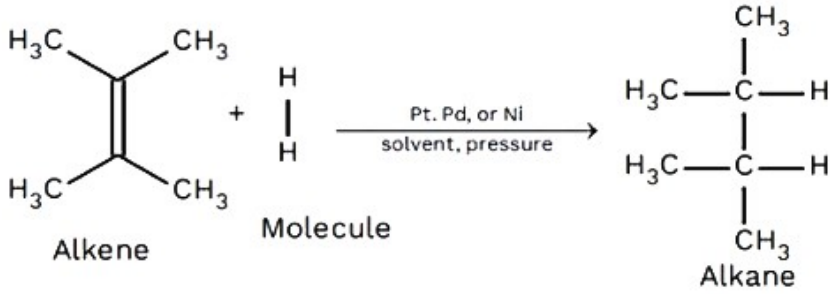
Sol.

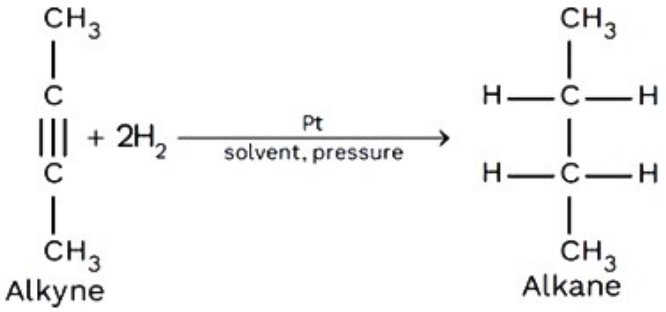


(1, 4 addition)

Hydrogenation of Functional Group:

General Reaction



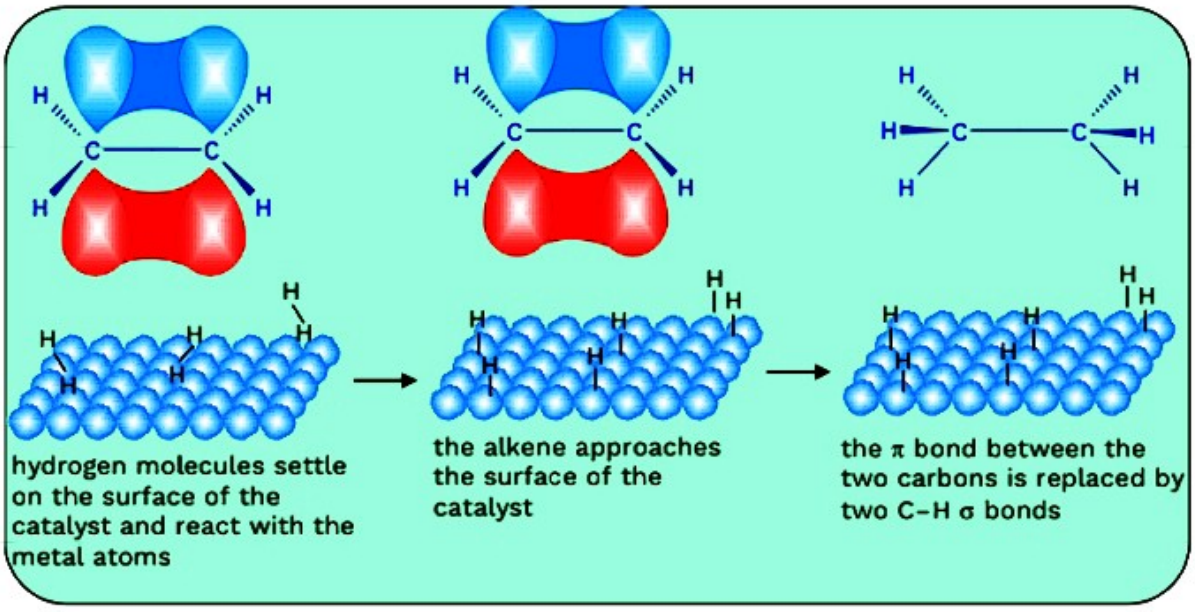


Required energy barrier to form 2'σ' bond, breaking one 'π' bond in hydrogenation full with the help of metal catalyst such as platinum, palladium or nickel.

Metal catalyst decreases energy barrier, lower activation energy required.

Adam's catalyst (PtO_2) mostly used.

Pt or Pd with charcoal (C), in finely divided state.



The details of the mechanism of catalytic hydrogenation are not completely understood. As we know that hydrogen is adsorbed on the surface of the metal and that the alkene form complexes with the metal by overlapping its own p orbitals with vacant orbitals of the metal. Breaking the π bond of the alkene and the σ bond of H_2 and forming the C–H σ bonds all occur on the surface of the metal. The alkane product diffuses away form the metal surface as it is formed

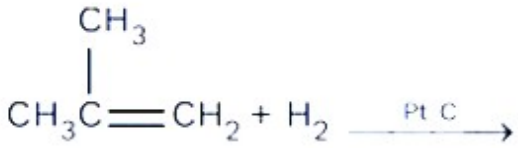
Q90 $CH_3CH{=}CHCH_3 + H_2 \xrightarrow{Pt\ C}$

2-butene

Sol. $CH_3CH{=}CHCH_3 + H_2 \xrightarrow{Pt/C} CH_3CH_2CH_2CH_3$

2-butene → butane

Q91



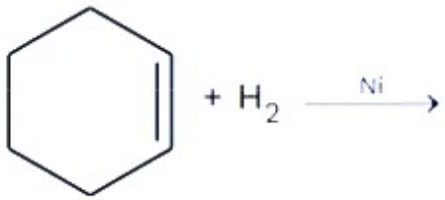
2-methylpropene

Sol.



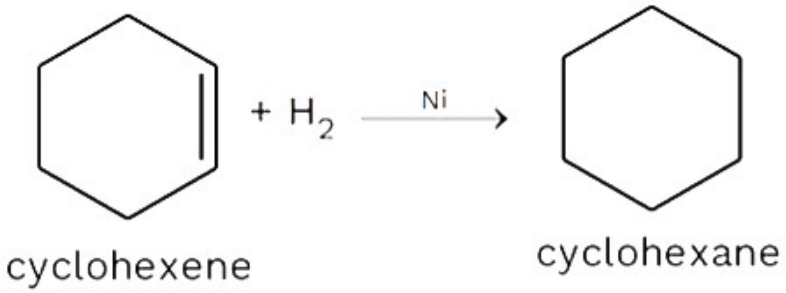
2-methylpropene → 2-methylpropane

Q92



cyclohexene

Sol.



Hydrogenation: The Function of the Catalyst

Hydrogenation of an alkene is an exothermic reaction $\left(\Delta H^{\circ} \cong -120 \text{ kJ mol}^{-1}\right)$:

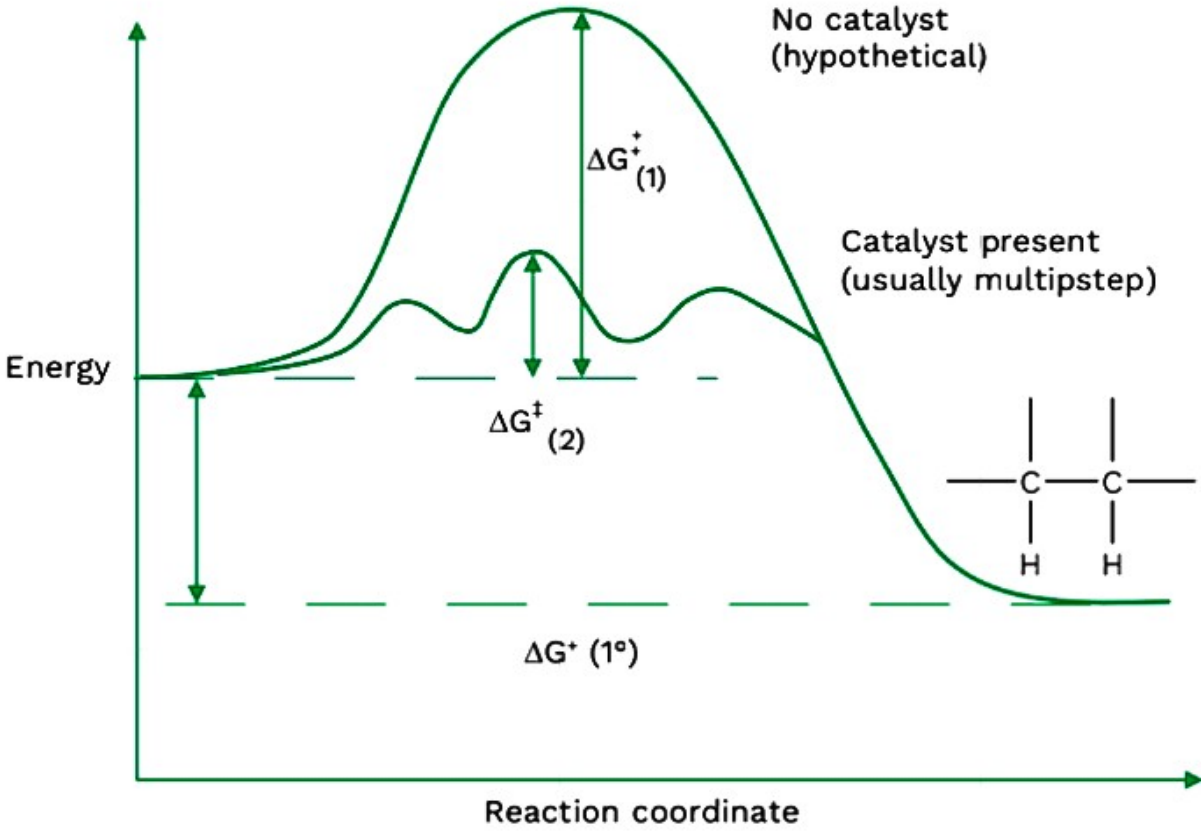
$$R - CH = CH - R + H_2 \xrightarrow{\text{hydrogenation}} R - CH_2 - CH_2 - R + \text{heat}$$

(i) Hydrogenation process is exothermic reaction, bond formation of two "σ" bonds.

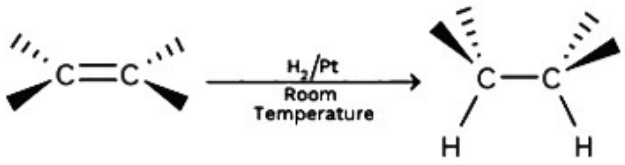
(ii) At room temperature, uncatalyzed reaction is not possible.

(iii) Catalyst provide new pathauy for the reaction that involve a lower free energy of activation.

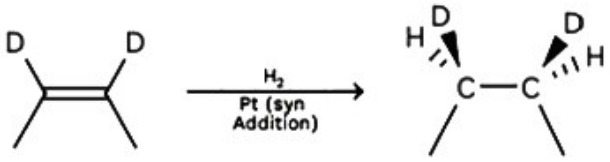
.



Stereochemistry of Hydrogen Addition

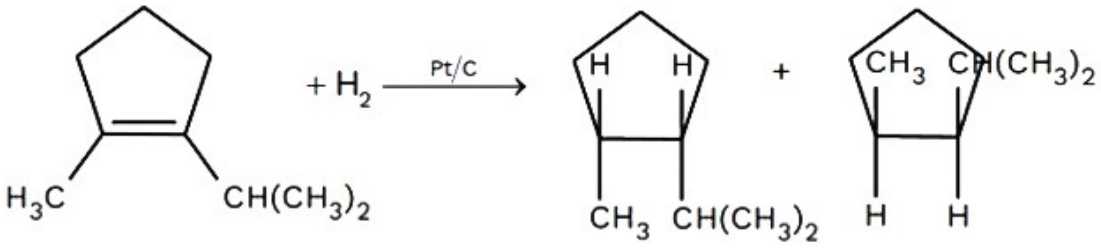


Example



In a catalytic hydrogenation, the alkene sits on the surface of the metal catalyst and both hydrogen atoms add to the same of the double bond.
The collision of an alkene with the surface bearing adsorbed hydrogen causes adsorption of the alkene as well as a consequence, both hydrogen atoms usually add from the same side of the molecule. This mode of addition is called a Syn addition.
Therefore, the addition of H_2 to a cyclic alkene with fewer than eight ring atoms forms only the cis enantiomers.

Example :

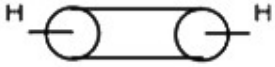


1-isopropyl-2-methyl
-cyclopentene

Heat of Hydrogenation

The heat evolved in a hydrogenation reaction is called the heat of hydrogenation. It is necessary to give it a positive value. Hydrogenation reactions, however, are exothermic (they have negative $\Delta H°$ values). So the heat of hydrogenation is the positive value of the $\Delta H°$ of value of the $\Delta H°$ of the reaction.

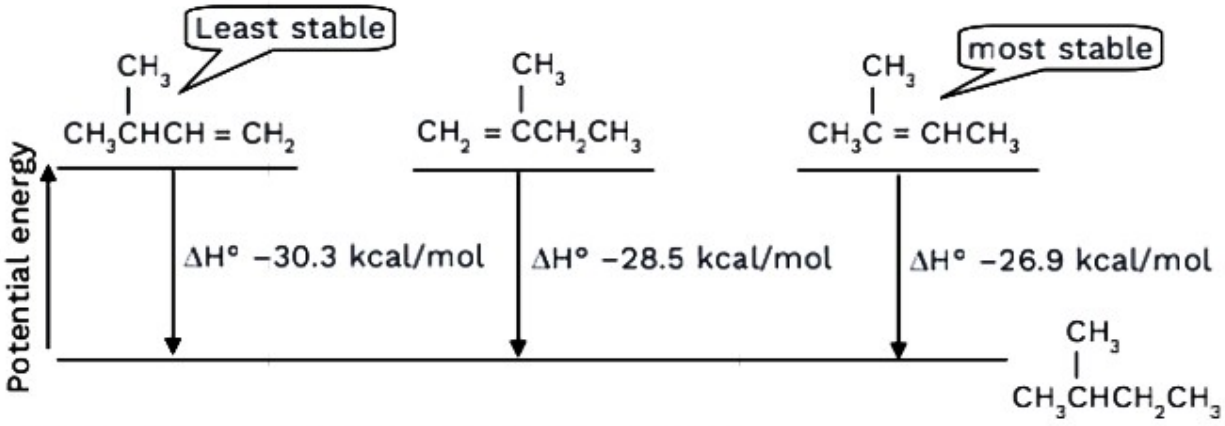
Alkyne is more reactive compare to alkene for hydrogenation.



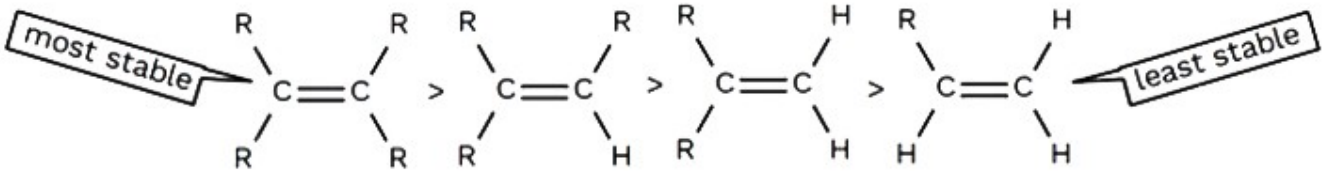
Points to remember

Due to Cylinder shape of electron cloud.

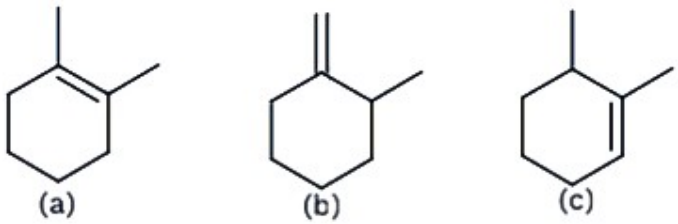
Due to cylindrical shape approach of H_2 molecule to π electron cloud is quite easy.



Relative Stabilities of Alkyl-Substituted Alkenes



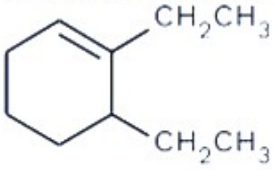
Q93 **Rank the following cycloalkenes in order of increasing stability.**

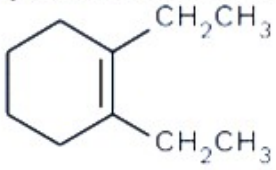


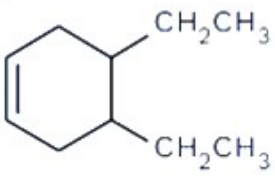
Sol. a > c > b

(a) = 10αH, (b) = 3αH, (c) = 6αH

Q94 **Which of the following compounds is the most stable?**



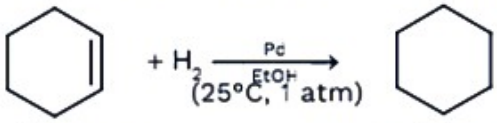




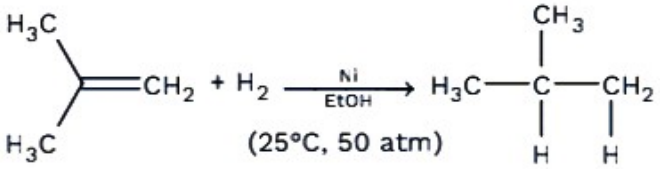
Sol. Maximum αH, Most stable alkene.

(b > a > c)

Q95 **Which has the smallest heat of hydrogenation?**



Cyclohexene Cyclohexane



2-Methylpropene Isobutane

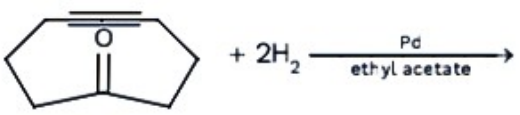
Sol.

Stability of alkene $\alpha \frac{1}{\text{H.O.H}}$

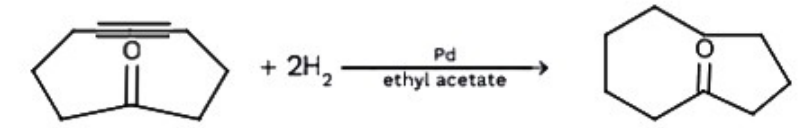
Stability order = B > A

H.O.H = A > B

Q96



Cyclononyn-

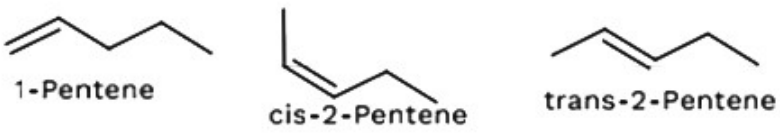


Cyclononyn-

Cyclononanone

Q97 **Draw the structures of three pentenes that would all yield pentane on hydrogenation.**

Sol.



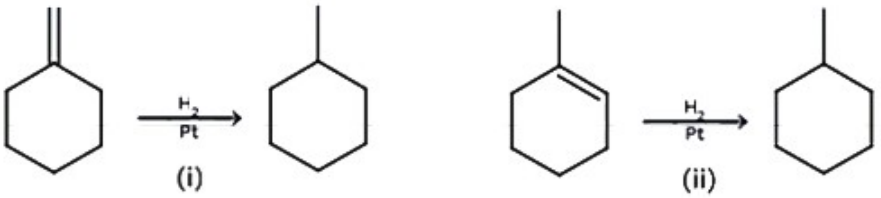
1-Pentene

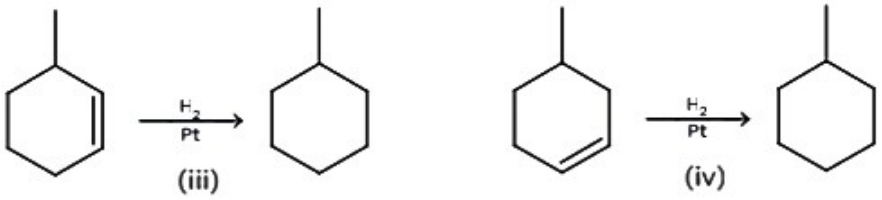
cis-2-Pentene

trans-2-Pentene

Q98 **Four different cycloalkenes will all yield methyl cyclopentane when subjected to catalytic hydrogenation. What are their structures? Show that reactions.**

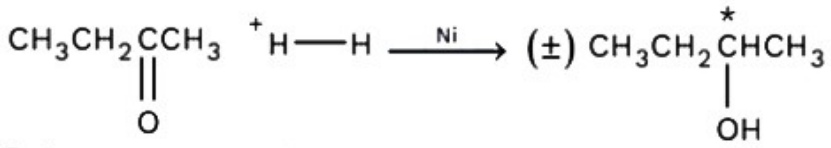
Sol.





Q99 **Example is the synthesis of 2-butanol by the nickel-catalysed hydrogenation of butanone. In this reaction the hydrogen molecule adds across the carbon-oxygen double bond in much the same way that it adds to a carbon-carbon double bond. Show mechanical approach of Hydrogenation.**

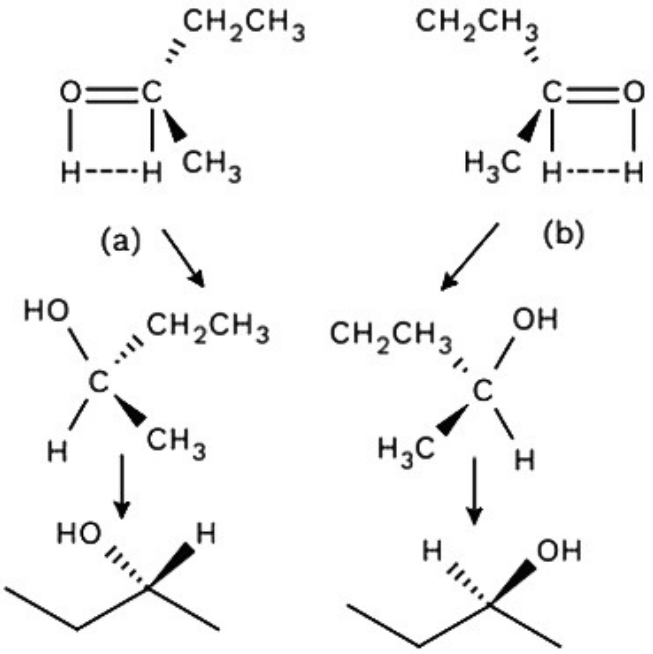
Sol.



Butanone (achiral molecules)

Hydrogen (achiral molecules)

(±)-2-Butanol [chiral molecules but 50:50 mixture (R) and (S)]

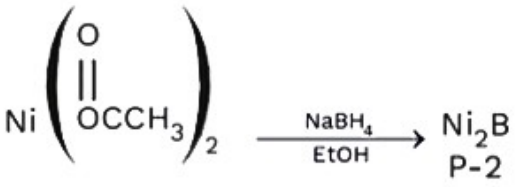


(R)-(-)(2)-Butanol (50%)

(S)-(+)(2)-Butanol (50%)

Synthesis of cis-Alkenes (P-2 Catalyst)

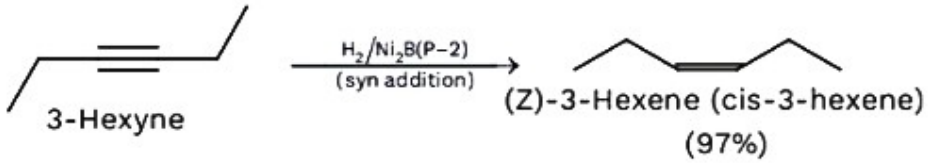
A heterogeneous catalyst that permits hydrogenation of an alkyne to an alkene is the nickel boride compound called the P-2 catalyst. The P-2 catalyst can be produced by the reduction of nickel acetate with sodium borohydride:



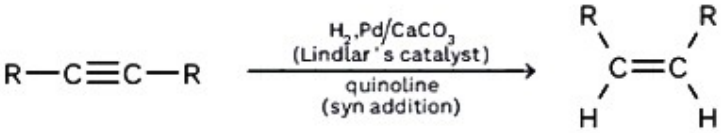
The hydrogenation of alkynes in the presence of P-2 catalyst causes syn addition of hydrogen. The alkene obtained from an internal alkyne has the (Z) or cis configuration.

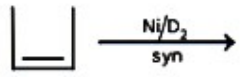
Example

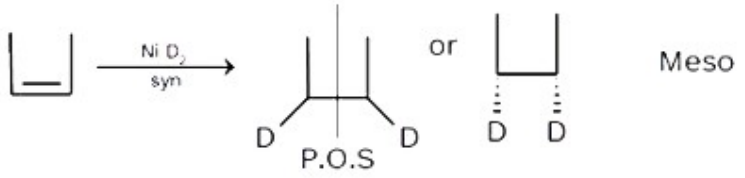
Syn addition of hydrogen to an alkyne



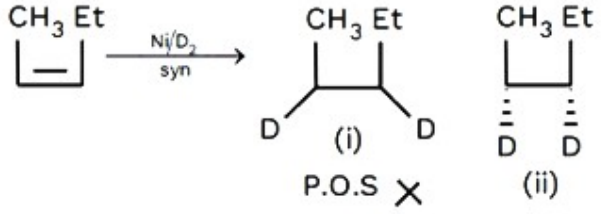
Other specially conditioned catalysts can be used to prepare cis-alkenes from disubstituted alkynes. Metallic palladium deposited on calcium carbonate can be used in this way after it has been conditioned with lead acetate and quinoline special catalyst is known as Lindlar's catalyst:



Q100 

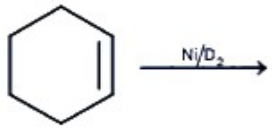
Sol. 

Q101 

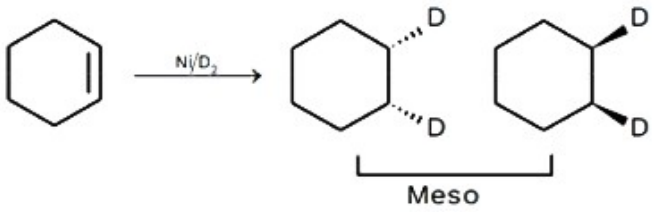
Sol. 

'2' enantiomers

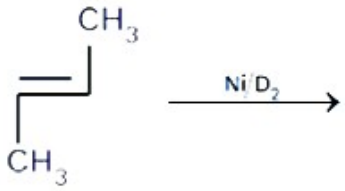
Q102



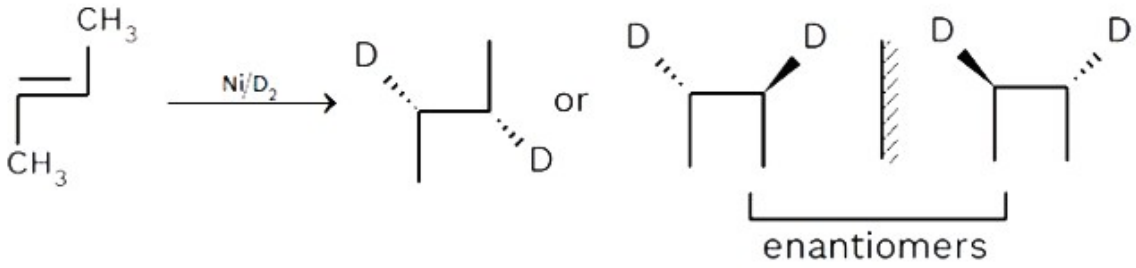
Sol.



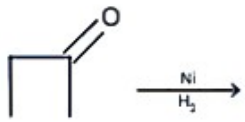
Q103



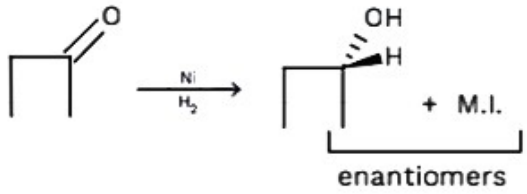
Sol.



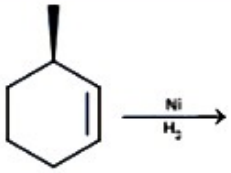
Q104



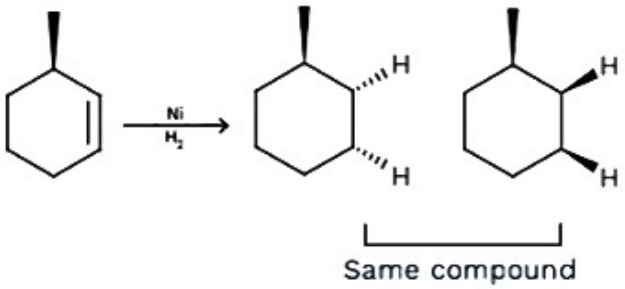
Sol.



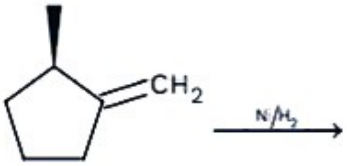
Q105



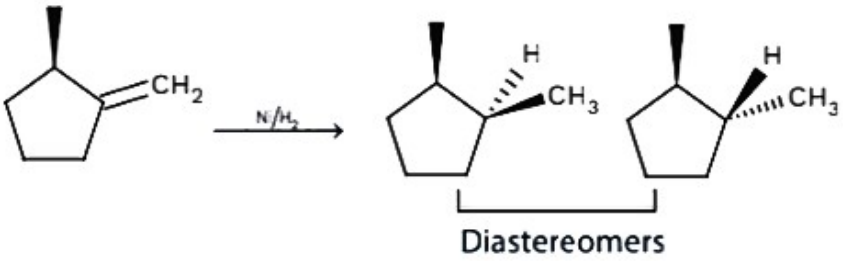
Sol.



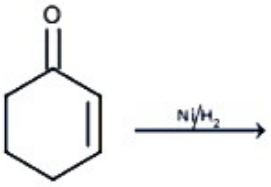
Q106



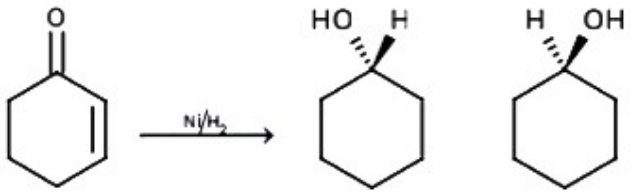
Sol.



Q107

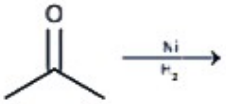


Sol.

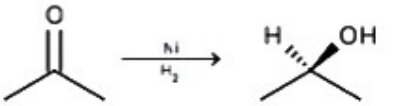


(Same compound)

Q108



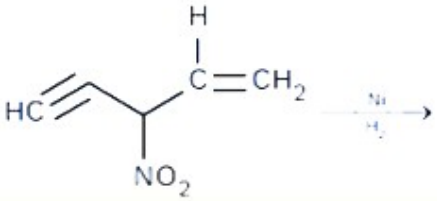
Sol.



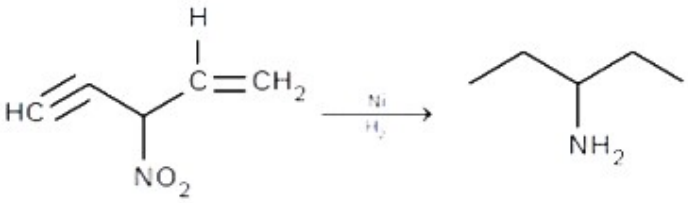
Single Product

During Reduction n-mole H_2 is available for reduction due to large surface area of void.

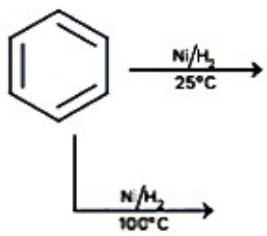
Q109



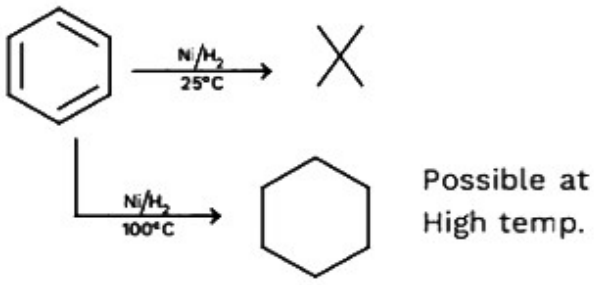
Sol.



Q110

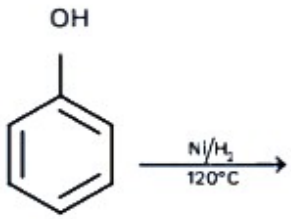


Sol.

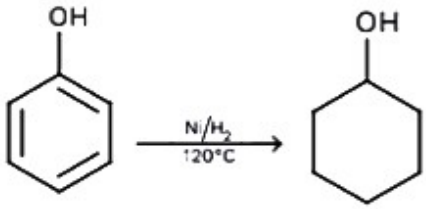


Possible at
High temp.

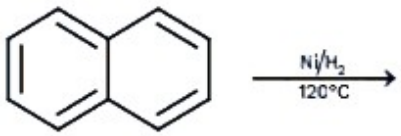
Q111



Sol.



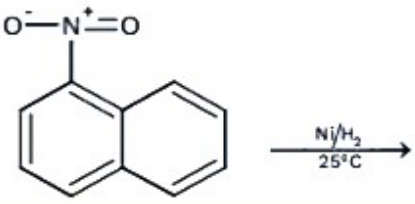
Q112



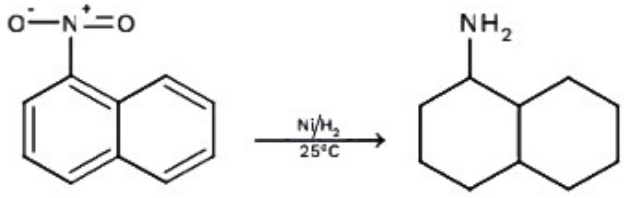
Sol.



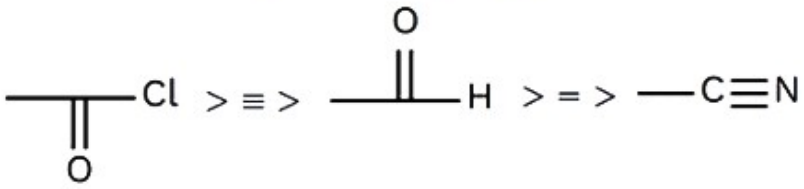
Q113



Sol.



Reactivity Order of Functional Groups towards hydrogenation.



Know the Concept

Rate of reaction is high for unstable and/or less steric hydrocarbons.

Compare rate of Hydrogenation of alkene for following molecules.

Q114



(a) (b)

Sol. a > b (High angle strain, High rate of Hydrogenation)

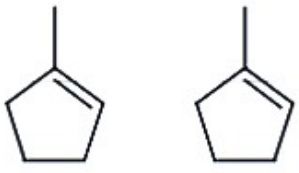
Q115



(a) (b)

Sol. a > b (Smaller ring, High strain)

Q116



(a) (b)

Sol. a > b (Smaller ring, High rate of Hydrogenation)

Q117



(a)



(b)

Sol. a > b

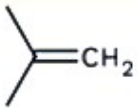
$$\text{H.O.H} \propto \frac{1}{\text{stablility}}$$

Trans alkene is more stable, H.O.H decreases.

Q118

$CH_2C{=}CH_2$

(a)



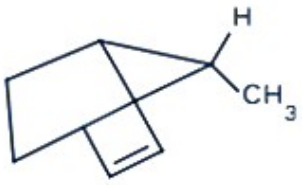
(b)

Sol. b > a

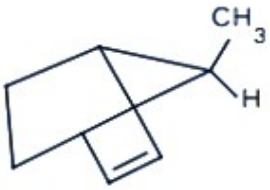
$$\text{H.O.H} \propto \frac{1}{\text{stablility}}$$

(Number of $\propto$-H↑ H.O.H↓)

Q119



(a)

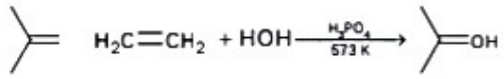


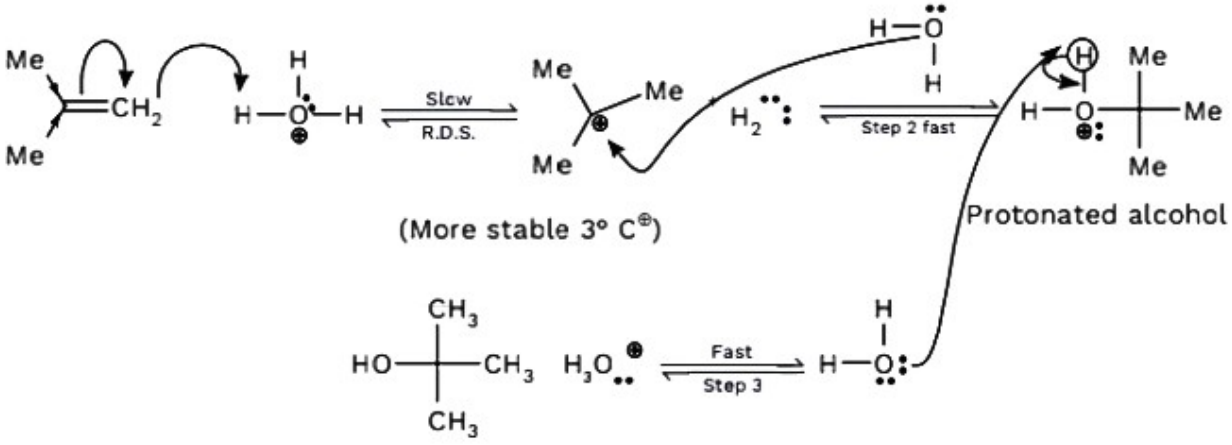
(b)

Sol. b > a (Steric Hindrance) (Synaddition of H_2 molecule will hinder due to CH_3 in path in compound a.)

Acid-Catalysed Reaction with Water

Most commonly used acids to catalyse the hydration of alkenes are dilute solutions of H_2SO_4 and H_3PO_4. These reactions follow Markovnikov's rule with rearrangement (if possible) and are usually regioselective. (or stereoselective i.e., exclusively one product is formed in major amount.)





In the hydration of alkenes, it is best to used dilute acids so that concentration of water remains high (it also helps in lowering the temperature) and the carbocation produced in step 1 must react weak nucleophile.

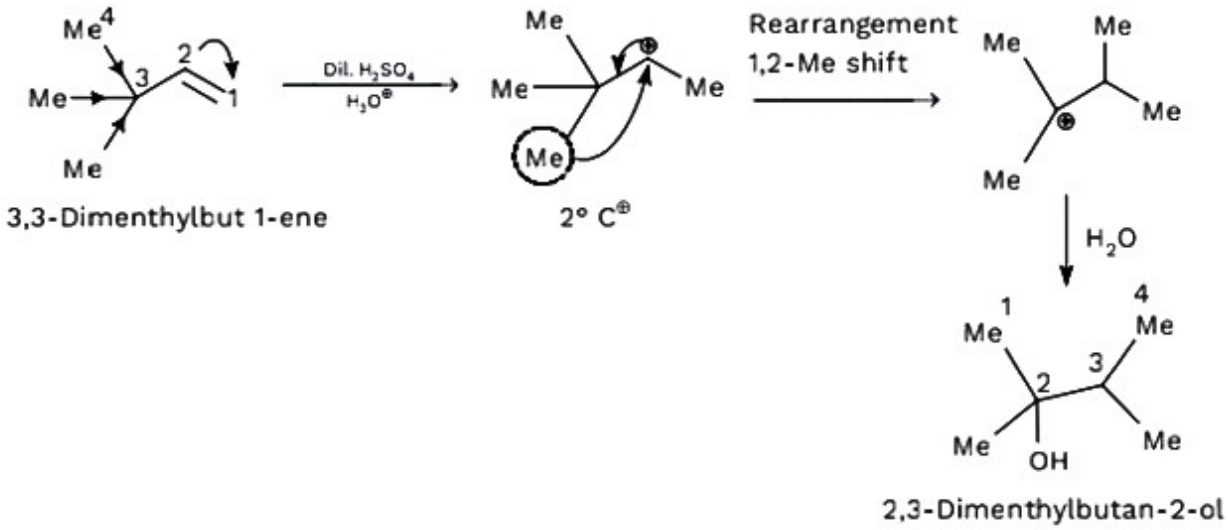
Q120 **Why H_2SO_4 or H_3PO_4 is used instead of HCl, HBr, or HI?**

Sol. The reverse reaction, i.e. dehydration of alcohol to alkene, will take place if low concentration of H_2O (i.e. use of concentrated H_2SO_4) and higher temperature is used; all these conditions encourage elimination.

$$\underset{\text{Weak acid}}{HI} + H_2O \rightleftharpoons H_3O^{\oplus} + \underset{\substack{\text{Conjugate base}\\ \text{(Strong nucleophile)}}}{I^{\ominus}}$$

$$\underset{\text{Strong acid}}{H_2SO_4} + H_2O \rightleftharpoons H_3O^{\oplus} + \underset{\substack{\text{Conjugate base}\\ \text{(Weak nucleophile)}}}{HSO_4^{\ominus}}$$

Rearrangement in the Hydration Reaction
Example 1

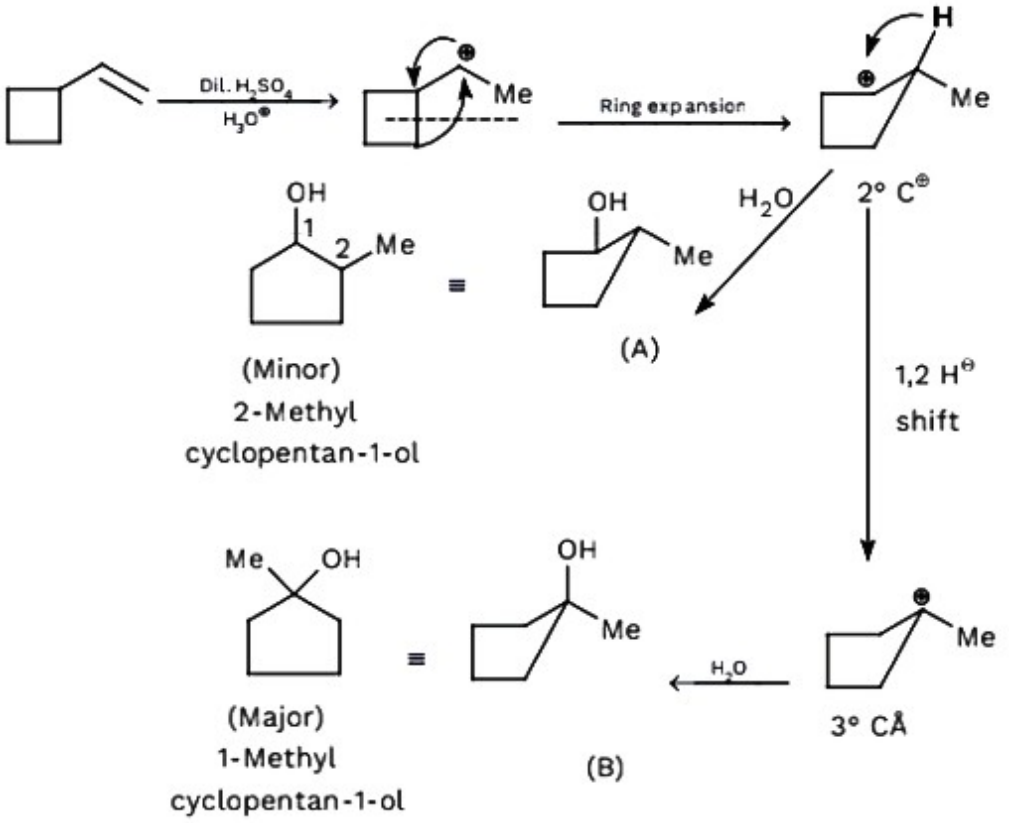


3,3-Dimenthylbut 1-ene

$2° C^{\oplus}$

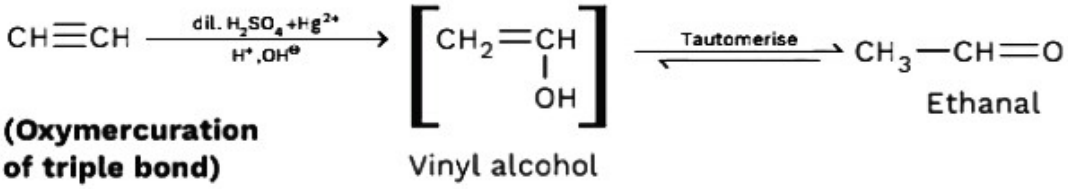
2,3-Dimenthylbutan-2-ol

Example

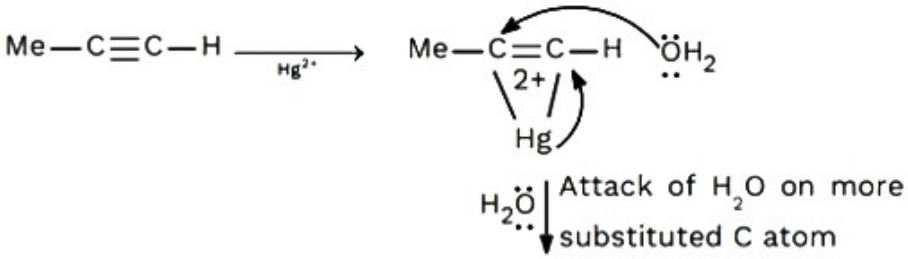


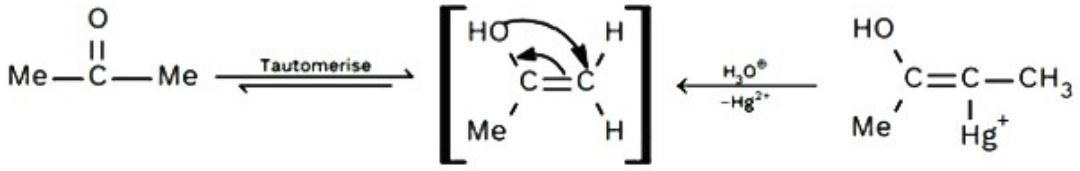
Addition of H_2O (Kucherov's Reaction)

Alkynes add to H_2O in the presence of 60% H_2SO_4 and 1% $HgSO_4$ (Hg^{2+}) as catalyst at 333 K to give unstable vinyl alcohol that rearranges (tautomerise) to more stable carbonyl compounds.



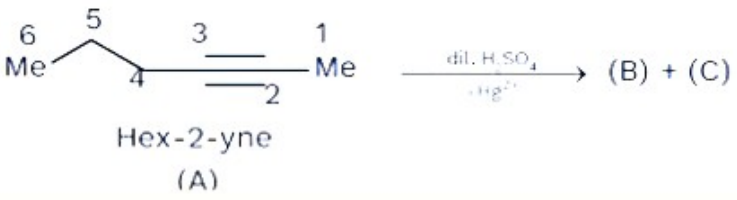
Mechanism



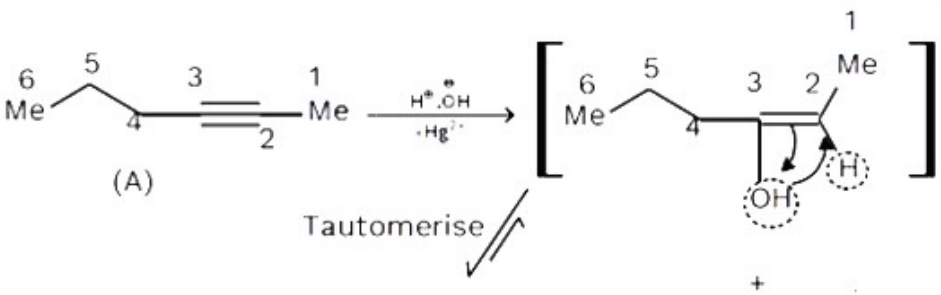


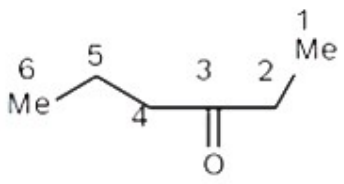
Complete the following reactions:

Q121

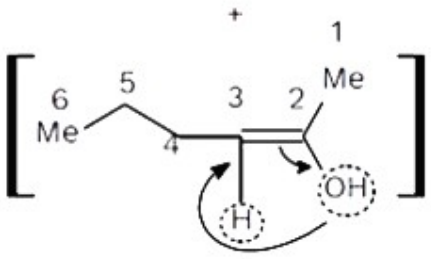


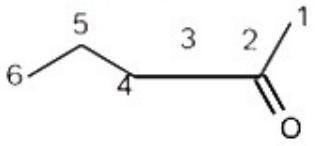
Sol.



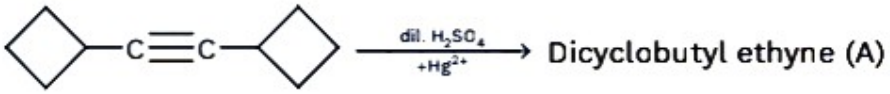


(B)
Hexan-3-one
(Major)



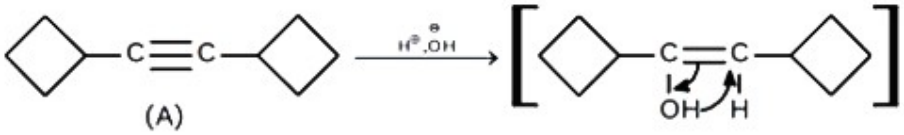


(C)
Hexan-2-one
(Minor)

Q122 

Dicyclobutyl ethyne (A)

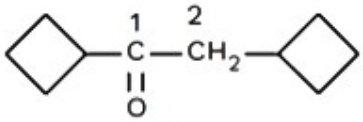
Sol.



(A)

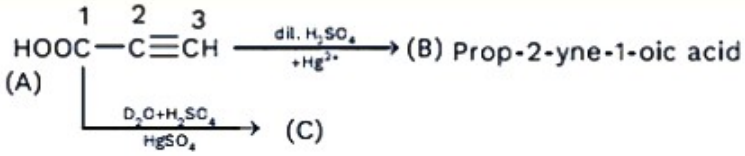


Tautomerise



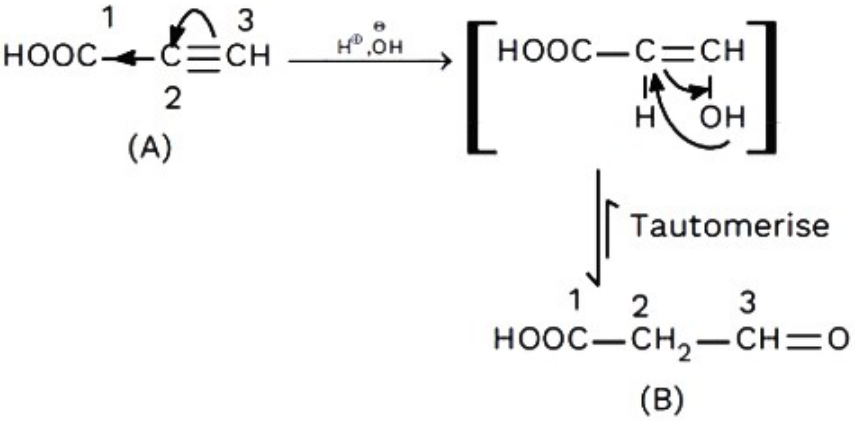
(B)
1,2-Dicyclobutylethan-1-one

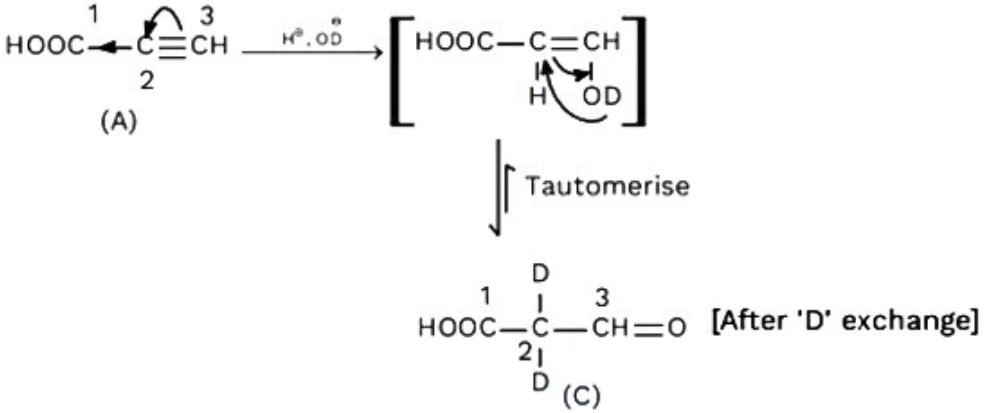
Q123



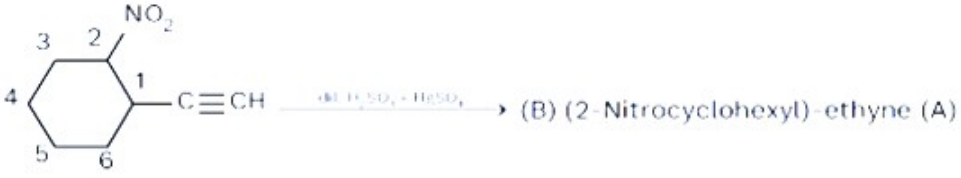
Due to EWG (–COOH group), electromeric effect takes place from C-3 to C-2.

Sol.

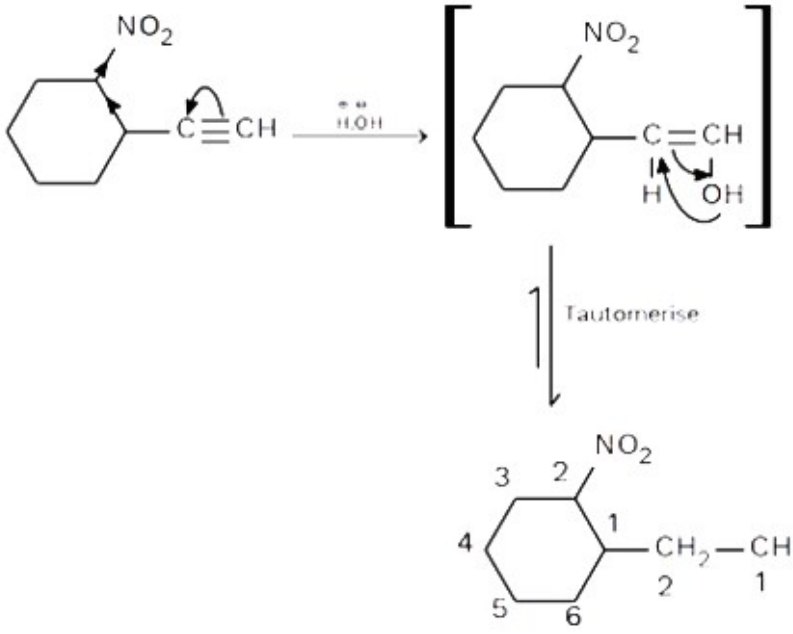




Q124



Sol. Due to EWG (–NO_2 group), electromeric effect takes place from C-2 to C-1.



2-(2-Nitrocyclohexyl)-ethanal

Oxymercuration Demercuration (OMDM)

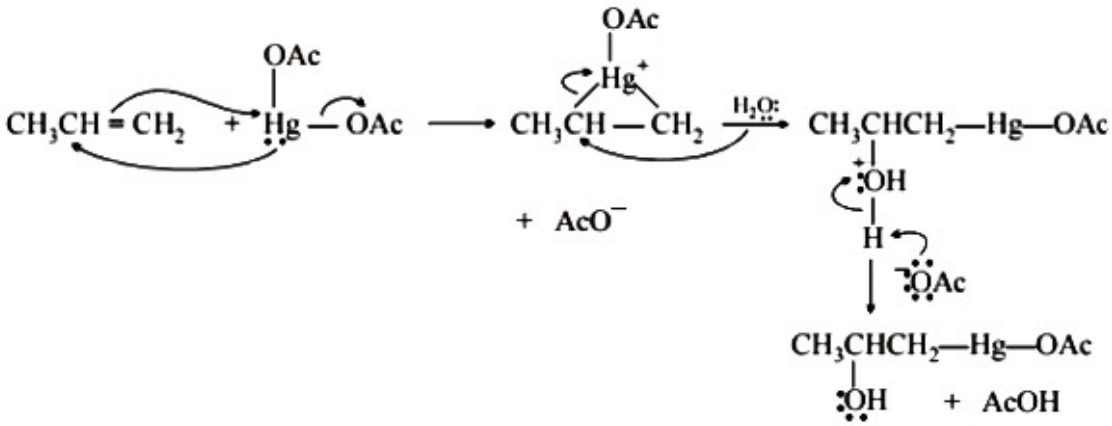
When reaction with that reagent is complete, sodium borohydride and hydroxide ion are added to the reaction mixture.

$$R-CH=CH_2 \xrightarrow[(ii)NaBH_4HO^-]{(i)Hg(OAc)_2,H_2O,THF} R-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3$$

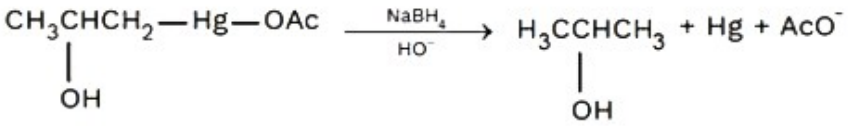
Points to remember

OMDM is a hydration process of alkene according to Markovnikov's rule with no rearrangement of cyclic mercurinium ion. In oxymercuration, the alkene is treated with mercuric acetate in aqueous tetrahydrofuran (THF).

Mechanism for Oxymercuration:

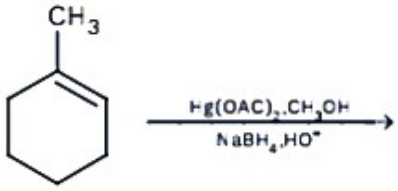


Sodium borohydride ($NaBH_4$) converts the carbonmercury bond into a carbonhydrogen bond. Because the reaction results in the loss of mercury, it is called demercuration.

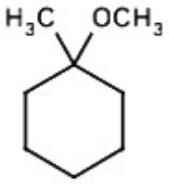


Identify the product of the following reaction.

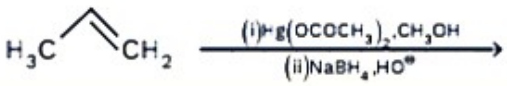
Q125



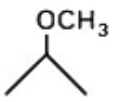
Sol.



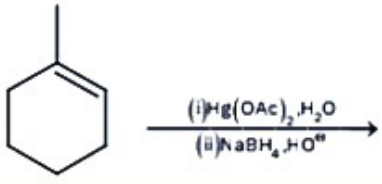
Q126



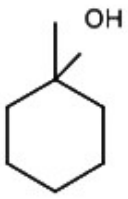
Sol.



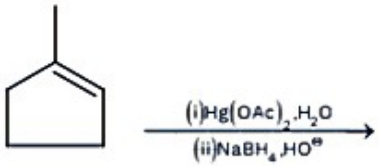
Q127



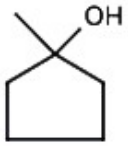
Sol.



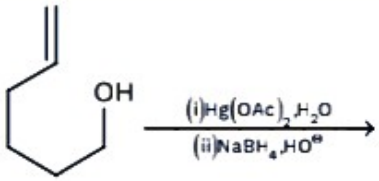
Q128



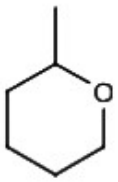
Sol.



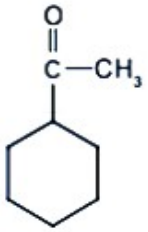
Q129



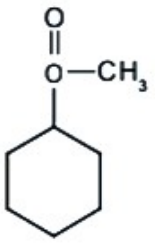
Sol.



Q130



Sol.

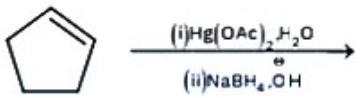


Identify possible alkene used to synthesis alcohols :

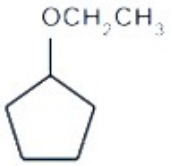
Q131



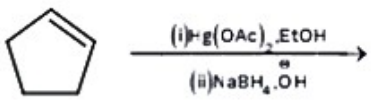
Sol.



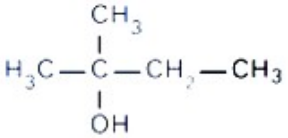
Q132



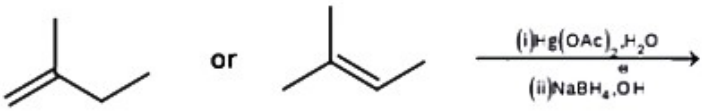
Sol.



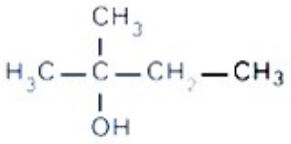
Q133



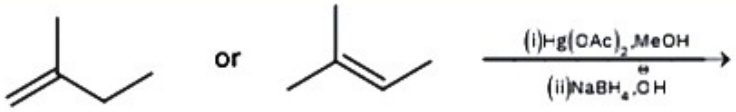
Sol.



Q134

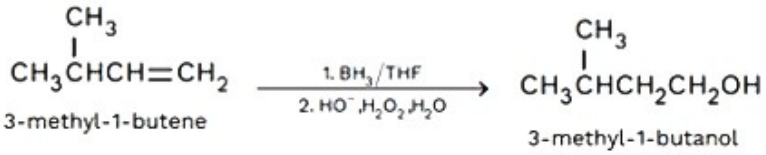


Sol.



Hydroboration

Hydroboration involves addition of a β–H bond of borane (BH_3) to an alkene to yield organoborane intermediates, RBH_2, R_2BH and R_3B. Since BH_3 has three hydrogens, addition occurs in three steps forming R_3B as the final product.



(i) The reaction procedure is simple and convenient leading to excellent yield.

(ii) Hydroboration is carried out in ether, viz. tetrahydrofuran or diglyme (diethylene glycol methyl ether).

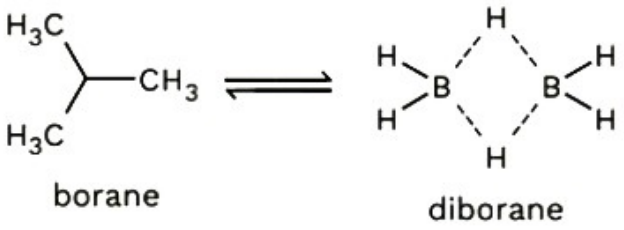
(iii) Alkylboranes are important intermediates for the synthesis of alcohols and alkanes. These are not isolated and converted in situ to the corresponding product.

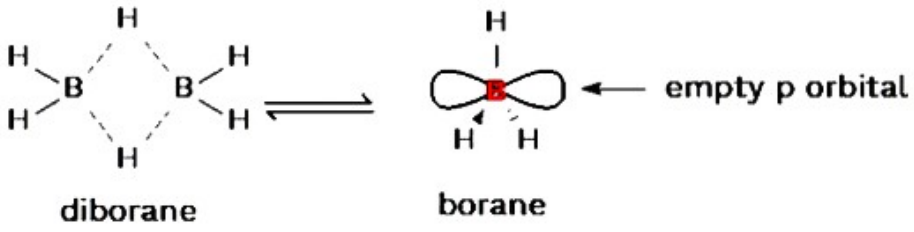
Borane and Diborane

Borane exists primarily as a colorless gas called diborane electron deficient borane dimerizes into diborane (gas), $6e^{(-)}$ containing electron deficient borane Joined by Unusual half bonds (dotted lines), made up of fewer than the normal two electrons.

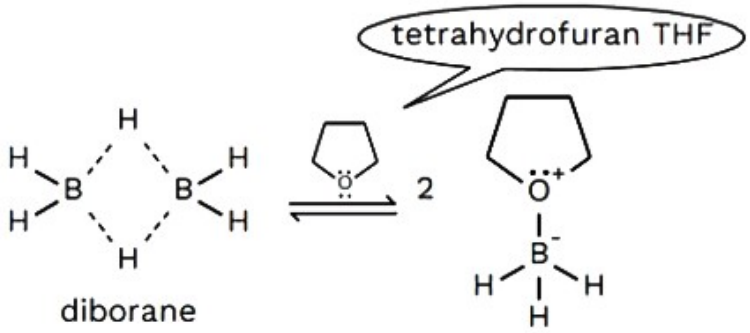
Tips and Trick

Di Borane is used as Electrophile in reaction that's why it is electrophilic addition





Commercially diborane is available in tetrahydrofuran solution, where it exists as the monomer in the form of acid-base complex with the solvent.

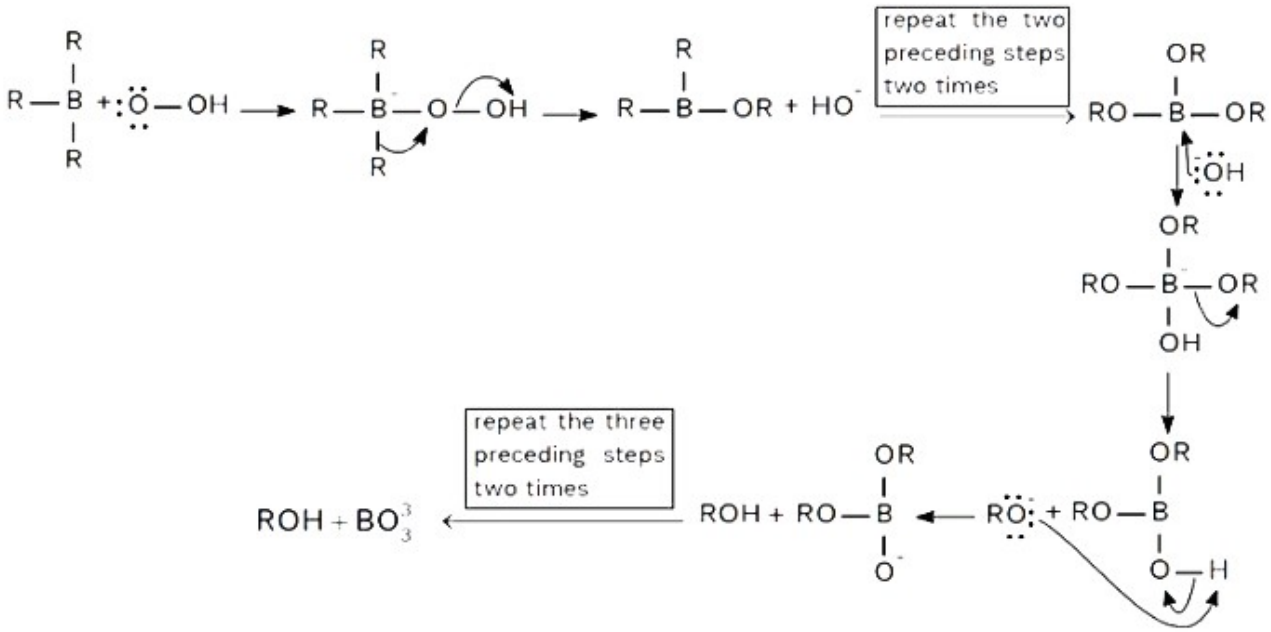


Use-Preparation of Alcohols:

Alkylboranes formed by hydroboration of alkenes can be easily oxidised by alkaline hydrogen peroxide in which boron is replaced by –OH forming alcohol as the final product

$$3CH_2 = CH_2 + BH_3 \xrightarrow{THF} \underset{\text{Triethylborane}}{(CH_3CH_2)_3B} + 3H_2O_2 \xrightarrow{OH^-} \underset{\text{Ethyl alcohol}}{3CH_3CH_2OH} + \underset{\text{Boric acid}}{B(OH)_3}$$

Hydroboration:

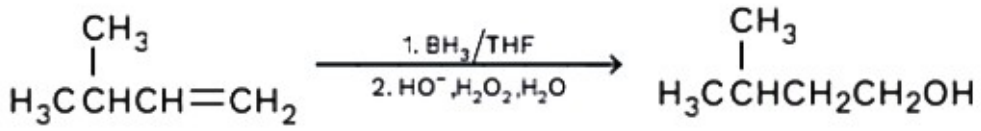


Points to remember

1. vacant orbital on boron, borane may be regarded as an electrophile and thus attacks π electrons of the alkene; positive charge on the alkene develops on that carbon atom which is best able to support it.
2. Steric hindrance favours addition of boron to the less hindered carbon of the double bond.
3. Because of the four-centre mechanism, hydroboration is a cis-addition i.e., boron and hydrogen add across the double bond on the same side of the molecule.
4. During the oxidation step, the boron is replaced by an –OH with the same stereochemistry, overall resulting in syn non-Markovnikov addition of water.
5. The rate of hydroboration decreases with the increase in the number of alkyl substituents on the double bond.

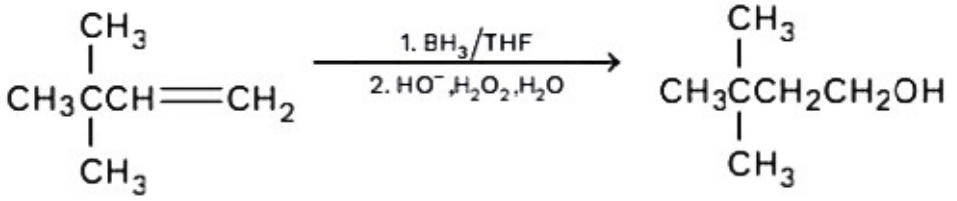
No Rearrangement in Reaction :

Because carbocation intermediates are not formed in the hydroboration reaction, carbocation rearrangements do not occur.



3-methyl-1-butene

3-methyl-1-butanol

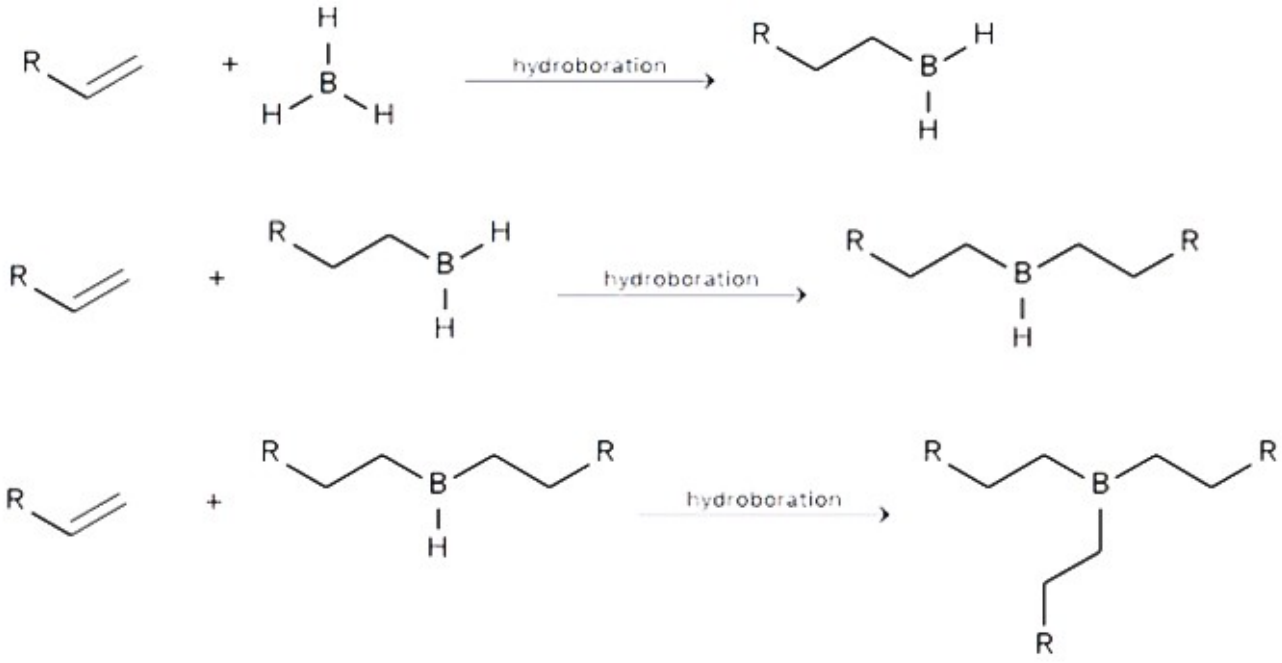


3,3-dimethyl but-1-ene

3,3-dimethyl butanol

Points to remember

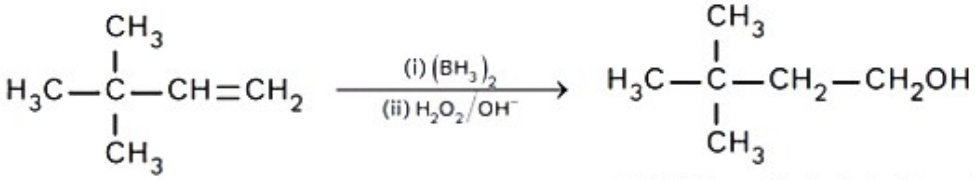
In hydroboration-oxidation reaction, three moles of alkene react with one mole of BH_3 to form three moles of alcohol. (hydration of alkenes)



Stereochemistry :

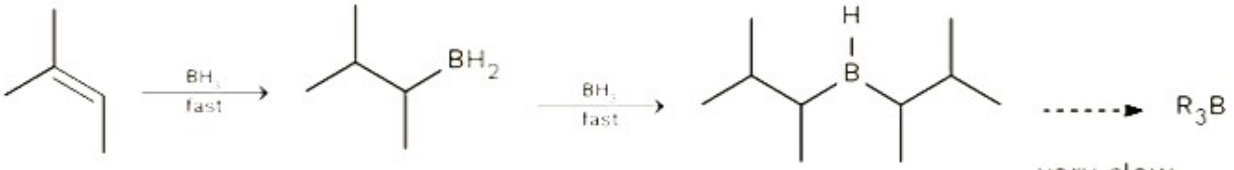
Points to remember

Hydration of alkenes through hydroboration is highly regioselective, alcohol formed corresponds to anti-Markovnikov's addition of water.

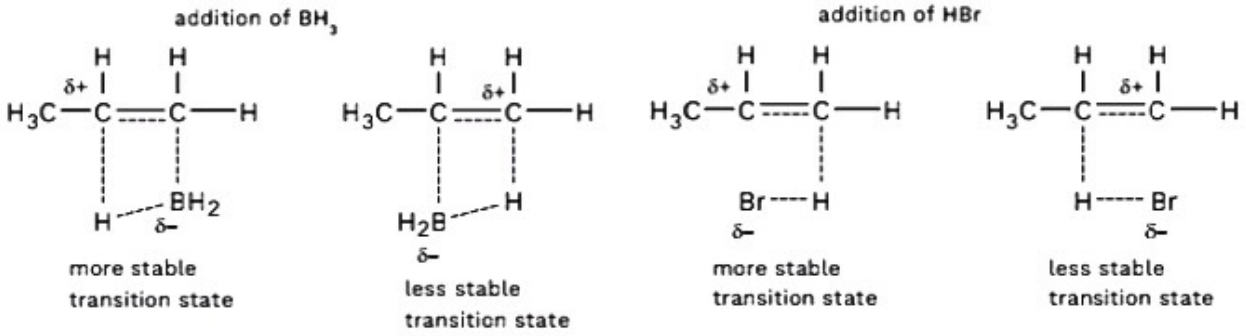


3,3-Dimethyl-1-butanol(1°)

Bulky substituents improve the selectivity of hydroboration



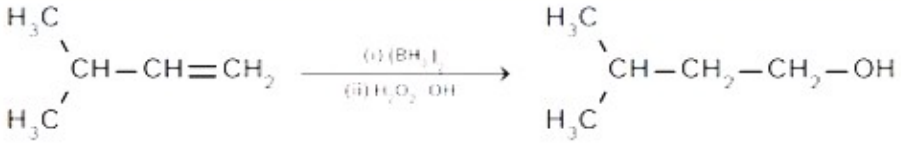
Reason behind anti mark product formation in HBO reaction :



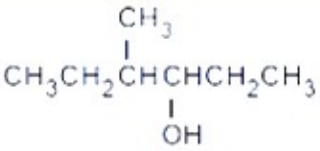
What alkenes might be used for preparing the following alcohols by hydroboration-oxidation?

Q135 $(CH_3)_2CHCH_2CH_2OH$ **Expected alkene may be?**

Sol.

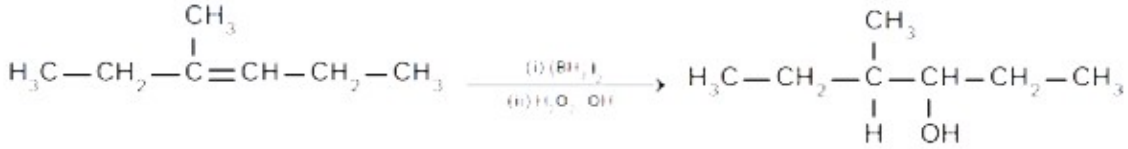


Q136



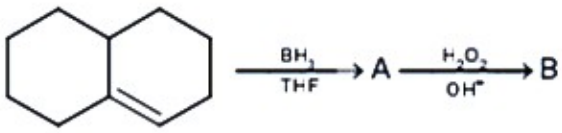
Expected alkene may be?

Sol.



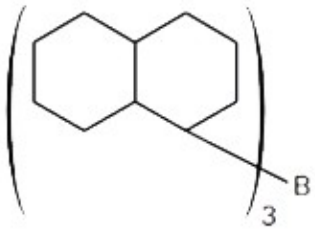
Give the product of

Q137

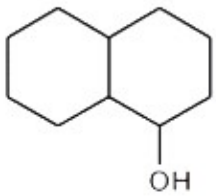


Sol.

A:

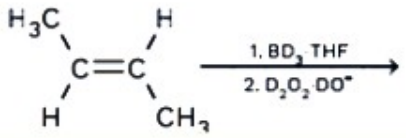


B:

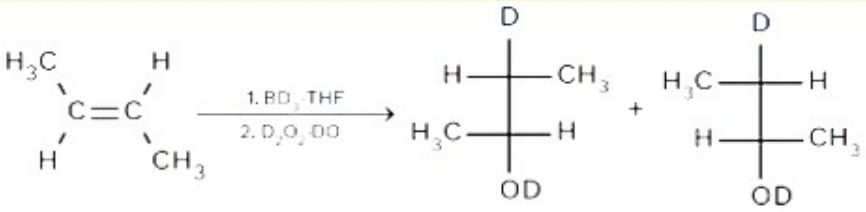


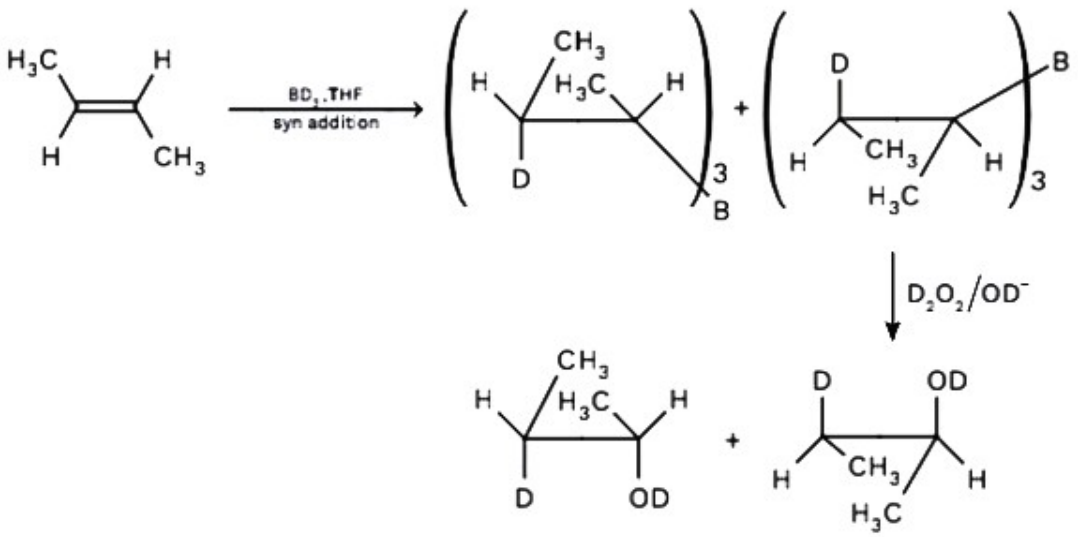
(3 mole)

Q138

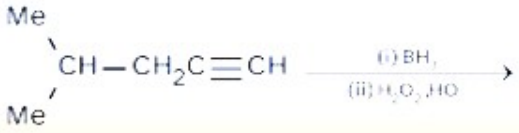


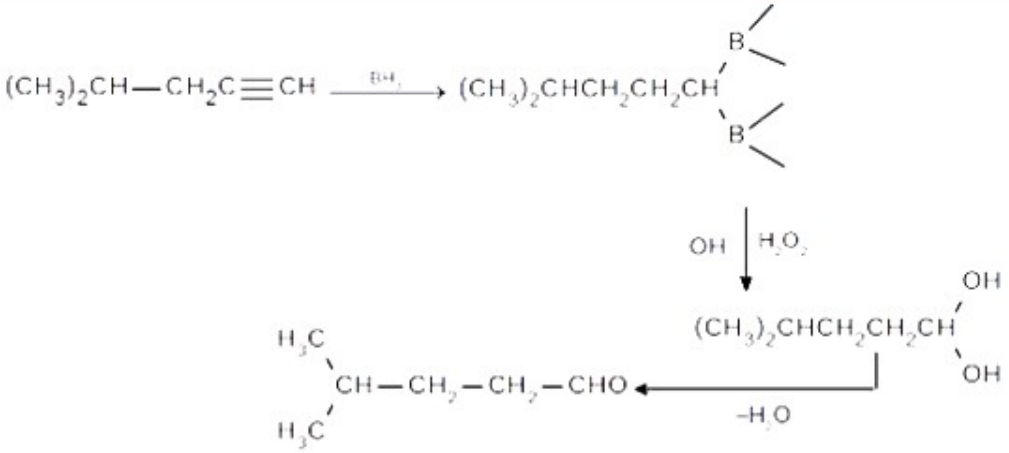
Sol.



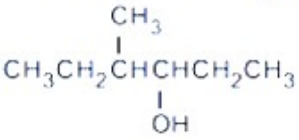


Q139

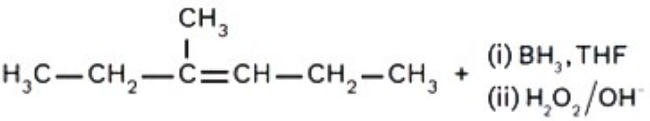




Q140 **Predict the starting material for preparing following alcohol by hydration method so that a quantitative yield can be obtained.**



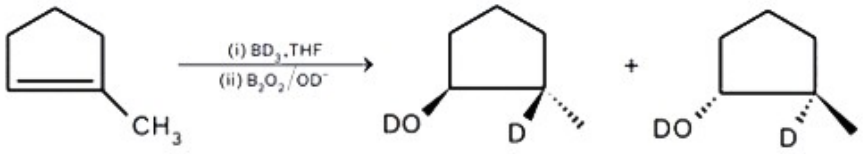
Sol.



The product formed is result of antimarkovnikov addition on alkene which can be obtained by HBO.

Q141 **What product will be formed from hydroboration-oxidation of 1-methylcyclopentene with deuterated borane, BD_3? Show both the stereochemistry and regiochemistry of the product.**

Sol.



(1-methylcyclopentene)

→ Anti Mark Product Formation

→ syn addition

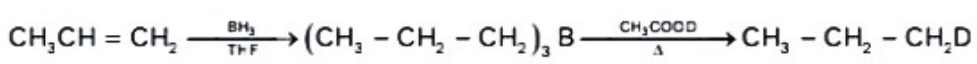
As reaction is syn addition two products will be formed as enantiomeric pairs of each other.

Preparation of Alkanes:

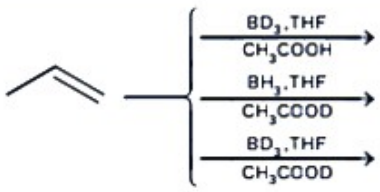
Q142 **Alkylboranes are readily cleaved to alkanes on boiling with acetic or propanoic acid.**

Convert $CH_3CH = CH_2$ to $CH_3CH_2CH_2D$.

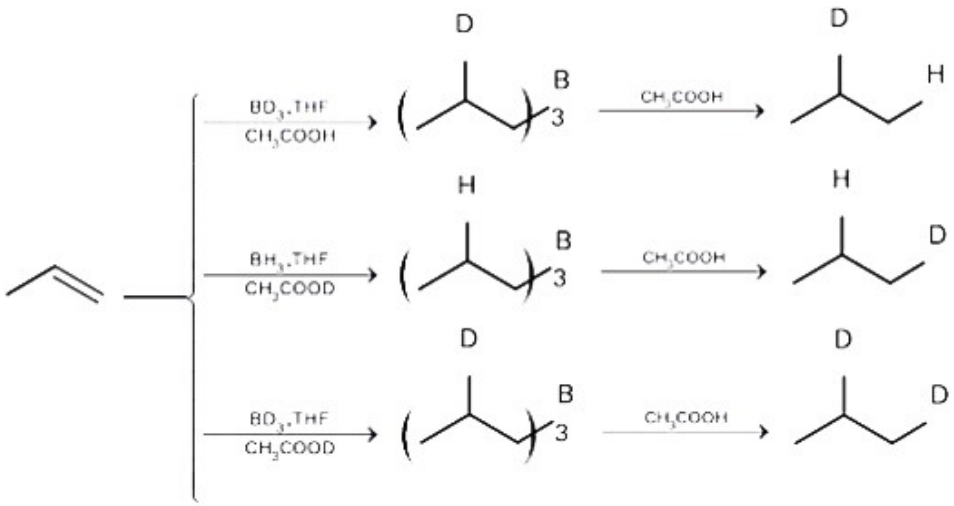
Sol.



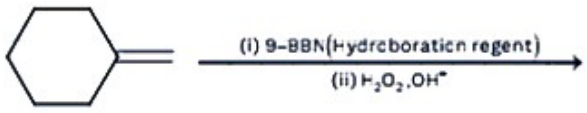
Q143



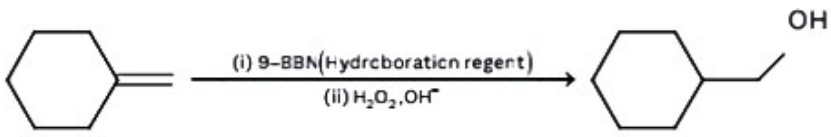
Sol.

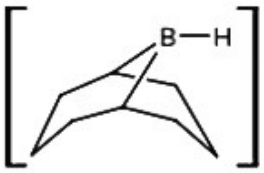


Q144

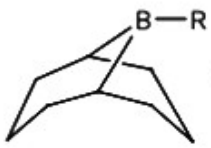


Sol.



9-BBM 

with 9-BBN the 1st step of HBO will form



instead of R_3B.

Ozonolysis:

Ozonolysis is example of electrophilic addition reaction. Bond cleavage using peroxide E^+ leads two possible way of formation of products.

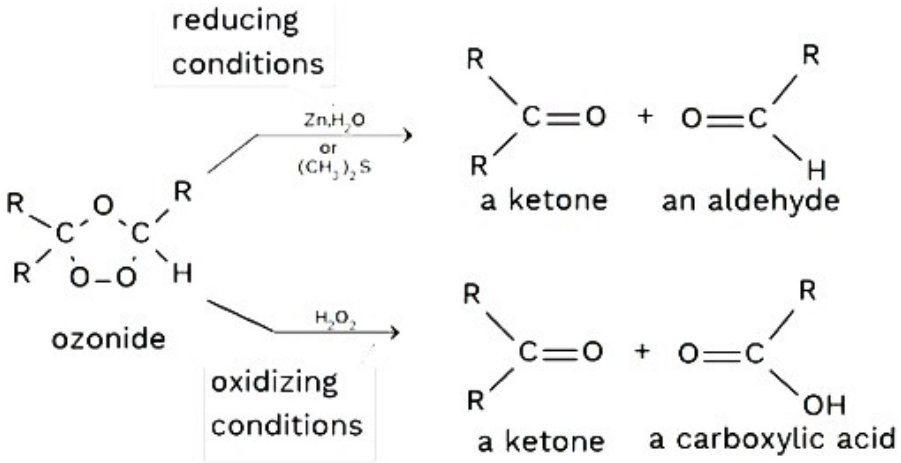
(i) Oxidative product

(ii) Reductive product

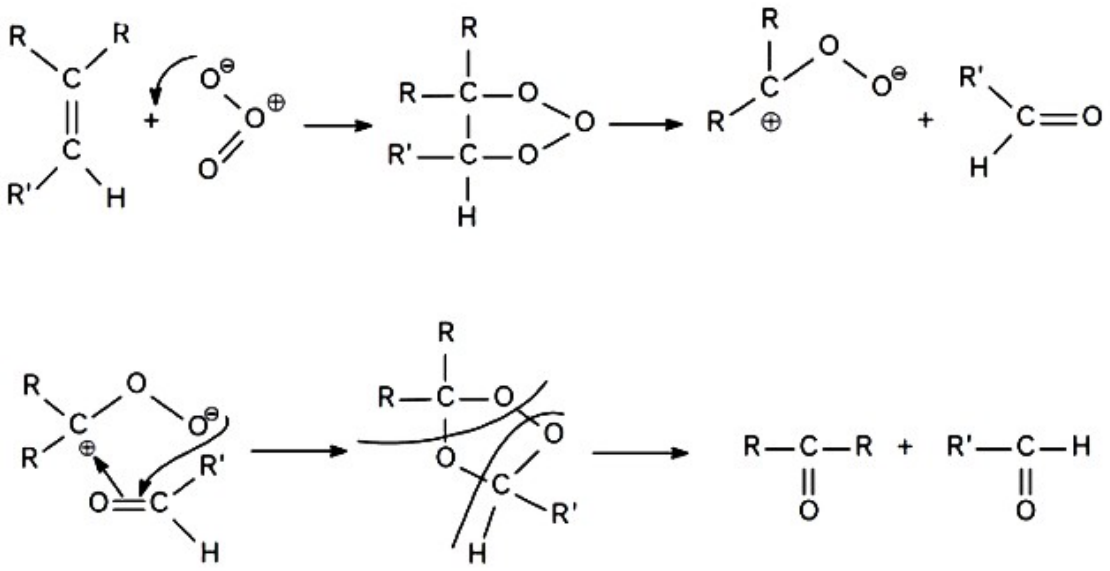
Reagent Used

(i) For Reductive cleavage Zn/H_2O or Zn/CH_3COOH

(ii) For oxidative cleavage O_3 or RCOOOH (peroxide) or Ag_2O



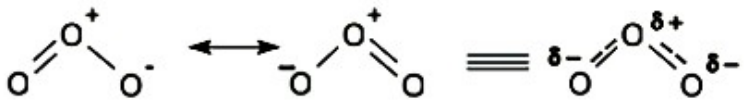
Ozone and the alkene undergo a concerted cycloaddition reaction- the oxygen atoms add to the two sp^2 carbons in a single step,. The addition of ozone to the alkene should remind you of the electrophilic addition reactions of alkenes



Points to remember

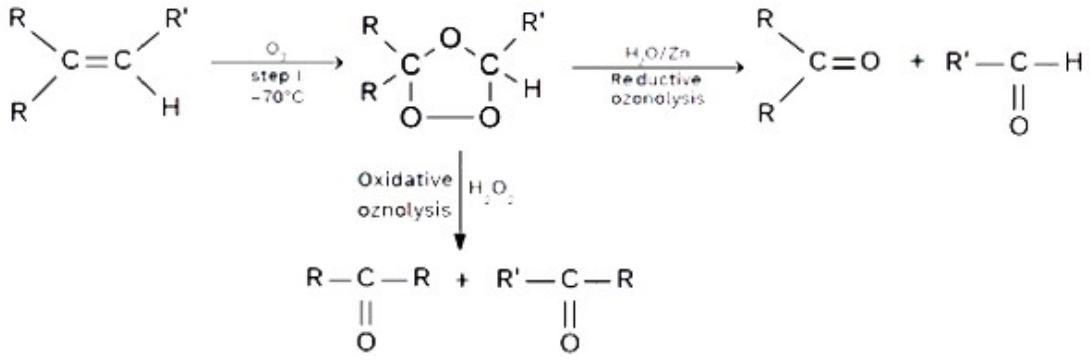
In mechanism of ozonolysis electrophile add to one of sp^2 carbon and nucleophile adds to other. Oxygen in ozone act as electrophile and nucleophile both. Product of ozone addition to an alkene is first molozonide, Unstable due to O–O bond. Rearrange to more stable ozonide

Ozone is produced by passing oxygen gas through an electric discharge. The structure of ozone can be represented by the following resonance contributors :



Resonance contributors of ozone

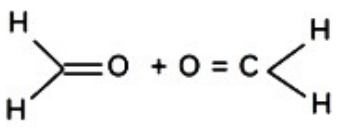
Example



In case of oxidative ozonolysis aldehyde (not ketone) further undergoes oxidation which gives acid as product.

Give the product of the following reaction.

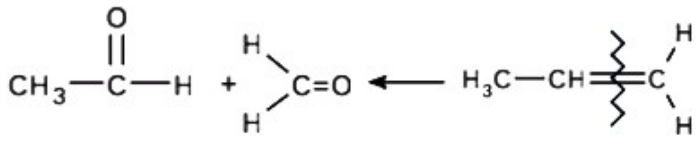
Q145 $H_2C = CH_2 \xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3}$

Sol. 

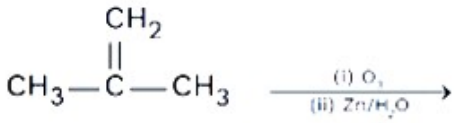
(Reductive ozonolysis)

Q146 $CH_3-CH=CH_2 \xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3}$

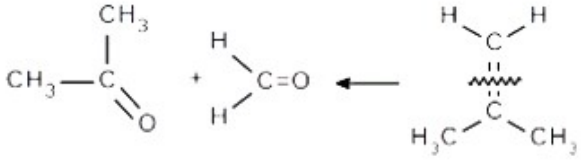
Sol.



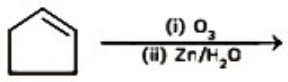
Q147



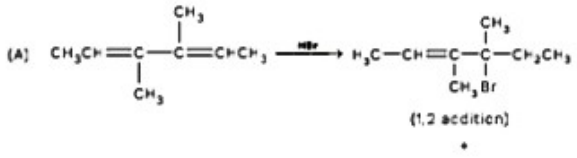
Sol.



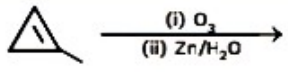
Q148



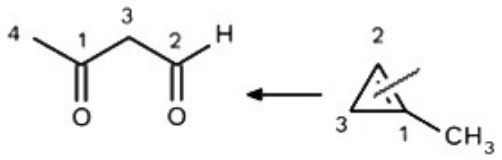
Sol.



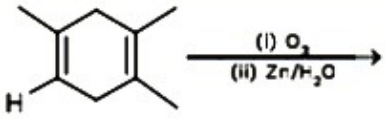
Q149



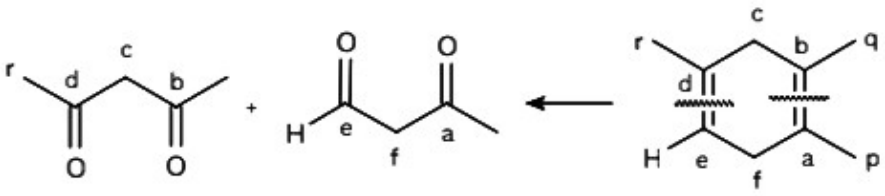
Sol.



Q150



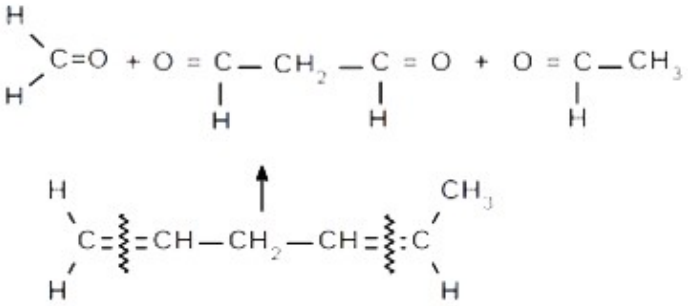
Sol.



Q151

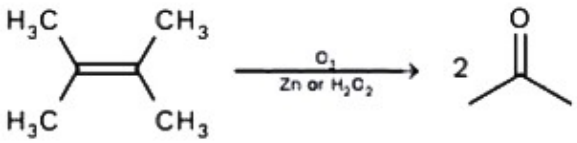
$H_2C{=}CH{-}CH_2{-}CH{=}CH{-}CH_3 \xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3}$

Sol.



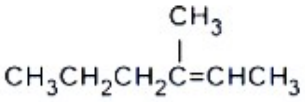
Q152 **Give an example of an alkene that will form the same ozonolysis products, regardless of whether the ozonide is worked up under reducing conditions (Zn, H_2O) or oxidizing conditions (H_2O_2).**

Sol. Alkenes where double bond is substituted by only alkyl groups will give same produces in reductive as well as oxidative ozonolysis.



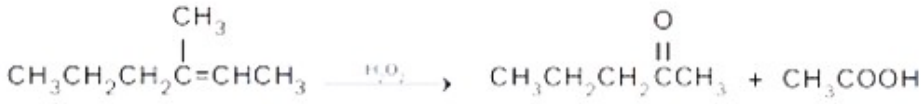
Direction: Products that you would expect to obtain when the following compounds are treated with ozone, followed by work-up with :
(a) Zn, H_2O: (b) H_2O_2:

Q153



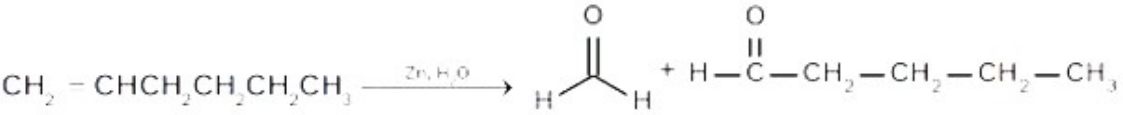
Sol.





Q154 $CH_2=CHCH_2CH_2CH_2CH_3$

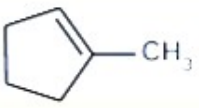
Sol. **(i)**



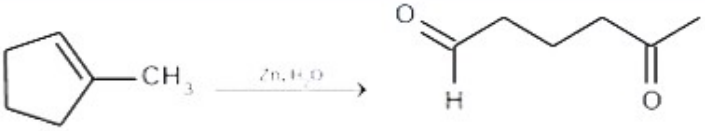
(ii)



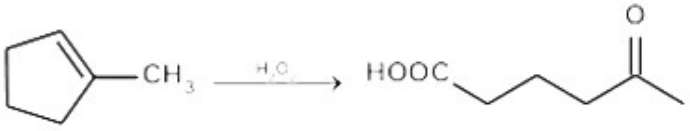
Q155



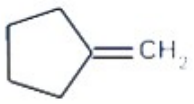
Sol. **(i)**



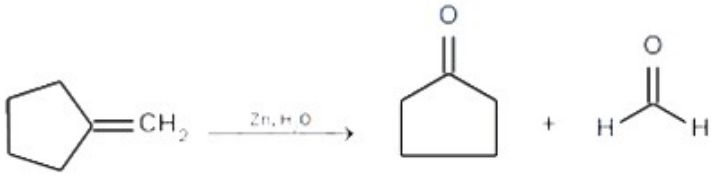
(ii)



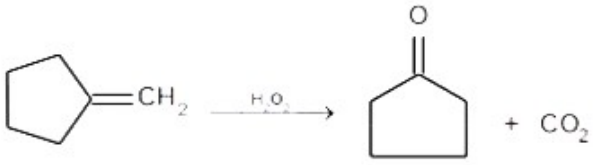
Q156



Sol. **(i)**



(ii)



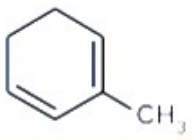
Q157 $CH_3CH_2CH_2CH = CHCH_2CH_2CH_3$

Sol.

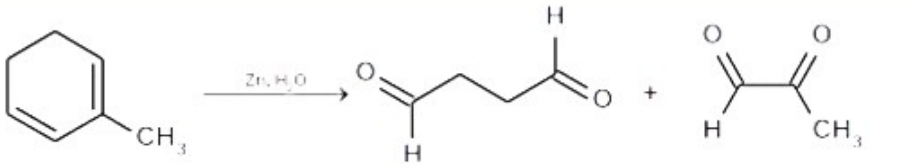
$$CH_3CH_2CH_2CH = CHCH_2CH_2CH_3 \xrightarrow{Zn,H_2O} 2CH_3CH_2CH_2\overset{\overset{O}{\|}}{C}-H$$

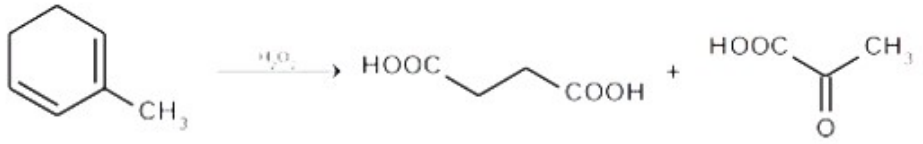
$$CH_3CH_2CH_2CH = CHCH_2CH_2CH_3 \xrightarrow{H_2O_2} 2CH_3CH_2CH_2COOH$$

Q158



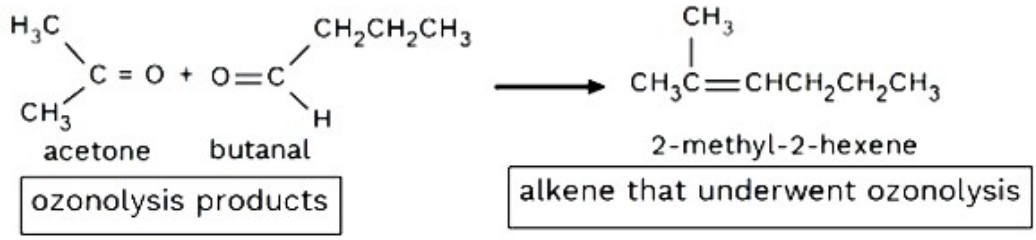
Sol.





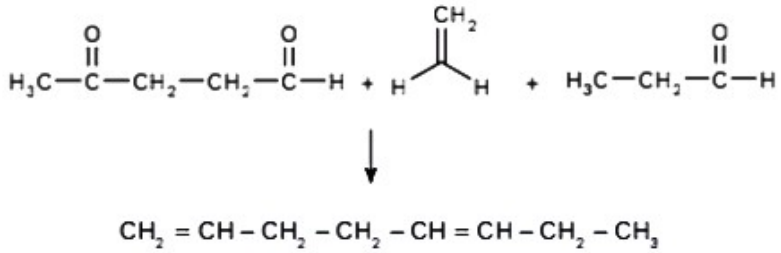
Workup in Ozonolysis:

Ozonolysis can be used to determine the structure of an unknown alkene. If you known what carbonyl compounds are formed by ozonolysis, you can mentally work backward to deduce the structure of the alkene.



Q159 **The following products were obtained from ozonolysis of a diene followed by work-up under reducing conditions. Give the structure of the diene.**

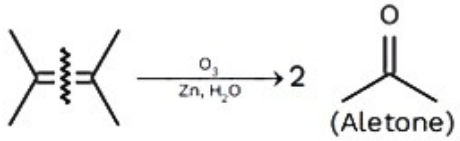
Sol.



$$CH_2 = CH - CH_2 - CH_2 - CH = CH - CH_2 - CH_3$$

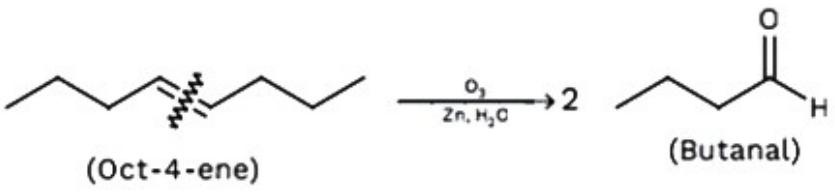
Q160 **What alkene would give only acetone as a product of ozonolysis?**

Sol.



Q161 **What alkenes would give only butanal as a product of ozonolysis?**

Sol.

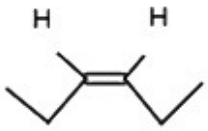


(Oct-4-ene) (Butanal)

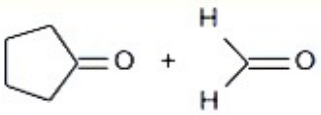
Find out the structure of reactant.

Q162 $X \xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3} CH_3-CH_2-\underset{\underset{O}{\|}}{C}-H$

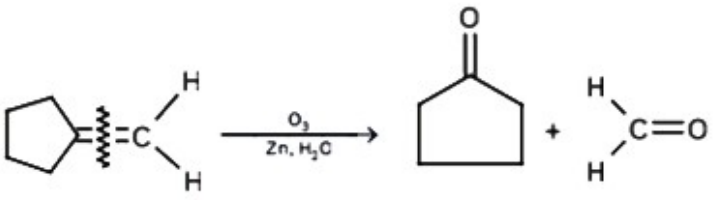
Sol.



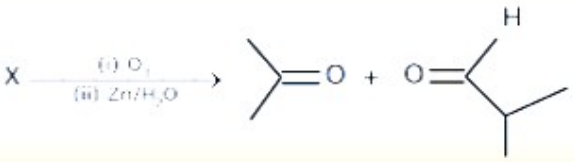
The product is propanal hence alkene must be (Hex-3-ene)

Q163 $X \xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3}$ 

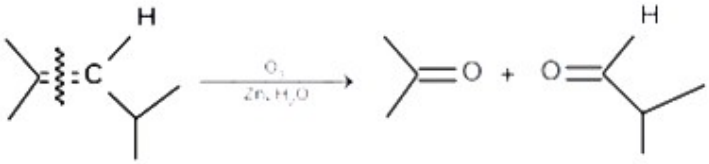
Sol.



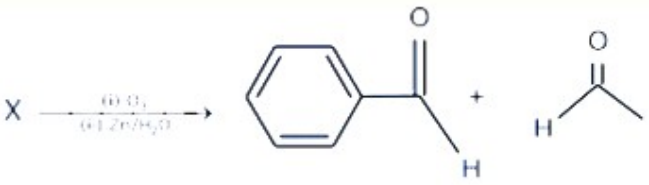
Q164



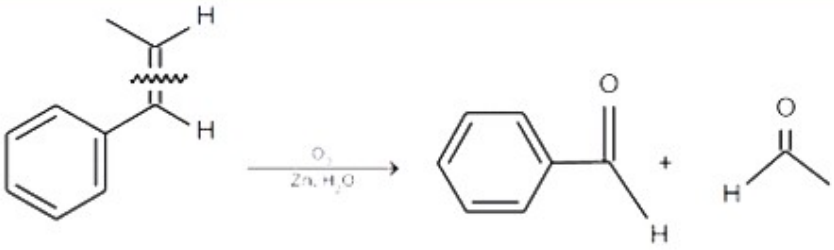
Sol.



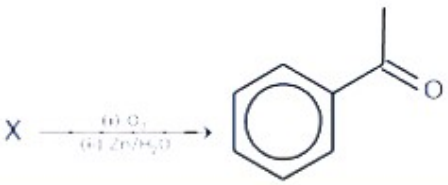
Q165



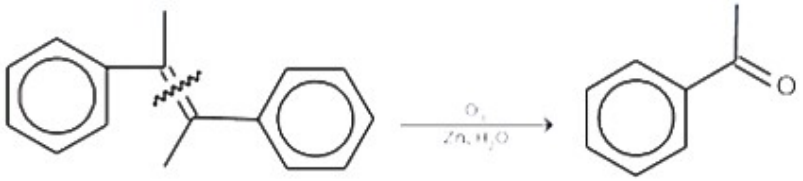
Sol.



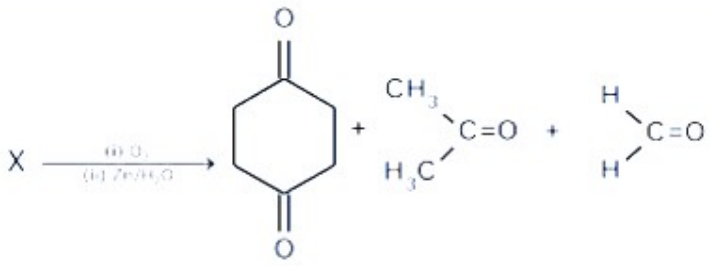
Q166



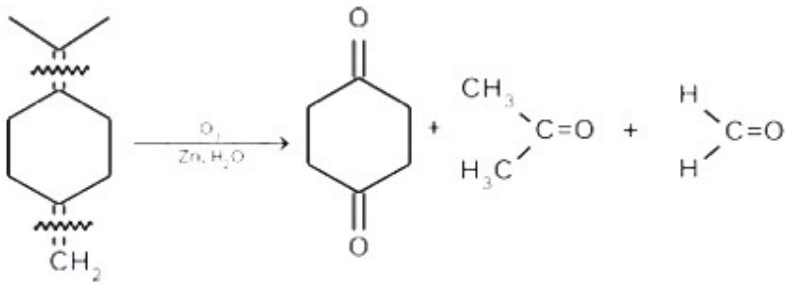
Sol.



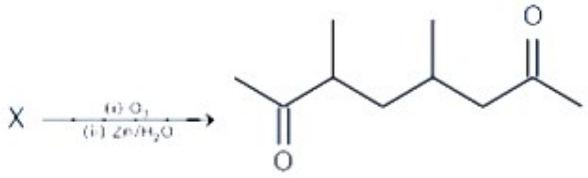
Q167



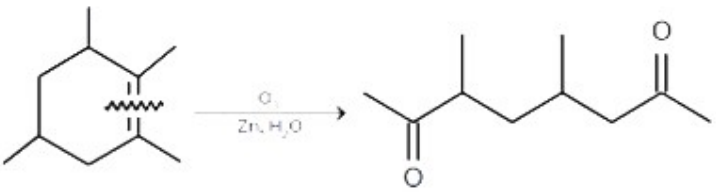
Sol.



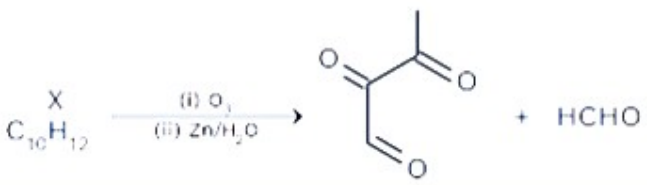
Q168



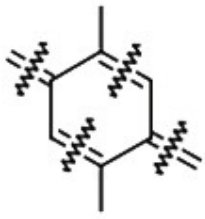
Sol.



Q169



Sol.

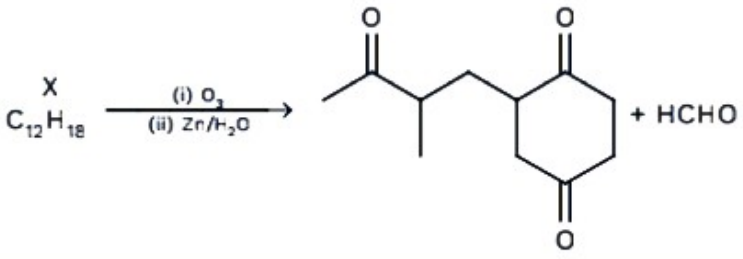


Q170 $\underset{(X)}{C_6H_4} \xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3} C_3H_2O_3$

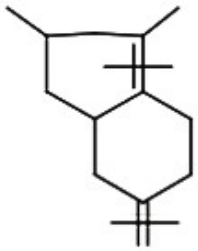
Sol.



Q171

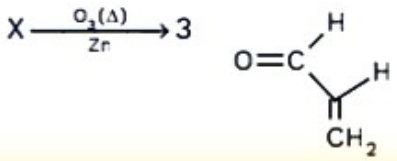


Sol.



Give the ozonolysis product of the following.

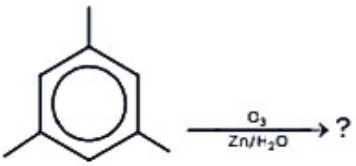
Q172



Sol.

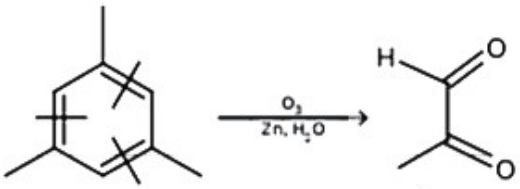


Q173



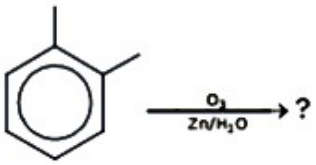
How many species.

Sol.



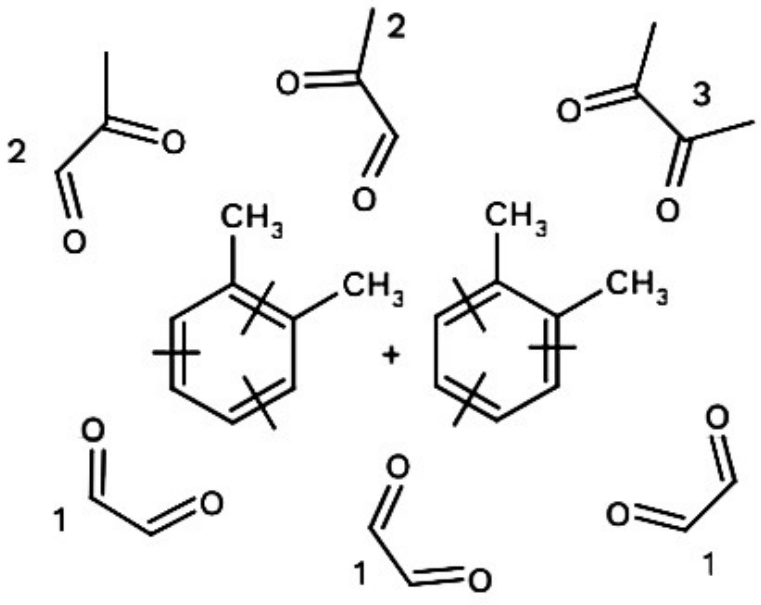
only one

Q174



How many species are found.

Sol.

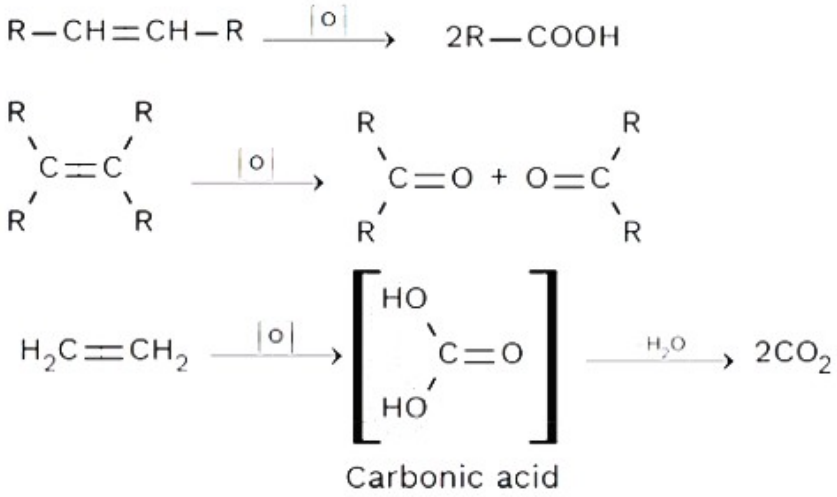


Oxidative Cleavage

Alkenes are oxidatively cleaved by any of the following reagents:

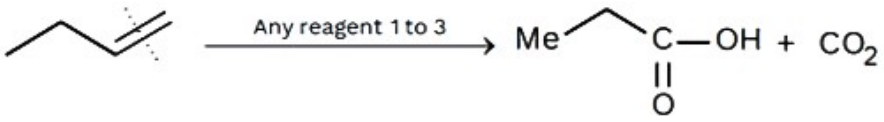
1. Hot alkaline $KMnO_4$ solution followed by acidification
2. Hot acidic $KMnO_4$ solution
3. Hot acidic $K_2Cr_2O_7$ solution

Tricks for Oxidation



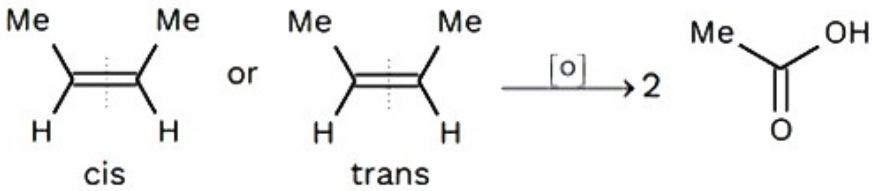
Carbonic acid

The oxidative cleavage of alkenes is used to establish the position or location of the double bond and the structure of the alkene chain or ring. (Rules)

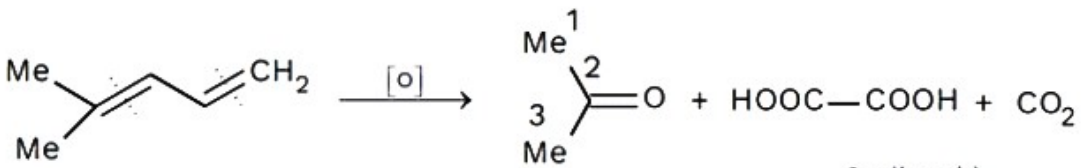


Examples

1.



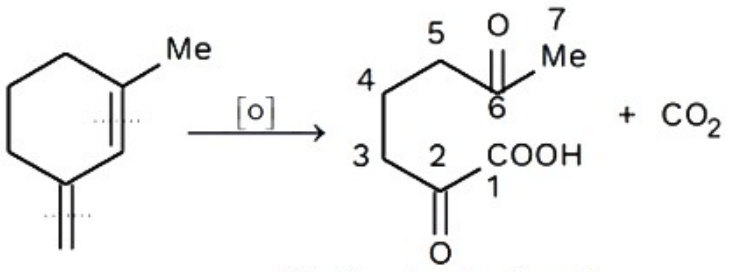
2.



Acetone
or
Propan-2-one

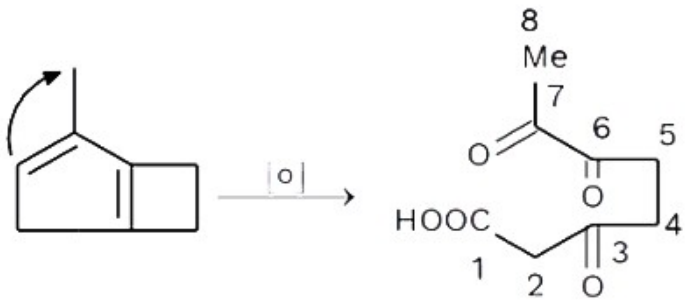
Oxalic acid
or
Ethane-1,2-dioic acid

3.



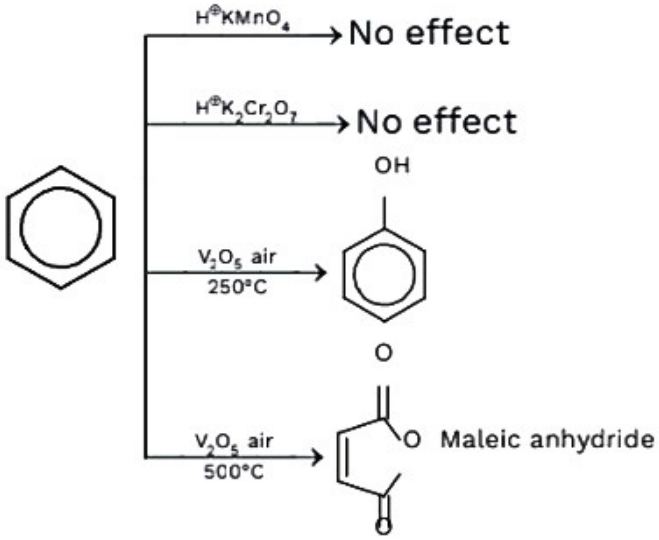
2,6-Dioxoheptanoic acid

4.

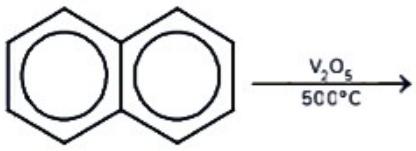


3,6,7-Trioxo octanoic acid

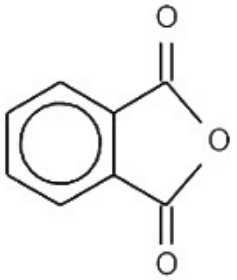
Oxidation of Aromatic Compounds



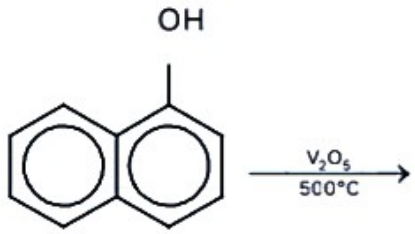
Q175



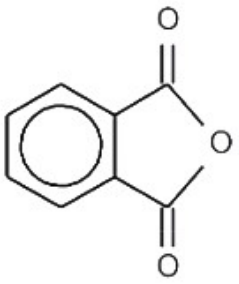
Sol.



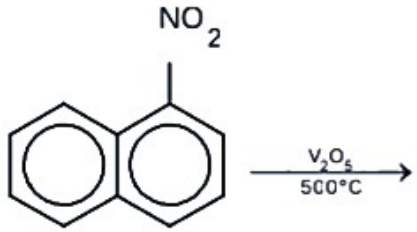
Q176



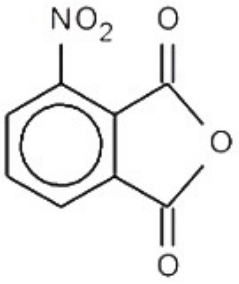
Sol.



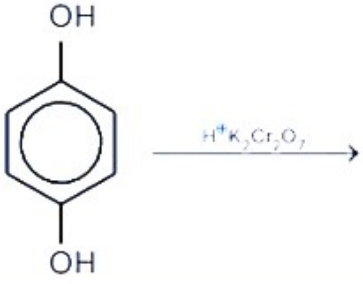
Q177



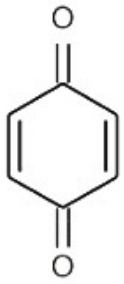
Sol.



Q178

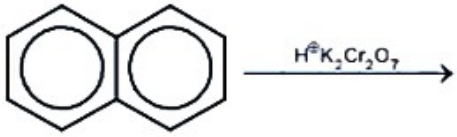


Sol.

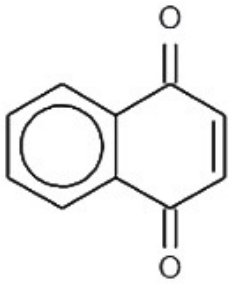


Benzoquinone

Q179

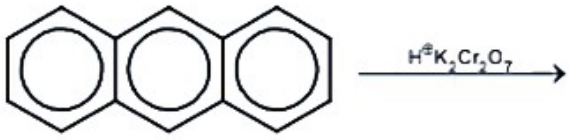


Sol.

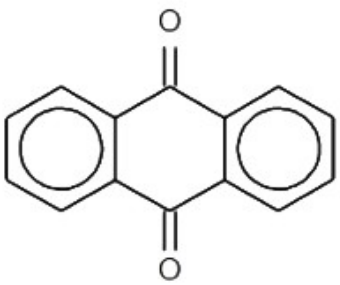


Naphthoquinone

Q180

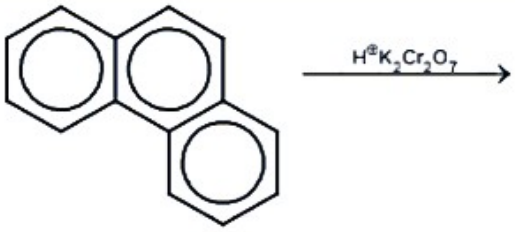


Sol.

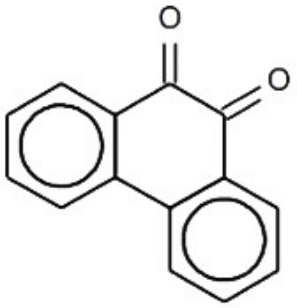


Anthraquinone

Q181

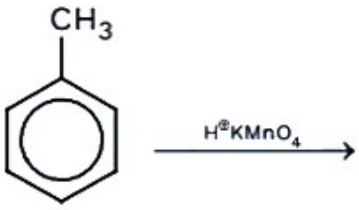


Sol.

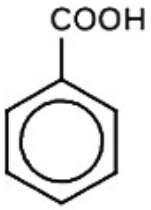


Phenanthroquinone

Q182

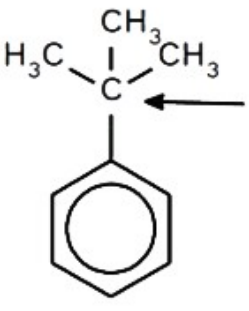


Sol.



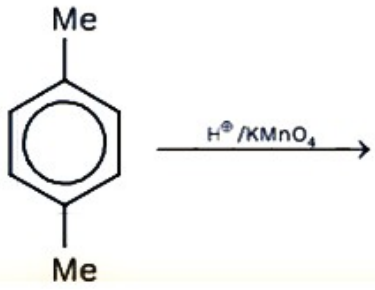
Q183 $Ph - CMe_3 \xrightarrow[\Delta]{H^{\oplus}KMnO_4}$

Sol. $Me_3C - COOH$

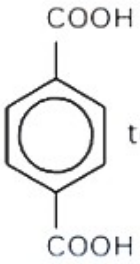


Being 3° C-atom, won't get oxidized into -COOH. Hence cleavage of benzene will take place to form -COOH.

Q184

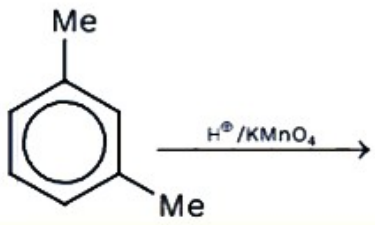


Sol.

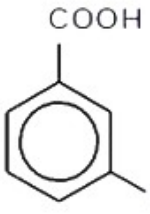


terephthalic acid

Q185

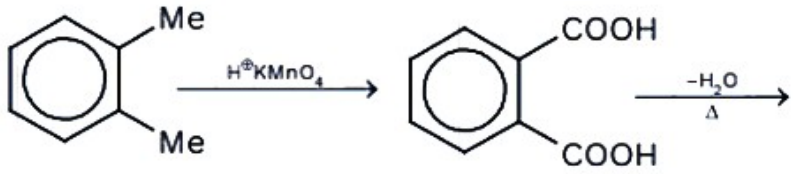


Sol.



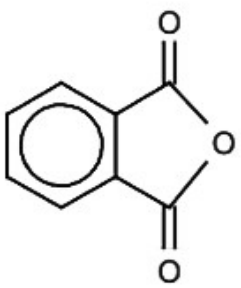
isophthalic acid

Q186



Phthalic acid

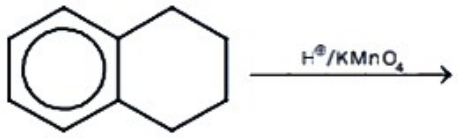
Sol.



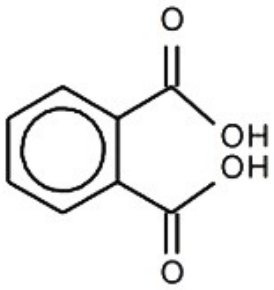
(anhydride of dioic acid)

Phthalic anhydride

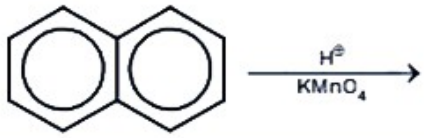
Q187



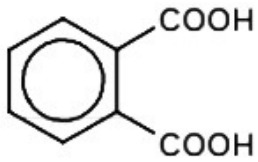
Sol.



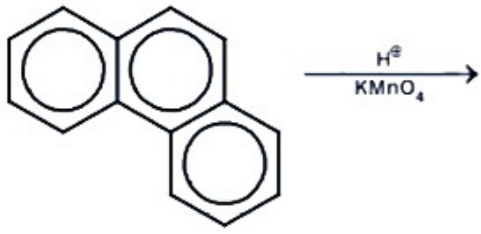
Q188



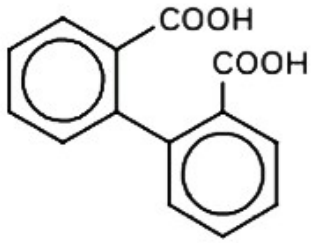
Sol.



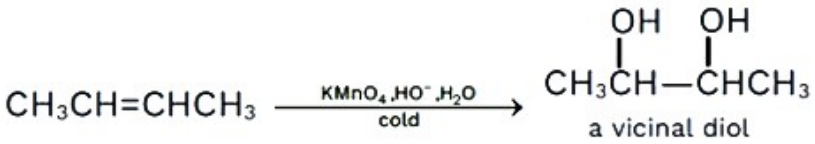
Q189

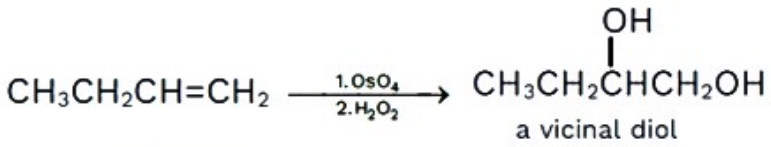


Sol.



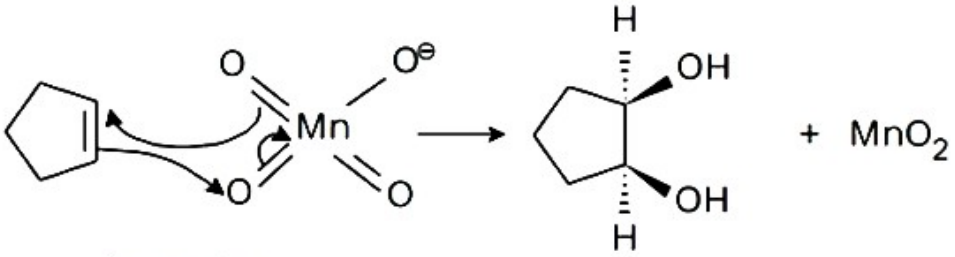
Oxidation of Alkenes:
Syn Addition
1,2-dihydroxylation





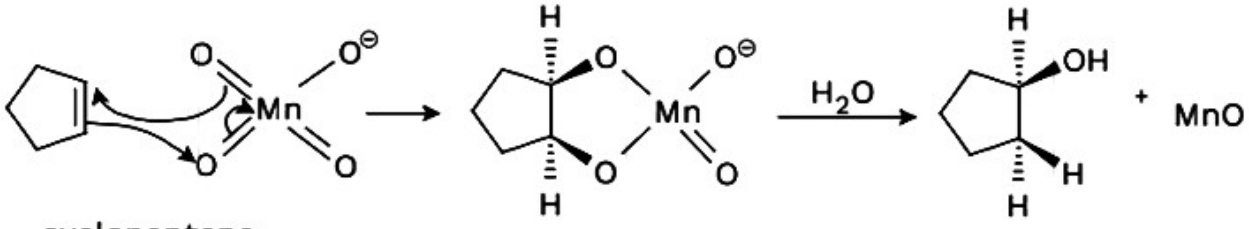
Reactants Products

Cold alkaline $KMnO_4$



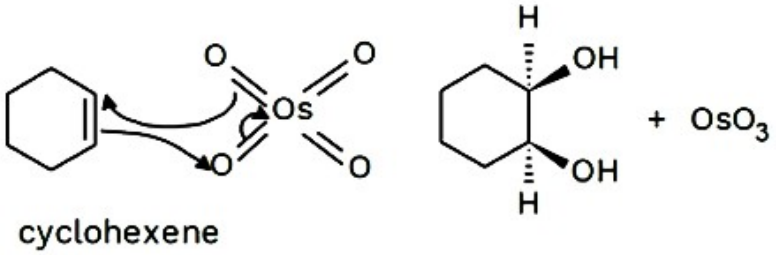
cyclopentene

Mechanism for cis glycol formation



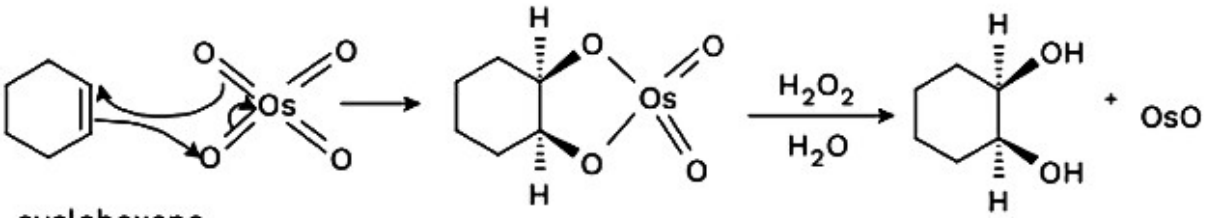
cyclopentene a cyclic manganate intermediate cis-

Syn Addition of Osmium Tetraxoide

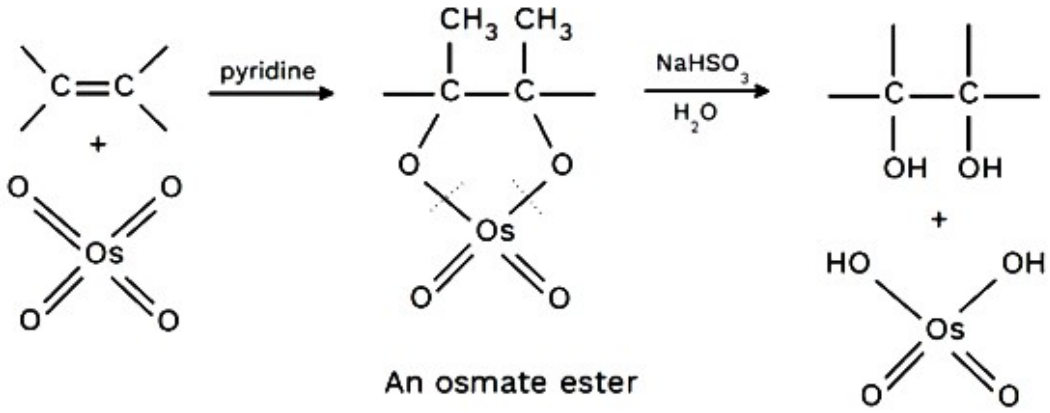


cyclohexene

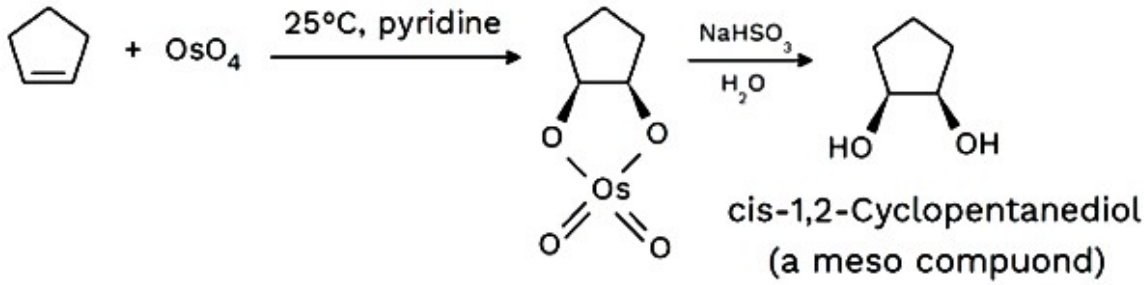
Mechanism for Cis Glycol formation



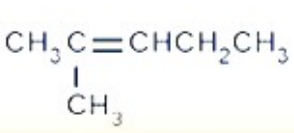
After formation of the cyclic intermediate with osmium, cleavage at the oxygen-metal bonds takes place without altering the stereochemistry of the two new C–O bonds.

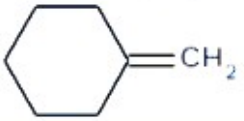


The syn stereochemistry of this dihydroxylation can readily be observed by the reaction of cyclopentene with osmium tetraoxide. The product is cis-1,2-cyclopentanediol.

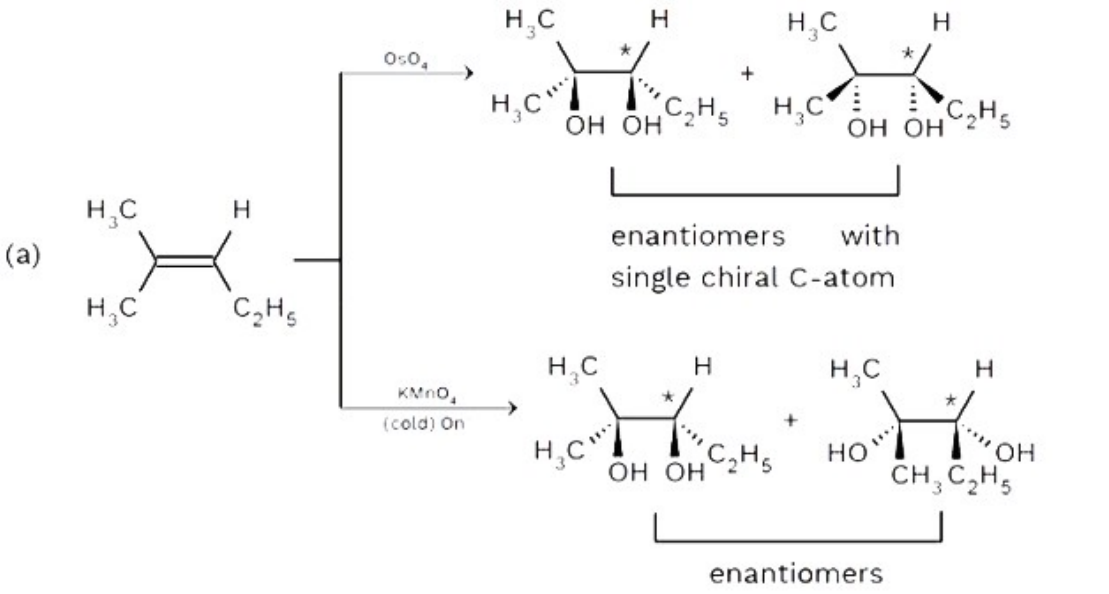


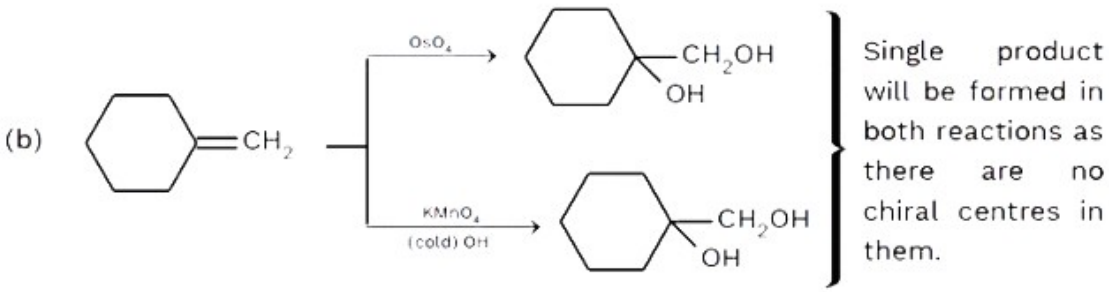
Q190 **Given the products that would be formed from the reaction of each of the following alkenes with OsO_4 and KMO_4 followed by hydrolysis.**

(a) 

(b) 

Sol.





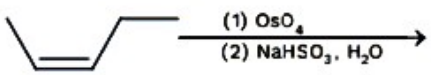
Single product will be formed in both reactions as there are no chiral centres in them.

Q191 **What stereoisomers would be formed from hydroxylation reaction of each of the following Alkenes with Oso_4 followed by H_2O_2:**

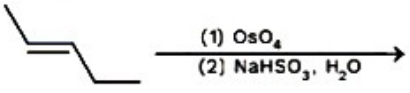
(a) trans-2-butene

(b) Cis-2-butene

(c) Cis-2-pentene

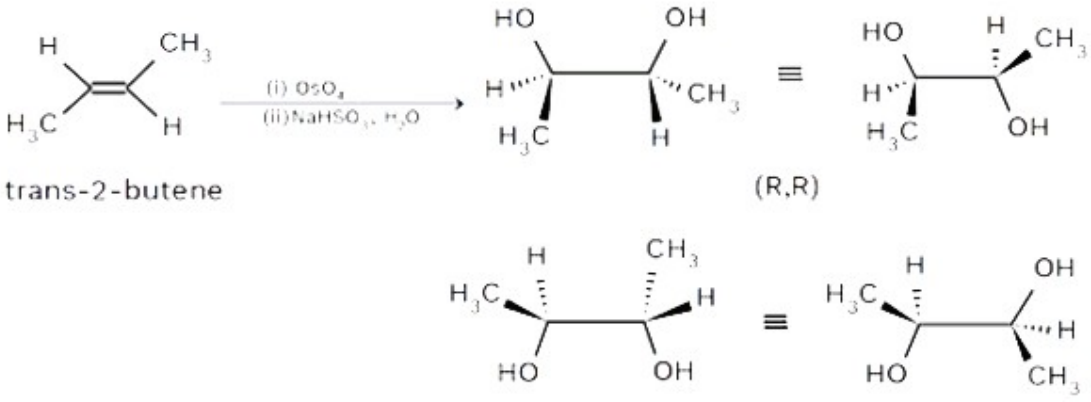


(d) trans-2-pentene

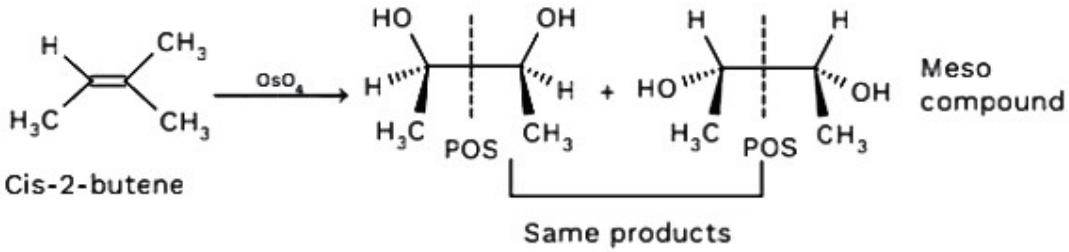


Sol.

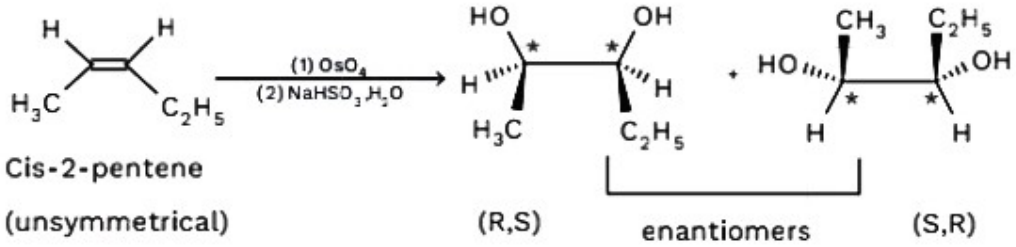
(a)


trans-2-butene
(i) OsO4
(ii) NaHSO3, H2O
(R,R)
(S,S)
Enantiomeric 2,3-butanediols

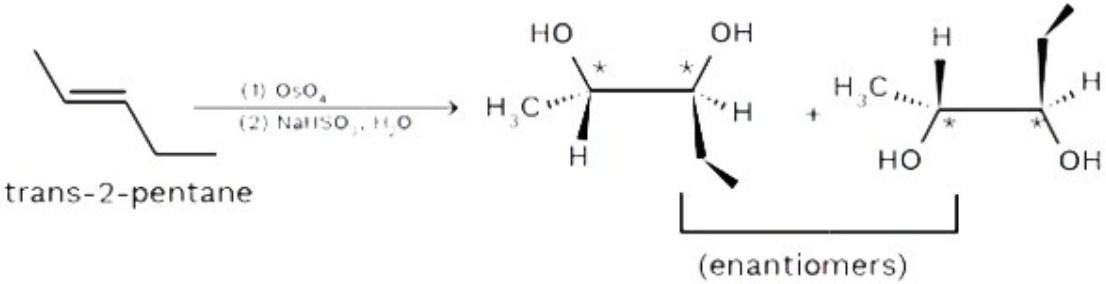
(b)


Cis-2-butene
OsO4
POS
POS
Meso compound
Same products

(c)

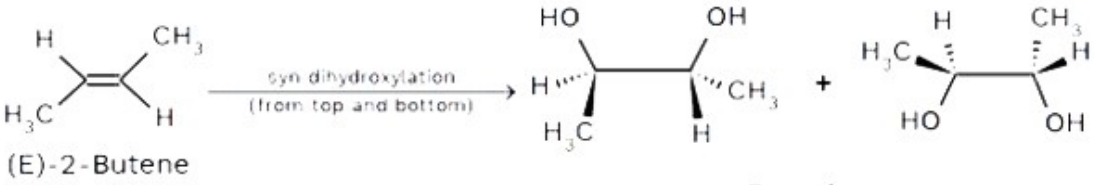

Cis-2-pentene
(unsymmetrical)
(1) OsO4
(2) NaHSO3, H2O
(R,S)
enantiomers
(S,R)

(d)


trans-2-pentane
(1) OsO4
(2) NaHSO3, H2O
(enantiomers)

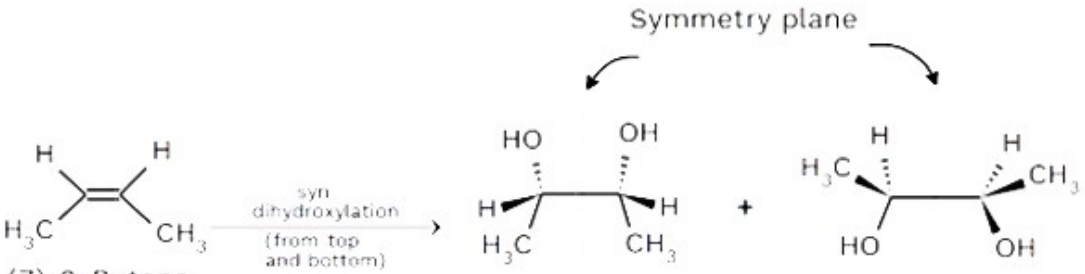
Q192 **Explain the following facts : Treating (Z)-2-butene with OsO_4 in pyridine and then $NaHSO_3$ in water gives a diol that is optically inactive and cannot be resolved. Treating (E)-2-butene with the some reagents gives a diol that is optically inactive but can be resolved into enantiomers.**

Sol.



Enantiomers
(separable and when separated, optically active)

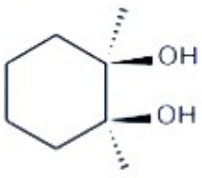
***optically inactive**



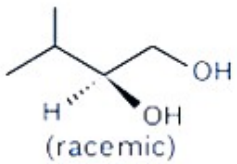
Identical (meso compound)

Q193 **Specify the alkene and reagents needed to synthesize each of the following diols.**

(a)

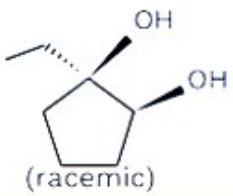


(c)



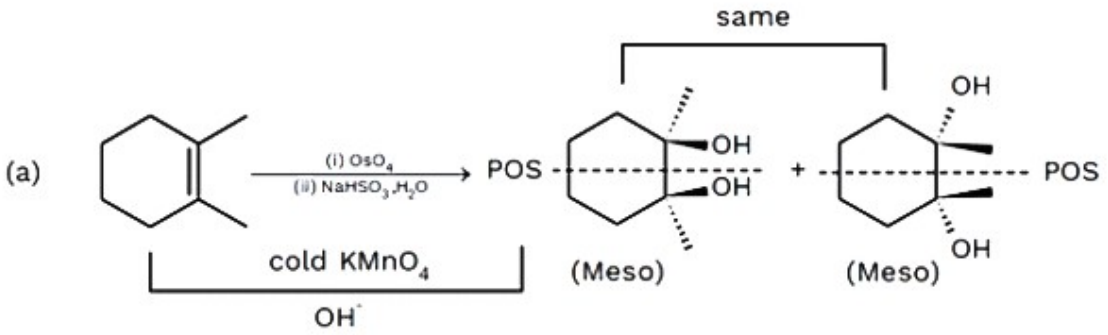
(racemic)

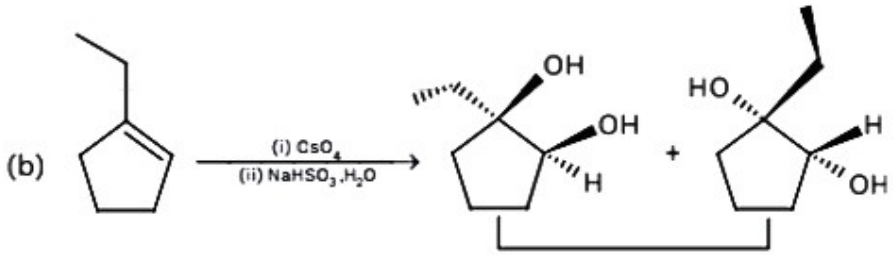
(b)



(racemic)

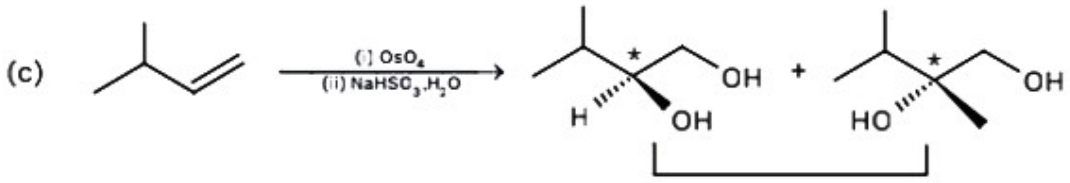
Sol.





enantiomers

⇒ Racemization will take place



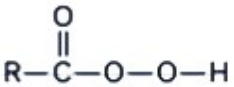
enantiomers

⇒ Racemic mixture

Points to remember

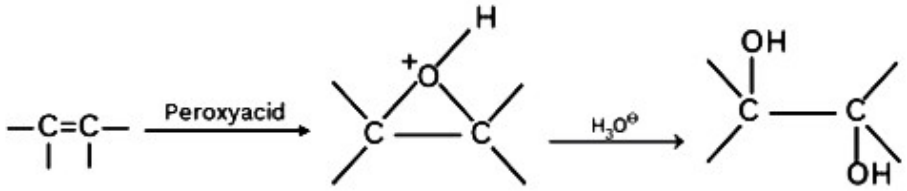
Anti-hydroxylation of Alkenes

1. Peroxyacid have general structure as



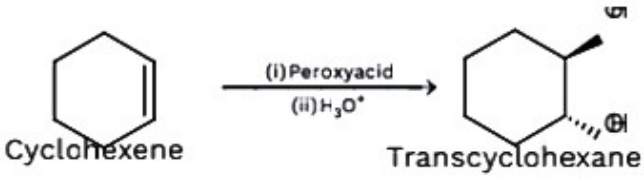
2. The are relatively unstable and easily gives nascent O hence, acting as oxidant. It converts alkene into epoxides.
3. A combination of RCO_3H followed by hydrolysis [$RCO_3H(H_3O^+)$] further breaks epoxide to anti hydroxylation product.

General Reaction:



Epoxide

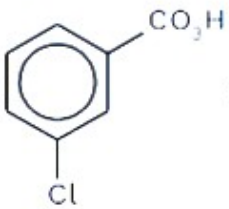
1. It is anti-addition. Both OH groups add on the opposite side of pi-bond.
2. Double bond of aromatic ring cannot react with these reagents.



Cyclohexene

Transcyclohexane 1,2-diol

Q194 Peroxyacid may be any one of these:

(A) 

= m-CPBA (Metachloroperbenzoic acid)

(B) CH_3CO_3H = PAA (Peracetic acid)

(C) $PhCO_3H$ = PBA (Per benzoic acid)

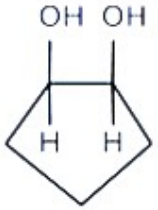
(D) CF_3CO_3H = TFPAA (Trifluoro peracetic acid).

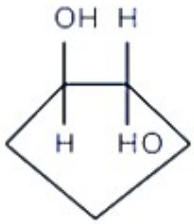
Sol. A, B, C, D

Q195 

x, x is :

(A) 

(B) 

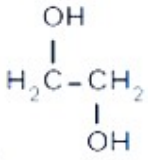
(C) 

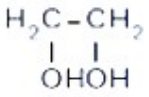
(D) 50% B and 50% C

Sol. (A)

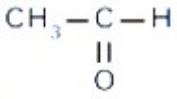
As reaction uses only peroxyacid; Diol will be formed when hydrolysis is done after treating alkene with peroxyacid.

Q196 $CH_2 = CH_2 \xrightarrow{\text{Peroxyacid}} X$, X is :

(A) 

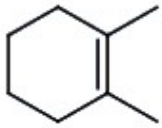
(B) 

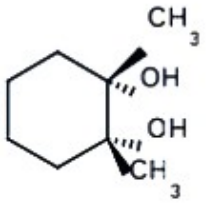
(C) 

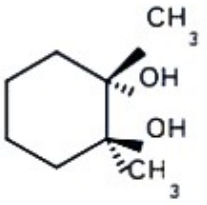
(D) 

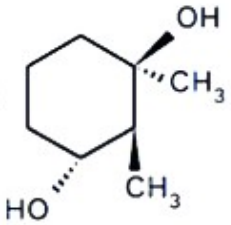
Sol. (C)

peroxy acid converts alkene to epoxide.

Q197  $\xrightarrow[(2)H_3O^{\oplus}]{(1)\text{Peroxyacid}}$ P Product is :

(A) 

(B) 

(C) 

(D) None of these

Sol. (B)

Treatment with peroxyacid followed by hydrolysis on alkene is anti addition of 2-OH groups on alkene only (B) is anti product.

Q198 **Anti hydroxylation of alkene takes place by :**

(A) m-CPBA/$H_3O^{\oplus}$ **(B) $CH_3COOOH/H_3O^{\oplus}$**

(C) $F_3CCOOOH/H_3O^{\oplus}$ **(D) All of these**

Sol. (D)

Follow the theory given for hydroxylation above.

Wurtz Reaction:

This reaction involves the condensation of two molecules of alkyl halides in the presence of sodium and dry ether.

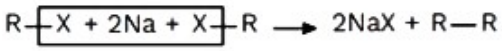
Free radical mechanism.

Product : Alkane (By coupling C–C)

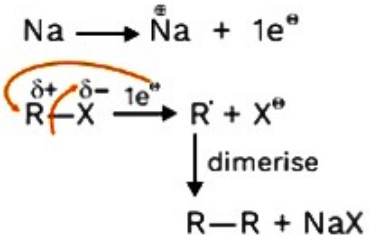
Reaction :

$$2R{-}X \xrightarrow[DE]{2Na} R{-}R + 2NaX$$

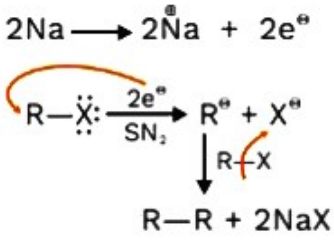
Short Trick :



Free Radical Mechanism

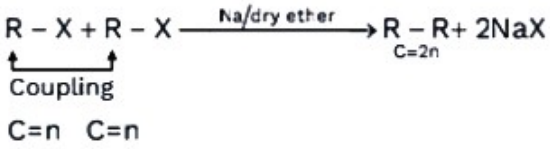


Ionic Mechanism



Different Possibility in Wurtz Reaction

Case-I: When both alkyl halides are same (Simple Wurtz):



Q199 $CH_3 - X + CH_3 - X \xrightarrow{Na/dry\ ether}$

Sol. $CH_3 - X + CH_3 - X \xrightarrow{Na/dry\ ether} CH_3 - CH_3$

Q200 $CH_3 - cH_2 - Br + CH_3 - CH_2 - Br \xrightarrow{Na/dry\ ether}$

Sol. $CH_3 - cH_2 - Br + CH_3 - CH_2 - Br \xrightarrow{Na/dry\ ether} CH_3 - CH_2 - CH_2 - CH_3 + (CH_2 = CH_2 + CH_3 - CH_3)$

(traces of products will be obtained)

Case II: When both alkyl halides are different (Mix Wurtz):

Q201 $R - X + R' - X \xrightarrow{Na/dry\ ether}$

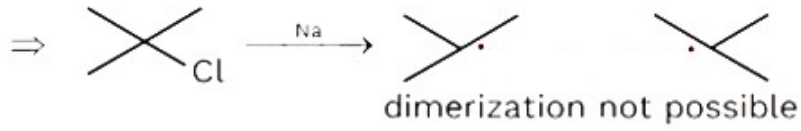
Sol. $R - X + R' - X \xrightarrow{Na/dry\ ether} R - R + R' - R' + R - R'$ (3-products will be obtained)

Q202 $CH_3 - CH_2 - Br + CH_3 - CH_2 - CH_2 - CH_2 - Br \xrightarrow{Na/dry\ ether}$

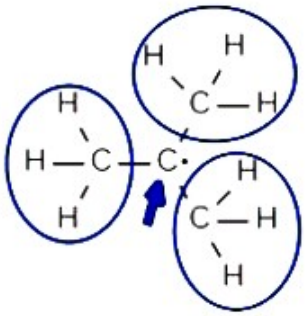
Sol. $\underset{n\text{-butane}}{CH_3CH_2CH_2CH_3} + \underset{n\text{-octane}}{CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_3} + \underset{n\text{-hexane}}{CH_3CH_2CH_2CH_2CH_2CH_3}$

Points to Remember

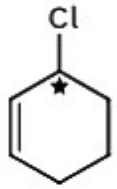
(a) Methane cannot be prepared by this method.
(b) Tertiary alkyl halides do not give this reaction.
(c) Best result is obtained when both alkyl halides are same, i.e., this method is suitable for preparation of alkanes having even number of carbons.
(d) Disproportionation takes place which leads the formation of alkene and alkane as the by products.



Reason

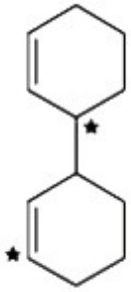


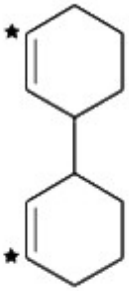
Approach towards Radical center not possible
Dimerization not possible

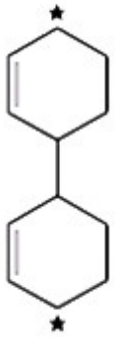
Case III: Ex.  $\xrightarrow[\text{Dry ether}]{Na}$ Possible products is/are

★ = C-14 Isotope

(A) 

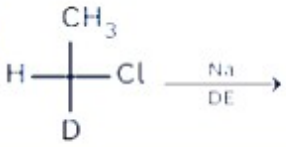
(B) 

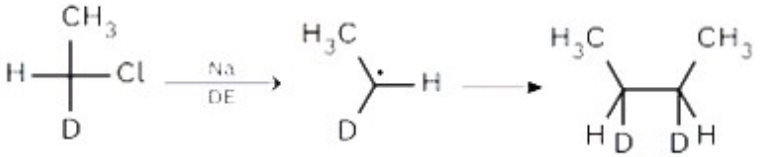
(C) 

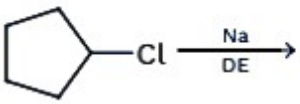
(D) 

Q203 $R-Cl \xrightarrow[DE]{Na}$

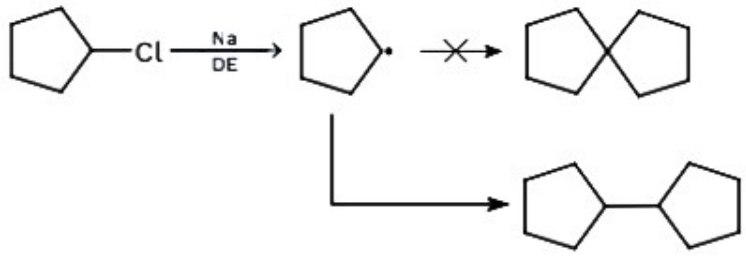
Sol. $R-Cl \xrightarrow[DE]{Na} R^{\bullet} \xrightarrow{dimer} R-R\,(Alkane)$

Q204 

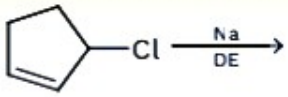
Sol. 

Q205 

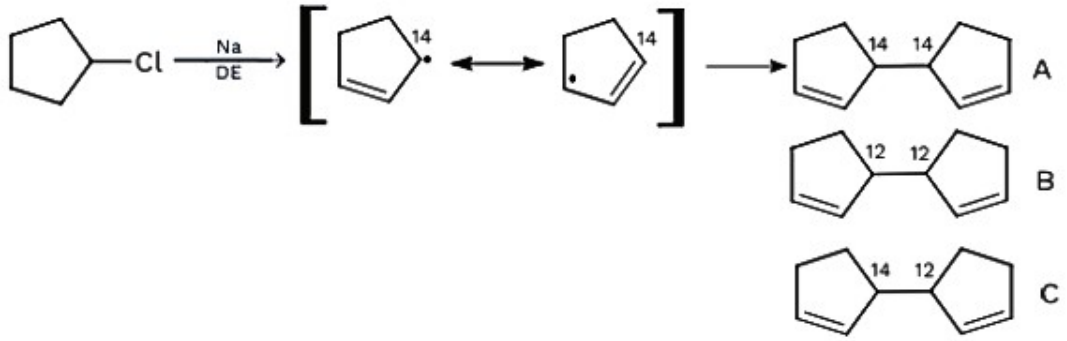
Sol.



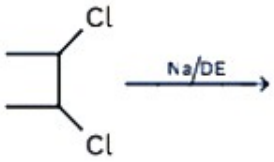
Q206



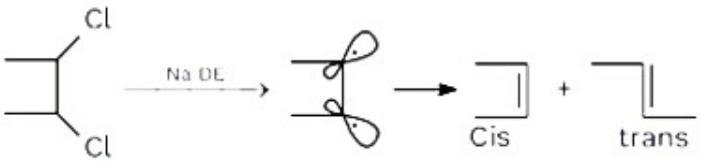
Sol.



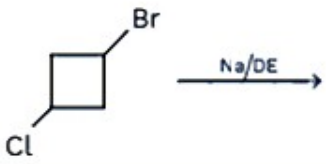
Q207



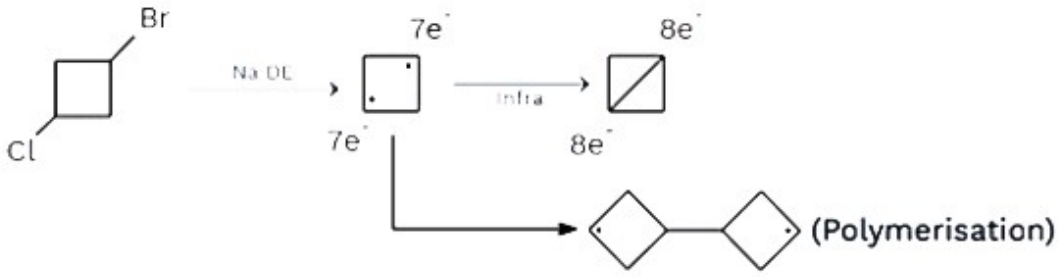
Sol.



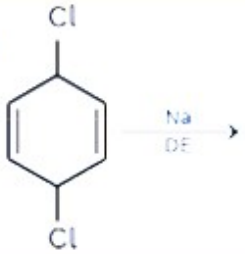
Q208



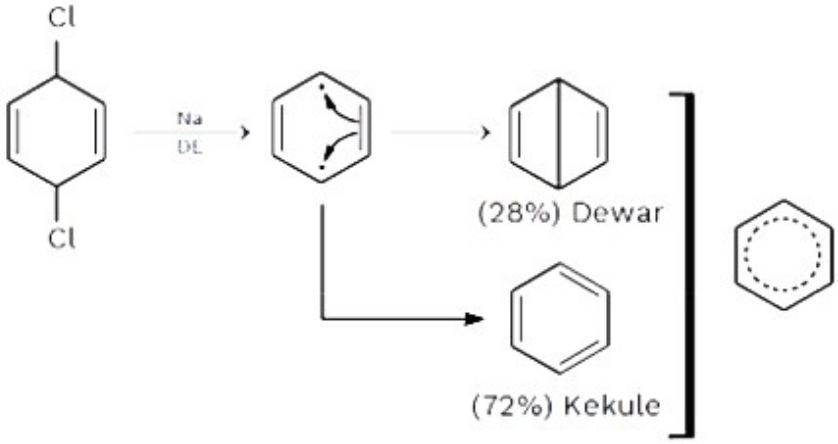
Sol.



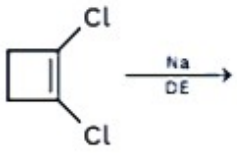
Q209



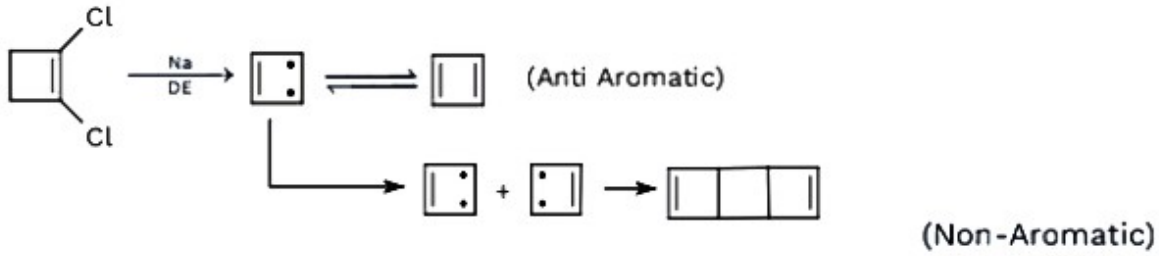
Sol.



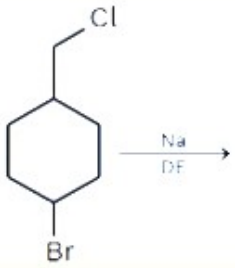
Q210



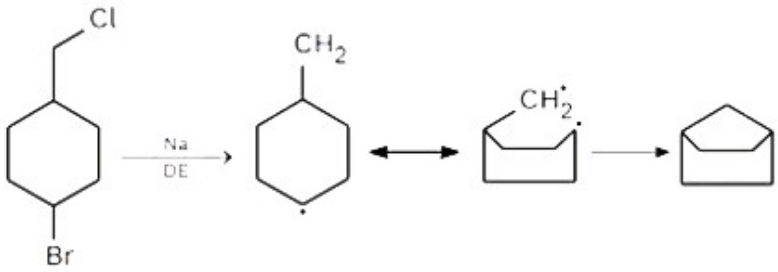
Sol.



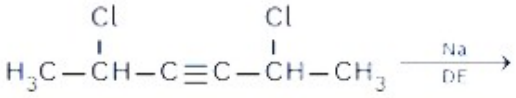
Q211



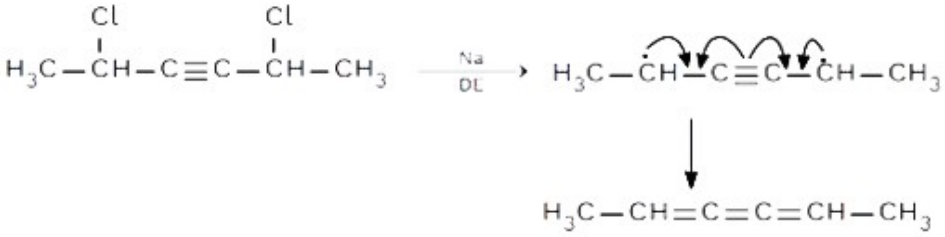
Sol.



Q212



Sol.

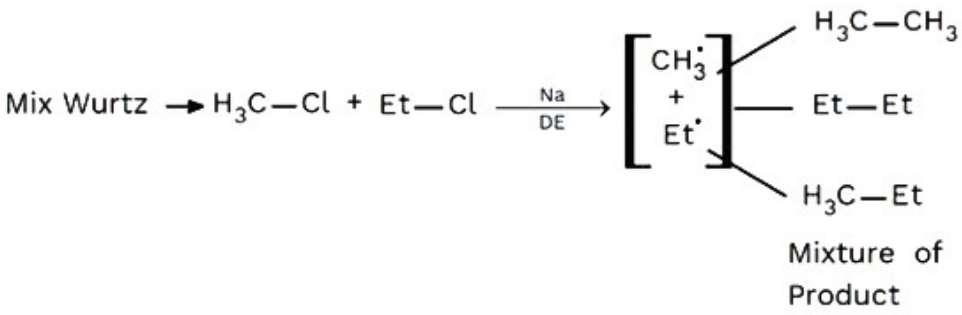


Q213 Simple Wurtz → $CH_3CH_2 - Cl \xrightarrow[DE]{Na}$

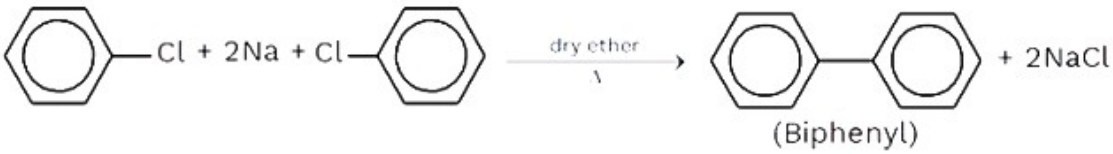
Sol. Simple Wurtz → $CH_3CH_2 - Cl \xrightarrow[DE]{Na}$ ⌐¬

Q214 Mixed Wurtz → $H_3C—Cl + Et—Cl \xrightarrow[DE]{Na}$

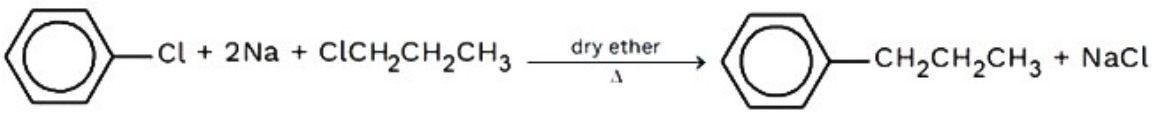
Sol.

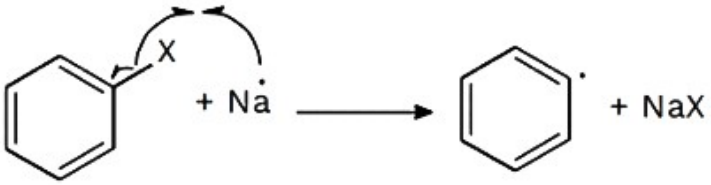


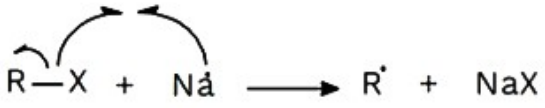
Fittig Reaction:

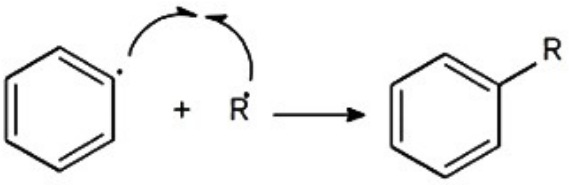


Wurtz Fittig Reaction:

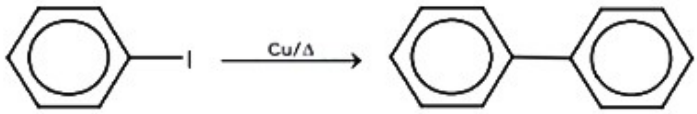


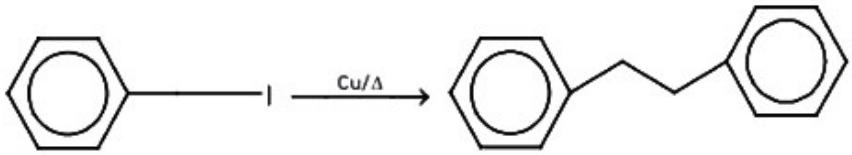


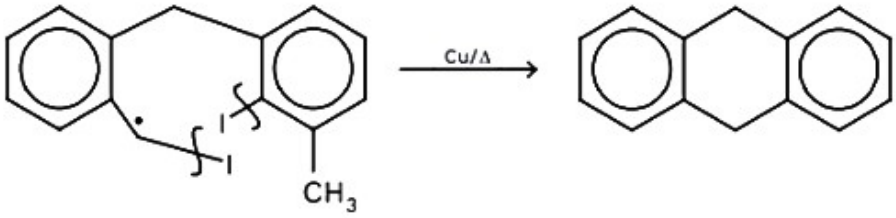




Ullmann Reaction :

1. 

2. 

3. 

Corey-House Synthesis

This reaction is used for the preparation of even, odd, symmetrical, and unsymmetrical alkanes because in Wurtz reaction, unsymmetrical and odd number of alkanes cannot be prepared efficiently.

$$RX + 2Li \xrightarrow{\text{Ether}} RLi + LiX$$

$$2RLi + CuI \rightarrow R_2CuLi + LiI$$

Dialkyl lithium cuprate

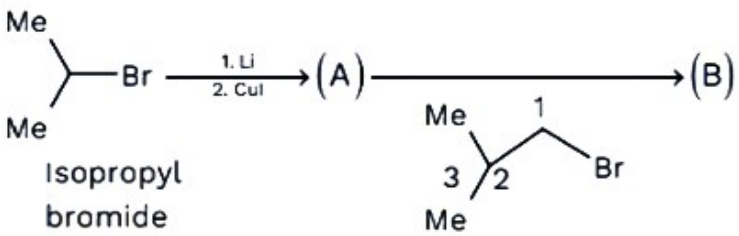
$$R_2CuLi \xrightarrow{R'X} R - R' + RCu + LiX$$

Example:

$$CH_3CH_2Cl \xrightarrow[2.\ CuI]{1.\ Li} (CH_3CH_2)_2\ CuLi \xrightarrow{CH_3CH_2Cl} CH_3CH_2CH_2CH_3$$

Note: The R'X that reacts with R_2CuLi should be 1° and C° not 3° R'X.

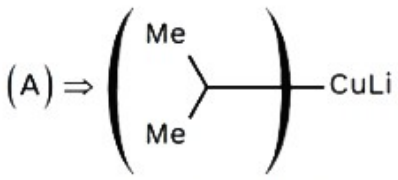
Q215



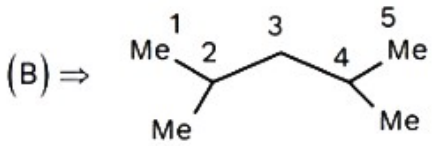
Isopropyl bromide

Isobutyl bromide or 1-Bromo-2-methyl propane

Sol.

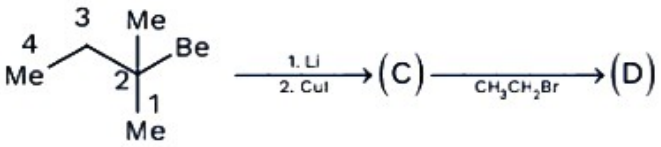


Di-isopropyl lithium cuperate



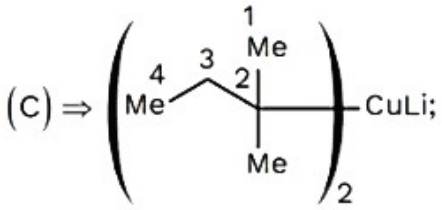
2,4-Dimethyl pentane

Q216

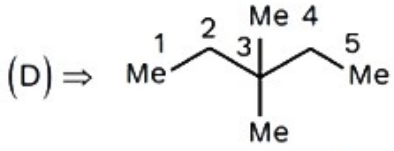


2-Bromo-2-methyl butane

Sol.

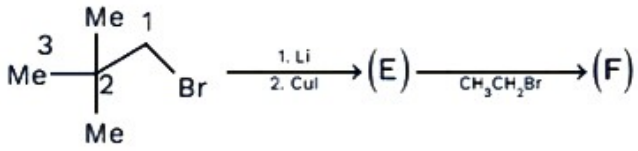


Di-(2-methyl butyl) lithium cuprate



3,3-Dimethyl pentane

Q217

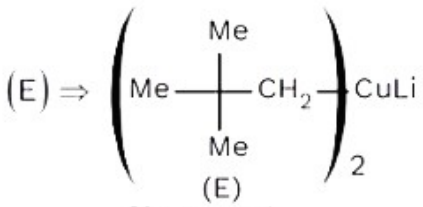


1-Bromo-2,2-dimethyl propane

or

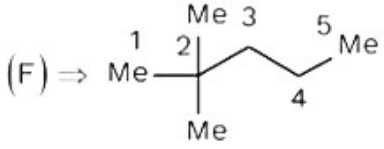
(Neopentyl bromide)

Sol.



(E)

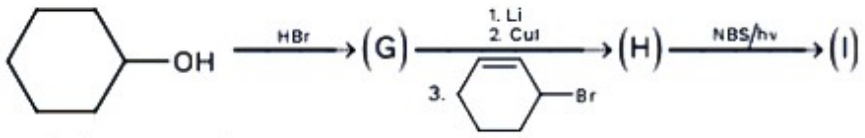
Di-neopentyl lithium cuperate



(F)

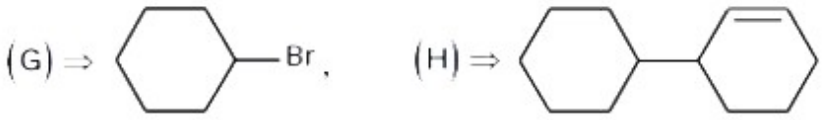
2,2-Dimethylpentane

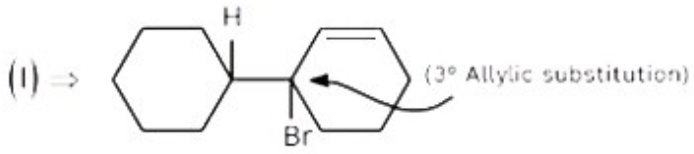
Q218



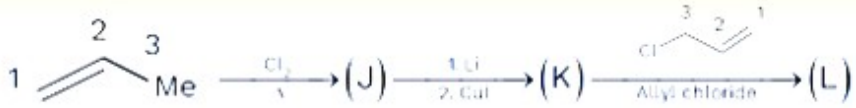
Cyclohexan-1-ol

Sol.

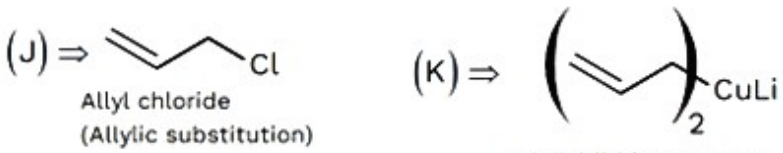




Q219

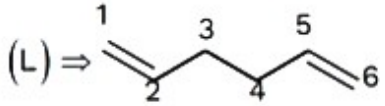


Sol.



Allyl chloride
(Allylic substitution)

Diallyl lithium cuprate



Hexa-1,5-diene

Decarboxylation

The process of removal of CO_2 from an organic compound is called decarboxylation. When a carboxylic acid is heated with soda lime [NaOH + CaO (3:1)] at 630 K, an alkane with one less C atom is formed.

$$RCOOH + NaOH \rightarrow RCOONa + H_2O$$

$$RCOONa + NaOH \xrightarrow{CaO,\ 630K} R - H + Na_2CO_3$$

The purpose of CaO is soda lime is to keep NaOH dry because it is quite hydroscopic in nature.

Q220 **Sodium salt of which acid will be needed for the preparation of propane? Write chemical equation for the reaction.**

Sol. Butanoic acid, $CH_3CH_2CH_2COO^{\oplus}Na^{\oplus} + NaOH \xrightarrow{CaO} CH_3CH_2CH_3 + Na_2CO_3$

Kolbe's Electrolytic Reaction

Electrolysis of an aqueous solution of sodium or potassium salt of a fatty acid gives higher alkane, and electrolysis of an aqueous solution of sodium or potassium salt of dibasic acid gives alkene, whereas the electrolysis of aqueous solution of sodium or potassium salt of dibasic unsaturated fatty acid gives alkyne.

$$2RCOONa \xrightarrow{\text{Electrolysis}} R - R + 2CO_2 + 2NaOH + H_2$$

Mechanism (Free Radical)

$$2RCOONa \xrightarrow{\text{Electrolysis}} 2RCOO^{\ddot{\ominus}} + 2Na^{\oplus}$$

At Anode (Oxidation)

$$2RCOO^{\ominus} \longrightarrow 2\overset{\ominus}{e} \longrightarrow 2RCO\dot{O} \longrightarrow 2R^{\cdot} + 2CO_2 \uparrow \rightarrow \underset{\text{Alkane}}{R - R}$$

At Cathode (Reduction)

Reduction of H_2O takes place since reduction potential of H_2O is greater than $Na^{\oplus}$.

$$2H_2O + 2\overset{\ominus}{e} \longrightarrow 2\overset{\ominus}{O}H + H_2 \uparrow$$

Points to remember

Methane cannot be prepared by this method.

Q221 $H\!:\!\ddot{C}\!\ddot{O}\!\ddot{O}\!:\!Na \xrightarrow{\text{Electrolysis}}$

Sodium formate

Sol. $H\!:\!\ddot{C}\!\ddot{O}\!\ddot{O}\!:\!Na \xrightarrow{\text{Electrolysis}} \underset{\text{At anode}}{\boxed{H_2(g) + CO_2(g)}} + \underset{\text{At cathode}}{\boxed{H_2(g) + 2\overset{\ominus}{O}H}}$

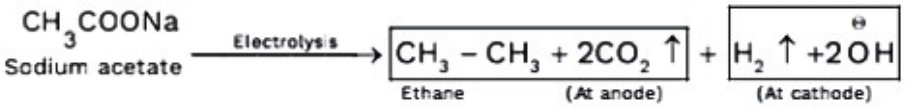
Sodium formate

Q222



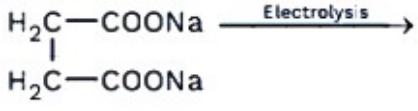
Sodium formate

Sol.



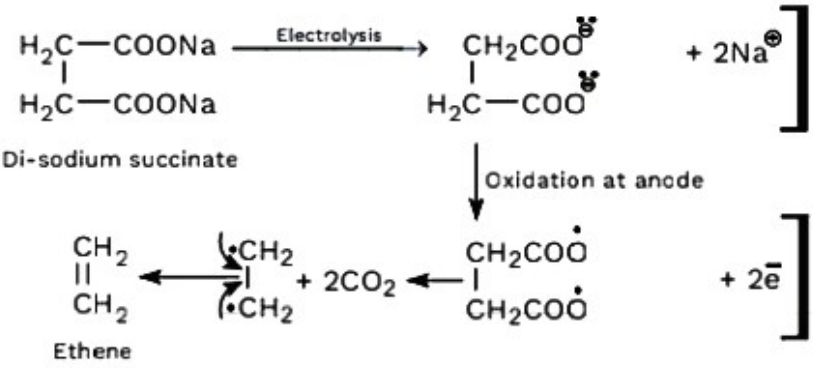
Sodium acetate

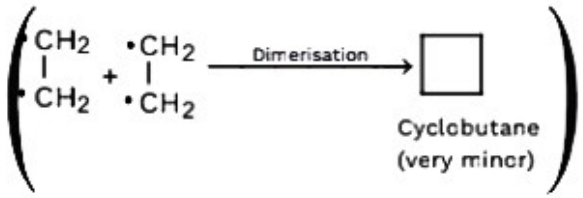
Q223

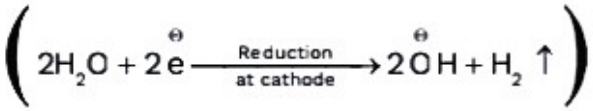


Di-sodium succinate

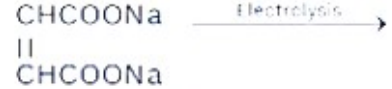
Sol.





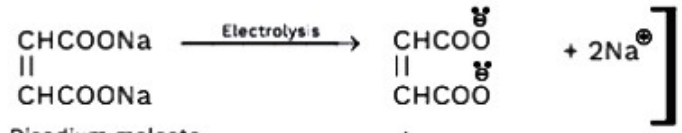


Q224



Disodium maleate
or Disodium fumerate

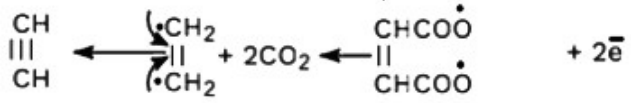
Sol.



Disodium maleate
or Disodium fumerate

Oxidation at anode

(i)



Ethene or
acetylene

(ii)

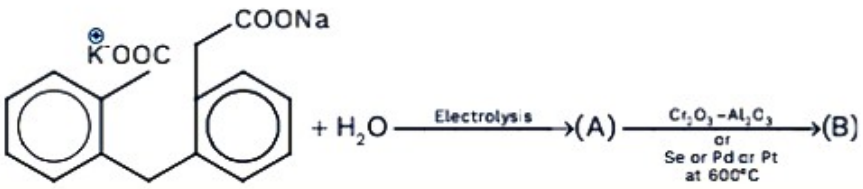


Cyclobutane
(very minor)

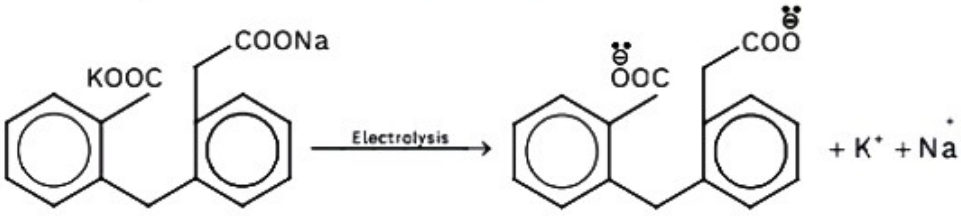


Complete the following reactions:

Q225

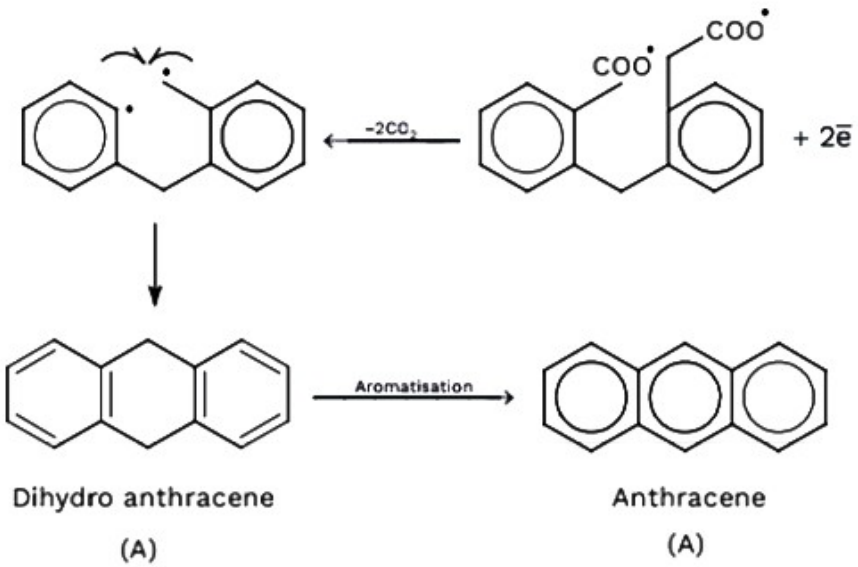


Sol. It is example of intramolecular Kolbe's reaction.

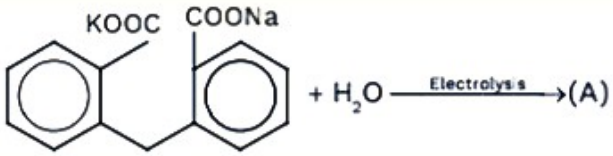


Oxidation at anode

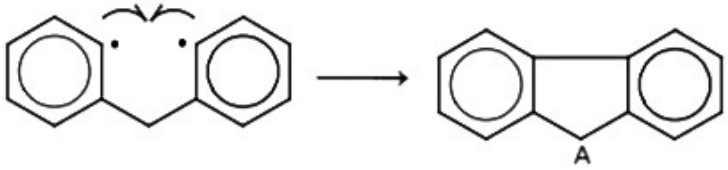
Sol.



Q226



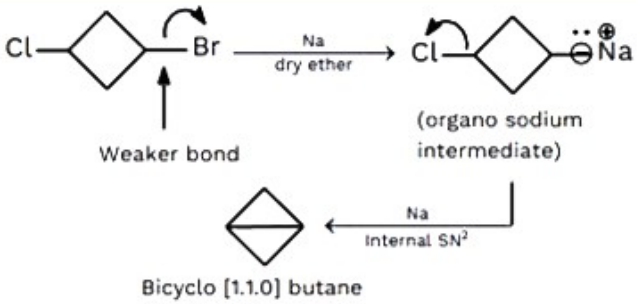
Sol. Proceed in the same way as in (a)



Q227

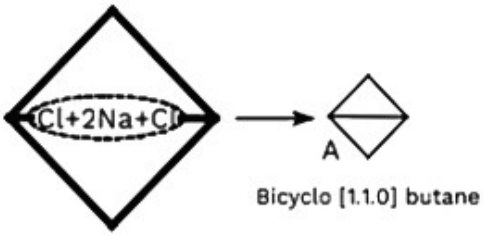


Sol.

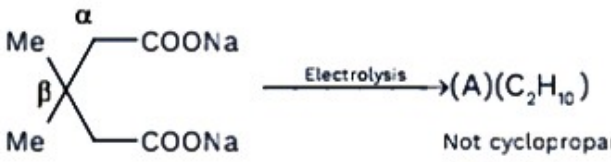


Q228 

Sol. Here, both (C–Cl) bonds are of equal strength and are broken simultaneously. Some explanation as in (c) above.

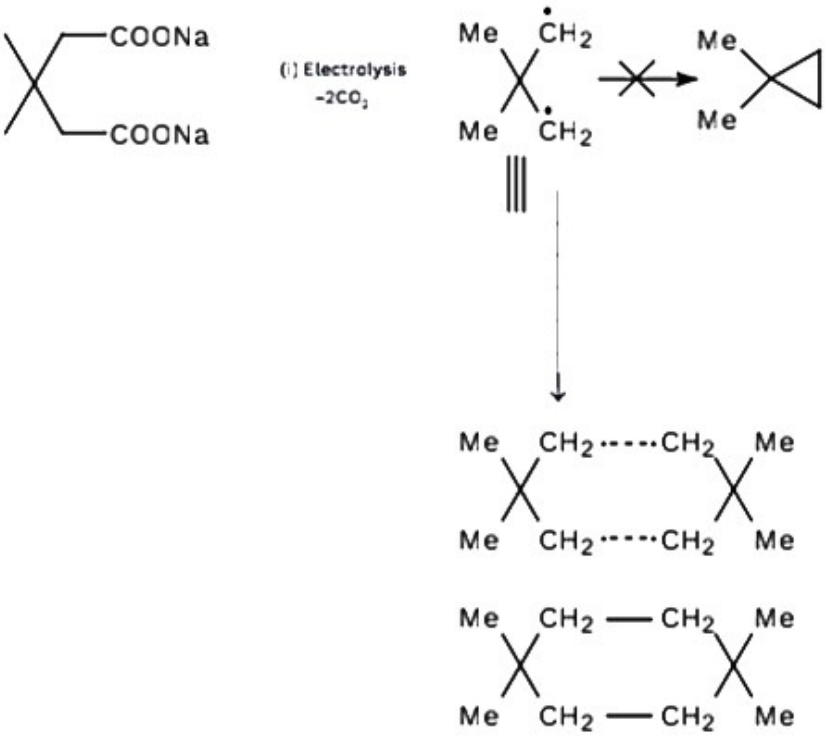


Bicyclo [1.1.0] butane

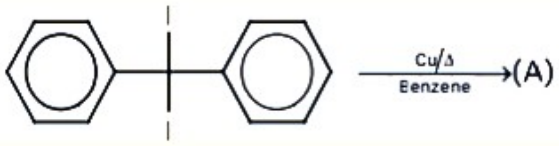
Q229 

β,β-Dimethyl glutaric acid

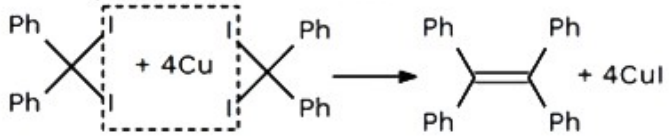
Not cyclopropane derivative

Sol. 

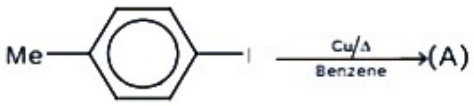
Q230



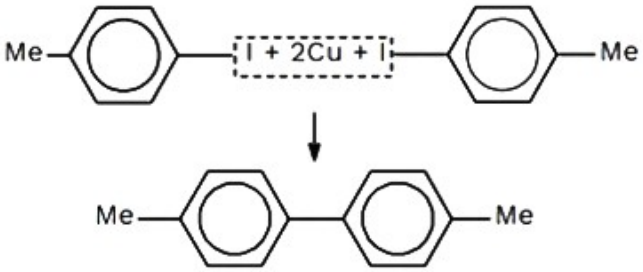
Sol. It is an example of Ullmann reaction.



Q231



Sol. Proceed in the same way as in (f).



4,4'-Dimethyl diphenyl
(A)